

CHEMICAL BONDING

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NEET SYLLABUS

Valence electrons, ionic bond, covalent bond, bond parameters, Lewis structure, polar character of covalent bond, valence bond theory, resonance, geometry of molecules, VSEPR theory, concept of hybridization involving s, p and d orbitals and shapes of some simple molecules, molecular orbital theory of homonuclear diatomic molecules (qualitative idea only). Hydrogen bond.

OBJECTIVES

After studying this unit, you will be able to :

- understand Kössel-Lewis approach to chemical bonding;
- explain the octet rule and its limitations, draw Lewis structures of simple molecules;
- explain the formation of different types of bonds;
- describe the VSEPR theory and predict the geometry of simple molecules;
- explain the valence bond approach for the formation of covalent bonds;
- predict the directional properties of covalent bonds;
- explain the different types of hybridisation involving s, p and d orbitals and draw shapes of simple covalent molecules;
- describe the molecular orbital theory of homonuclear diatomic molecules;
- explain the concept of hydrogen bond.

The meeting of two personalities is like the contact of two chemical substances; if there is any reaction, both are transformed.

Carl Jung

CHEMICAL BONDING

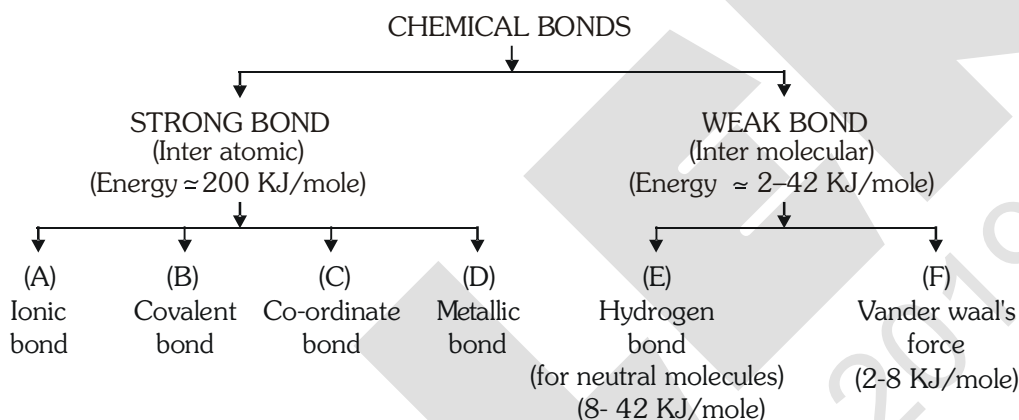
2.0 INTRODUCTION

- It is well known fact that except inert gases, no other element exists as independent atom under ordinary conditions.
- Most of the elements exist as molecules which are cluster of atoms. How do atoms combine to form molecules and why do atoms form bonds? Such doubts will be discussed in this chapter.
- A molecule will only be formed if it is more stable and has a lower energy, than the individual atoms.

Chemical Bond

- A force that acts between two or more atoms to hold them together as a stable molecule.
- It is union of two or more atoms involving redistribution of e^- among them.
- This process accompanied by decrease in energy.
- Decrease in potential energy (P.E.) $\propto$ Strength of the bond.
- Therefore molecules are more stable than atoms.

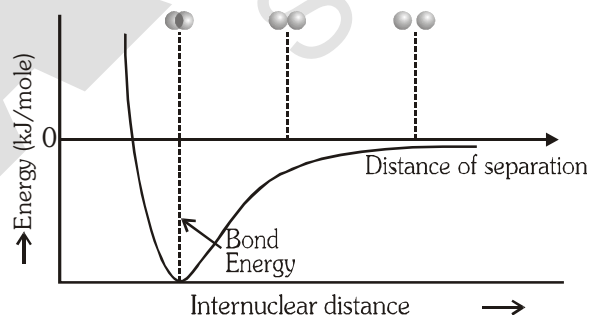
Classification of Bonds



Cause of Chemical Combination

(A) Tendency to acquire minimum energy

- When two atoms approaches to each other. Nucleus of one atom attracts the electron of another atom.
- Two nuclei and electrons of both the atoms repels each other.
- If net result is attraction, the total energy of the system (molecule) decreases and a chemical bond forms.
- So Attraction $\propto 1/\text{energy} \propto$ Stability.
- Bond formation is an exothermic process

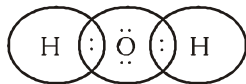


(B) Tendency to acquire noble gas configuration :

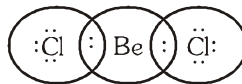
- Atom combines to acquire noble gas configuration.
- Only outermost electrons i.e. ns , np and $(n-1)d$ shells electrons participate in bond formation.
- Inert gas elements do not participate in bond formation, as they have stable electronic configuration hence will have minimum energy. (Stable configuration $1s^2$ or ns^2np^6)

2.1 KOSSEL - LEWIS APPROACH TO CHEMICAL BONDING

- Every atom has a tendency to complete its octet in outermost shell
- H has the tendency to complete its duplet.
- To acquire inert gas configuration atoms lose or gain electron or share electron.
- The tendency of atoms to achieve eight electrons in their outer most shell is known as Lewis octet rule.



Obeys octet rule



Doesn't obey octet rule

Exception of Octet Rule

(a) Incomplete octet molecules :- or (electron deficient molecules)

Compound in which octet is not complete in outer most orbit of central atom.

Example - Halides of IIIA groups, BF_3 , AlCl_3 , BCl_3 , hydride of III A/ 13^{th} group etc.

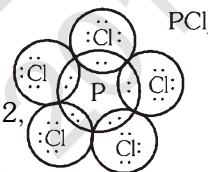


Other examples – BeCl_2 ($4e^-$), HgCl_2 ($4e^-$), $\text{Ga}(\text{CH}_3)_3$ ($6e^-$)

(b) Expansion of octet or (electron efficient molecules)

Compound in which central atom has more than $8e^-$ in outermost orbits.

Example - In PCl_5 , SF_6 and IF_7 the central atom P, S and I contain 10, 12, and 14 electrons respectively.



(c) Pseudo inert gas configuration :-

Cations which contain 18 electrons in outermost orbit

Ex. Ga^{+3} , Cu^+ , Ag^+ , Zn^{+2} , Cd^{+2} , Sn^{+4} , Pb^{+4} etc.

Electronic configuration of Ga - $1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^{10}, 4s^2 4p^1$

Electronic configuration of Ga^{+3} - $1s^2, 2s^2, 2p^6, \underbrace{3s^2 3p^6 3d^{10}}_{18e^-}$

(d) Cations having electron between 9 to 17 in their outer most shell

Ex. Mn^{+2} , Fe^{+2} , Fe^{+3} , Ti^{+2} etc.

Electronic configuration of Fe - $1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^6, 4s^2$

Electronic configuration of Fe^{+3} - $1s^2, 2s^2 2p^6, \overbrace{3s^2 3p^6 3d^5}^{\text{(less than } 18e^-)}}$

(e) Odd electron molecules :-

Central atom have an unpaired electron or odd no ($7e^-$, $11e^-$ etc) of electrons in their outer most shell.

e.g. NO, NO₂, ClO₂ etc.

VALENCY

It is defined as the combining capacity of the elements. The word valency is derived from an Italian word "Valentia" which means combining capacity.

Old concept : Given by : **Frankland**

Valency with respect to Hydrogen : Valency of H = 1

It is defined as the number of hydrogen attached with a particular element.

	IA	IIA	IIIA	IVA	VA	VIA	VIIA
	NaH	MgH ₂	AlH ₃	SiH ₄	PH ₃	H ₂ S	HCl
Valency	1	2	3	4	3	2	1

Note : Valency w.r.t. H across the period increases upto 4 and then again decreases to 1.

Valency with respect to oxygen : Valency of 'O' = 2

It is defined as twice the number of oxygen atoms attached with a particular atom.

	IA	IIA	IIIA	IVA	VA	VIA	VIIA
	Na ₂ O	MgO	Al ₂ O ₃	SiO ₂	P ₂ O ₅	SO ₃	Cl ₂ O ₇
Valency	1	2	3	4	5	6	7

Note : Valency with respect to oxygen increases from 1 to 7 across the period. Valency w.r.t. 'O' is equal to the group number.

New concept : This concept is based on the electronic configuration. According to this concept valency for IA to IVA group elements is equal to number of valence shell e⁻ and from VA to zero group, it is -

$$[8 - (\text{number of valence } e^-)]$$

$$\text{Valency} = \text{No. of valence } e^-$$

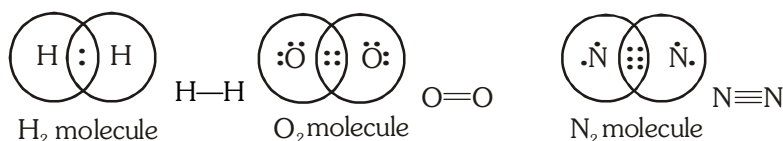
$$\text{Valency} = (8 - \text{no. of valence } e^-)$$

	Valency = No. of valence e ⁻				Valency = (8 - no. of valence e ⁻)			
	IA	IIA	IIIA	IVA	VA	VIA	VIIA	0
	ns ¹	ns ²	ns ² np ¹	ns ² np ²	ns ² np ³	ns ² np ⁴	ns ² np ⁵	ns ² np ⁶
Valence shell e ⁻	1	2	3	4	5	6	7	8
Valency	1	2	3	4	3	2	1	0
					(8 - 5) = 3			(8 - 8) = 0

Note : All the elements of a group have same valencies because they have same number of valence shell electrons.

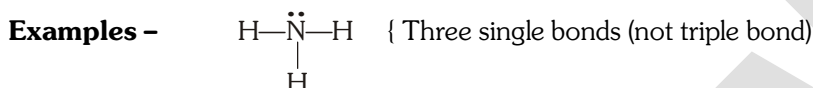
2.2 COVALENT BOND

- A covalent bond is formed by the mutual sharing of electrons between two atoms of electro negative elements to complete their octet. (Except H which completes its duplet)



- The shared pair of electrons should have opposite spins, and are localised between two atoms concerned.
- Sharing of electrons may occurs in three ways –

No. of electrons shared between two atoms	Electron pair	Bond.
2	1	Single bond (—)
4	2	Double bond (=)
6	3	Triple bond (≡)



$\text{N}\equiv\text{N}$ Triple bond. (not three single bonds) $\text{O}=\text{O}$ Double bond (Not two single bonds) $\text{H}-\text{O}-\text{H}$ (Two single bonds.)

Orbital Concept of Covalent Bond

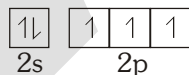
- One orbital can accommodate maximum 2 electrons with opposite spins like $\uparrow\downarrow$
- Half filled orbital or unpaired electron orbital share one electron from another atom, to complete its orbital.
- Tendency to complete orbital or to pair the electron is an essential condition of covalent bond. Completion of octet is not the essential condition of covalent bond.
- Covalency** : It is the number of covalent bonds which an atom makes in a molecule.
- If the outermost orbit has empty orbitals then covalent bonds are formed in excited state.

2.3 VARIABLE VALENCY IN COVALENT BONDS

- Variable valencies are shown by those elements which have empty orbitals in outermost shell.
- Lone pair electrons get excited in the subshell of the same shell to form the maximum number of unpaired electrons. Maximum covalency is shown in excited state.
- The energy required for excitation of electrons is called promotion energy.
- Promotion rule – Excitation of electrons in the same orbit.

Example –

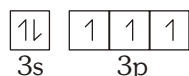
(a) Nitrogen → Ground state



Covalency 3 (NCl_3)

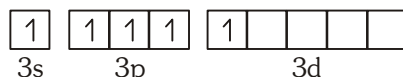
For Nitrogen → Excited states are not possible due to absence of vacant orbital that's why (NCl_5) does not exist

(b) Phosphorus → Ground state



Covalency 3 (PCl_3)

Phosphorus → Excited state



Covalency – 5 (PCl_5)

Note :

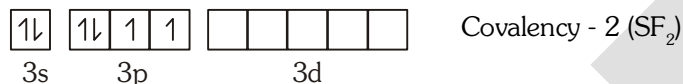
NCl_3 — exists

NCl_5 — doesn't exist (due to absence of d-orbitals in Nitrogen.) While PCl_3 and PCl_5 both exist because 3d orbitals are present in phosphorus.

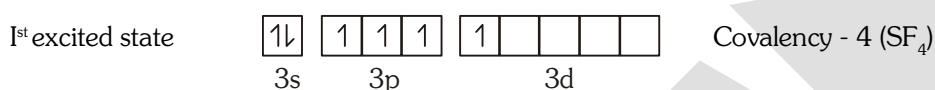
OF_2 — exists, but OF_4 and OF_6 doesn't exist due to absence of d-orbitals While SF_4 and SF_6 exist due to presence of d-orbital which are present in its valence shell.

- It can explain existence of molecules.

(c) Sulphur → Ground state.

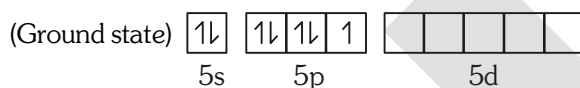


Sulphur → Excited state



So variable covalency of S is 2, 4, & 6.

(d) Iodine has three lone pair of electrons



So it shows three excited states – Maximum number of unpaired electrons = 7

Variable Valencies are 1, 3, 5 and 7

2.4 CHARACTERISTIC OF COVALENT COMPOUND

(a) Physical state :- Covalent compounds are found in all the three states - Gas, Solid & Liquid.

Separate molecules — In gaseous state

Associate molecules — In liquid & solid state

(Due to strong vander waal's force and hydrogen bonding among the molecules.)

As the molecular weight increases physical state changes -

eg.	F_2 and Cl_2	Br_2	I_2 , At_2
	gas	liquid	solid

—————→

Top to bottom in a group. Vander waal's force increases between the molecules.

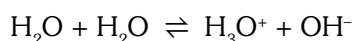
(b) Covalent solid :- Those solids in which atoms are linked together by covalent bonds, forms infinite three dimensional giant structure.

eg. Diamond, Graphite, AlN , SiC , SiO_2 etc.

Molecular solid :- Discrete (separate) molecules are formed by covalent bonds and then the molecules associates due to intermolecular force of attraction. (van der Waal force)

eg. Solid I_2 , dry ice (Solid CO_2) etc.

- (c) **Conductivity** :- Mostly covalent compounds are bad conductor of electricity. But few polar covalent compounds due to self ionisation can conduct electricity. eg. H_2O , liq. NH_3 etc.



Free ions are formed which can conduct electricity.

Exceptions :- Graphite, HCl in water.

- (d) **Solubility** :- Non polar compounds are soluble in non polar solvents. Non polar compounds forms vander waal bond with non polar solvent molecules.

Non polar solvents are C_6H_6 , CCl_4 etc.

- (e) **Isomerism** :- Covalent bond is rigid and directional, so it shows isomerism.

eg. Organic compounds.

- (f) **Reaction** :- Reaction between covalent compounds are slow. Because it involves breaking of old bonds and formation of new bonds.

BEGINNER'S BOX-1

- Which condition favours the bond formation:
 - (1) Maximum attraction and maximum potential energy
 - (2) Minimum attraction and minimum potential energy
 - (3) Minimum potential energy and maximum attraction
 - (4) None of the above
- Which one of the following element will never obey octet rule:
 - (1) Na
 - (2) F
 - (3) S
 - (4) H
- Which is not an exception to octet rule ?
 - (1) BF_3
 - (2) SiCl_4
 - (3) BeI_2
 - (4) ClO_2
- An oxide of chlorine which is an odd electron molecule is :
 - (1) ClO_2
 - (2) Cl_2O_6
 - (3) Cl_2O_7
 - (4) Cl_2O

2.5 VALENCE BOND THEORY (VBT) :

- (A) It was presented by Heitler & London to explain how a covalent bond is formed.

It was extended by Pauling & Slater.

- (B) The main points of theory are –

- To form a covalent bond overlapping occurs between half filled valence shell orbitals of the two atoms.
- Resulting bond acquires a pair of electrons with opposite spins to get stability.
- Orbitals come closer to each other from the direction in which there is maximum overlapping
- So covalent bond has directional character.
- Strength of covalent bond $\propto$ extent of overlapping.
- Extent of overlapping depends on two factors.
 - Nature of orbitals – p, d and f are directional orbitals $\rightarrow$ more overlapping
s-orbital $\rightarrow$ non directional – less overlapping

- (ii) Nature of overlapping – Co-axial overlapping - extent of overlapping more.
Collateral overlapping - extent of overlapping less

Order of strength of Co - axial overlapping – $p - p > s - p > s - s$



- (g) As the value of n increases, bond strength decreases.

$$1 - 1 > 1 - 2 > 2 - 2 > 2 - 3 > 3 - 3$$

$$1s - 2p > 2s - 2p > 3s - 3p$$

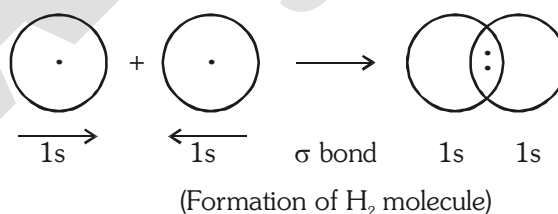
- (h) If n is same $2p - 2p > 2s - 2p > 2s - 2s$
- (i) Electron which is already paired in valency shell can enter into bond formation, if they can be unpaired first and shifted to vacant orbitals of slightly higher energy of the same energy shell.
- (j) This point can explain the trivalency of boron, tetravalency of carbon, pentavalency of phosphorus etc.
- (k) Two types of bonds are formed on account of overlapping.
(A) Sigma (σ) bond (B) Pi (π) bond

Sigma (σ) Bond

- (a) Bond formed between two atoms by the overlapping of half filled orbitals along their axis (end to end overlap) is called sigma bond.
- (b) σ bond is directional.
- (c) σ bond do not take part in resonance.
- (d) Free rotation is possible about a single σ bond.
- (e) Maximum overlapping is possible between electron clouds and hence it is strong bond.
- (f) There can be only one σ bond between two atoms.

Sigma bonds are formed by four types of overlapping

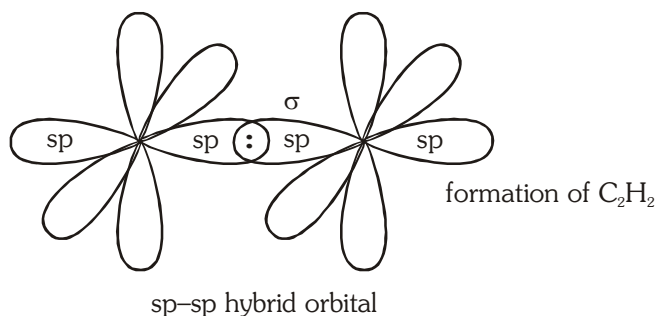
- (i) $s - s$ overlapping – Two half filled s -orbitals overlap along the internuclear axis. Ex. H_2 molecule.



- (ii) $s - p$ overlapping (Formation of HF) – When half fill s -orbital of one atom overlap with half filled p -orbital of other atom.

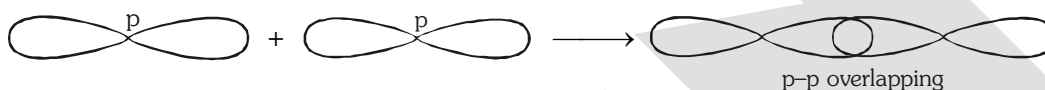


- (iii) Bond between two hybrid orbitals – $sp^3 - sp^3$, $sp^2 - sp^2$, $sp^3 - sp^2$, $sp^3 - sp$ etc.



Note : overlapping of hybrid orbitals form σ bond.

- (iv) p - p overlapping – (Coaxial) – It involves the coaxial overlapping between half filled p-orbitals of two different or same atoms.

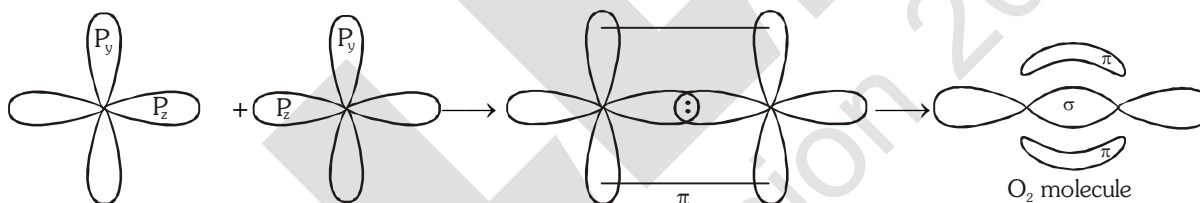


Ex. Formation of Cl_2 , F_2 , Br_2

Pi(π)-Bond

- (a) The bond formed by sidewise (lateral) overlapping are known as π bonds.
- (b) Lateral overlapping is only partial, so formed are weaker and hence more reactive than σ bonds (Repulsion between nucleus is more as orbitals have to come much close to each other for π bonds formation)

Example – Formation of O_2 molecule –



Note : Only two orbitals of oxygen atom have unpaired e^- in each orbital for bonding.

Electron configuration of oxygen is – $1s^2 2s^2 2p_x^2 2p_y^1 2p_z^1$

- (c) Free rotation about a π bond is not possible.
- (d) π bond is weaker than σ bond (Bond energy difference is 63.5 KJ or 15 K cal/mole)
- (e) π bonds are less directional, so do not determine the shape of a molecule.
- (f) π bond takes part in resonance.
- (g) π bond formed by pure or unhybrid orbitals.

Comparison between σ and π bond

σ bond	π bond
1. Formed by axial overlapping	1. Formed by side by side overlapping
2. Involves s-s, s-p, p-p (axial) & hybrid orbitals	2. Involve p-p, p-d & d-d orbital
3. Extent of overlapping is more so stronger	3. Extent of overlapping is less so weaker
4. Free rotation around σ bond is possible	4. Free rotation around π bond is not possible
5. Hybridized or unhybridized orbital forms σ bond	5. Hybridized orbital never forms π bond
6. Independent existence of σ -bond.	6. No independent existence.

BEGINNER'S BOX-2

- According to the valence bond theory, when a covalent bond is formed between two reacting atoms, the potential energy of the system becomes—
 (1) negative (2) positive (3) minimum (4) maximum
- The strongest covalent bond is formed by the overlap of— (If considering for same shell)
 (1) s and p orbitals (2) s and s orbitals
 (3) p and d orbitals (4) p and p collateral orbitals
- In a triple bond there is sharing of :—
 (1) 3—electrons (2) 4—electrons (3) Several electrons (4) 6—electrons
- Which of the following configuration shows second excitation state of Iodine:—
 (1)

1↓	1↓	1	1	1				
----	----	---	---	---	--	--	--	--

 (2)

1↓	1	1	1	1	1	1		
----	---	---	---	---	---	---	--	--

 (3)

1↓	1	1	1	1	1			
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 (4)

1	1	1	1	1	1	1		
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- Variable covalency is exhibited by:—
 (1) P and S (2) N and O (3) N and P (4) F and Cl
- Which of the following bonds will have directional character
 (1) Ionic bond (2) Metallic bond
 (3) Covalent bond (4) Both covalent & metallic
- Number of σ and π bonds present in $\text{CH}_3 - \text{CH} = \text{CH} - \text{C} \equiv \text{CH}$ are -
 (1) 10 σ , 3 π (2) 10 σ , 2 π (3) 9 σ , 2 π (4) 8 σ , 3 π
- Which of the following statements regarding covalent bond is not true ?
 (1) The electrons are shared between atoms
 (2) The bond is non-directional
 (3) The strength of the bond depends upon the extent of overlapping
 (4) The bond formed may or may not be polar
- Predict the nature of bond

Orbitals	Internuclear axis	Bond
s + s	any axis	
s + p _x	x-axis	
s + p _y	y-axis	
s + p _z	z-axis	
p _x + p _x	x-axis	
p _y + p _y	y-axis	
p _z + p _z	z-axis	
p _x + p _z	y or z-axis	
p _y + p _y	x or z axis	
p _z + p _z	x or y axis	

2.6 HYBRIDISATION

Consider an example of Be compound :-

If it is formed without hybridisation then $\text{Cl} \overset{\text{p}}{\text{---}} \text{Be} \overset{\text{p}}{\text{---}} \text{Cl}$

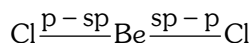
both the Be-Cl bonds should have different parameters and $\text{p-p bond strength} > \text{s-p bond strength}$. But practically bond strength and distance of both the Be-Cl bonds are same.

This problem may overcome if hybridisation of s and p-orbital occurs.

Hybridisation

- (1) It is introduced by Pauling, to explain equivalent nature of covalent bonds in a molecule.
- (2) **Definition** : Mixing of different shape and approximate equal energy atomic orbitals, and redistribution of energy to form new orbitals, of same shape & same energy. These new orbitals are called hybrid orbitals and the phenomenon is called hybridisation.

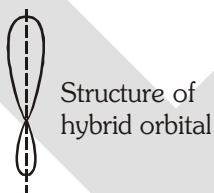
Now after considering s-p hybridisation in BeCl_2



(Bond strength of both the bonds will be equal)

Characteristic of Hybridisation

- (1) Hybridisation is a mixing of orbitals and not electrons. Therefore in hybridisation full filled, half filled and empty orbitals may take part.



- (2) Number of the hybrid orbitals formed is always be equivalent to number of atomic orbital which have taken part in the process of hybridisation.
- (3) Each hybrid orbital having two lobes, one is larger and other is smaller. Bond will be formed from large lobe.
- (4) The number of hybrid orbitals on central atom of a molecule or ion = number of σ bonds + lone pair of electron.
 - (i) The 1st bond between two atoms will be sigma.
 - (ii) The other bond between same two atoms will be pi bond.
 - (iii) The electron pair of an atom which do not take part in bond formation called as lone pair of electron.
- (5) One element can represent many hybridisation state depending on experimental conditions for example, C showing sp , sp^2 and sp^3 hybridisation in its compounds.
- (6) Hybrid orbitals are differentiated as sp , sp^2 , sp^3 etc.
- (7) The repulsion between $lp-lp > lp-bp > bp-bp$
- (8) The directional properties in hybrid orbital is more than atomic orbitals. Therefore hybrid orbitals form stronger sigma bond. The directional property of different hybrid orbitals will be in following order.

$$sp < sp^2 < sp^3 < sp^3d < sp^3d^2 < sp^3d^3$$

DETERMINATION OF HYBRIDISATION STATE

Method (I) : Number of hybrid orbital = number of σ bond + number of lone pair [surrounding the central atom]

Method (II) : To predict hybridisation following formulae may be used :

$$\text{No. of hybrid orbital} = \frac{1}{2} [Ve^- + SA \pm C]$$

[Ve^- = Total number of valence e^- in the central atom, SA = total number of monovalent atoms; C = charge]

eg. NH_4^+ $\frac{1}{2} [5 + 4 - 1] = 4$ sp^3 hybridisation.

SF_4 $\frac{1}{2} [6 + 4] = 5$ sp^3d hybridisation.

SO_4^{2-} $\frac{1}{2} [6 + 2] = 4$ sp^3 hybridisation.

('O' is divalent so add only charge on anion)

NO_3^- $\frac{1}{2} [5 + 1] = 3$ sp^2 hybridisation.

If such type of e^- pairs are –

two	–	sp	hybridisation
three	–	sp^2	hybridisation
four	–	sp^3	hybridisation
five	–	sp^3d	hybridisation
six	–	sp^3d^2	hybridisation
seven	–	sp^3d^3	hybridisation

S.No.	Formula	Total pair of e^-		Hybridisation	Geometry	Ex.
		bp	lp			
1.	AB_2	2	0	sp	Linear	$BeCl_2, CO_2$
2.	AB_3	3	0	sp^2	Trigonal Planar	BCl_3, BF_3
3.	AB_4	4	0	sp^3	Tetrahedral	CH_4, CCl_4
4.	AB_5	5	0	sp^3d	Trigonal bipyramidal	PCl_5
5.	AB_6	6	0	sp^3d^2	Octahedral (Square bipyramidal)	SF_6
6.	AB_7	7	0	sp^3d^3	Pentagonal bipyramidal	IF_7

Position of lone pair & multiple bond

(i) $sp/sp^2/sp^3$ = Any where

(ii) sp^3d = equatorial

(iii) sp^3d^2 = axial (defined first)

(iv) sp^3d^3 $\begin{cases} \text{Lone pair} = 1 \text{ then equatorial} \\ \text{Lone pair} = 2 \text{ then axial} \end{cases}$

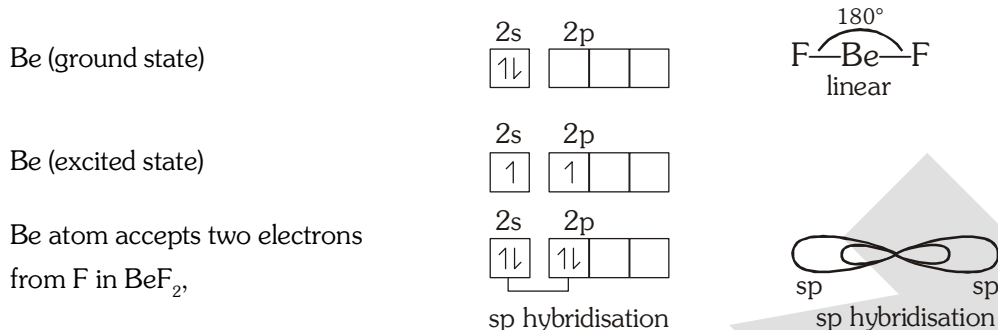
(v) sp^3d hybridisation Axial bond length > equatorial bond length

sp^3d^3 hybridisation Axial bond length < equatorial bond length

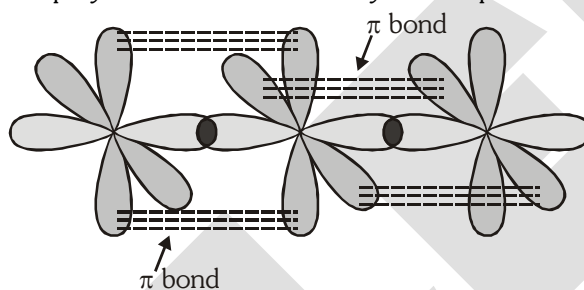
} terminal atom same

Types of Hybridisation**(A) sp hybridisation :**

- In this hybridisation one s- & one p- orbital of an atom are mixed to give two new hybrid orbitals which are equivalent in shape & energy known as sp hybrid orbitals.
- These two sp hybrid orbitals are arranged in a straight line & at bond angle 180° .
- s-character 50%

**CO₂ Molecule (O = C = O) :**

In CO₂ molecule, C has two sp hybrid orbitals & two unhybridised p orbitals.

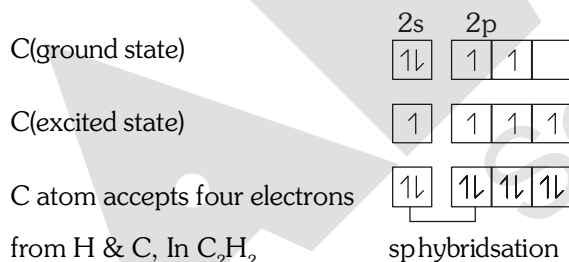
Molecular orbital picture of CO₂

Thus, CO₂ molecule is linear in shape & having 180° bond angle.

The bond length between C-O bond is reduced due to the presence of π bond.

CH $\equiv$ CH [H-C_A $\equiv$ C_B-H]

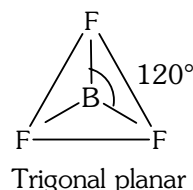
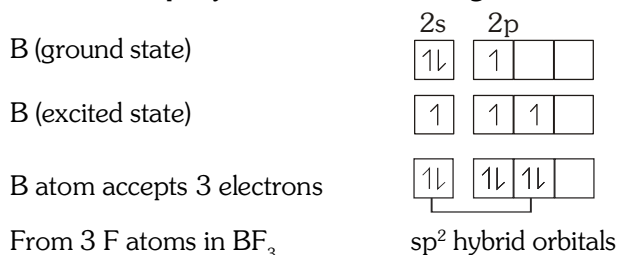
In CH $\equiv$ CH molecules, each C atom contains two sp hybrid orbitals & two unhybridised p orbitals



- sp hybrid orbital of each C overlaps to give sigma bond between C - C.
- The remaining one sp hybrid orbital of each C atom overlaps with s orbital of H, forming sigma bond between C - H.
- The two unhybridised p orbitals of each C atom (p_y and p_x) overlap laterally to form two $\pi(\pi)$ bonds.
- Therefore in H-C_A $\equiv$ C_B-H
 - sigma bond between C_A - C_B is formed sp - sp overlapping
 - sigma bond between C_A - H is formed sp - s overlapping
 - sigma bond between C_B - H is formed sp - s overlapping
 - π bond between C_A-C_B is formed : $p_y - p_y$, $p_x - p_x$ overlapping
- Each C atom forms two sigma bonds but in C₂H₂, total sigma bonds are 3
- Each C atom forms two π bonds. Total π bonds in C₂H₂ are two
- Total number of bonds in acetylene are : $3\sigma + 2\pi$ bond = 5 bonds.

(B) sp^2 Hybridisation :

- (a) In this hybridisation one s & two p orbitals are mixed to give three new sp^2 hybrid orbitals which are in same shape & equivalent energies.
 (b) These three sp^2 hybrid orbitals are at angle of 120° & giving trigonal planar shape.

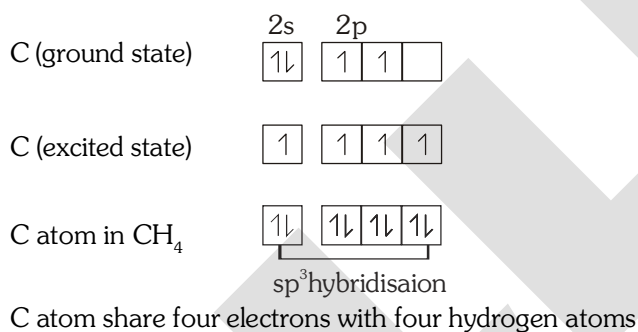


- (c) s - character 33.3 %

- SnX_2 having two σ bonds & one l.p. electron therefore hybridisation is sp^2
- The bond angle in SnX_2 will be less than 120° (due to presence of one l.p.e⁻).
- The shape of SnX_2 molecule is bent.

(C) sp^3 Hybridisation :

- (I) In this hybridisation one s orbital and three p orbitals of an atom of a molecule or ion, are mixed to give four new hybrid orbitals called as sp^3 hybrid orbitals.
 (II) The angle between hybrid orbitals will be $109^\circ 28'$ (6)

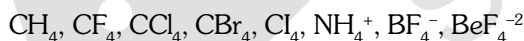


- (III) The shape obtained from these hybrid orbitals would be tetrahedron.

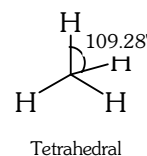
Three following examples represent this condition.

(a) Four sigma bonds with zero lone pair electron :

- (I) The following examples represent this condition.



- (II) In above compounds, bond angle is $109^\circ 28'$ & tetrahedron shape.



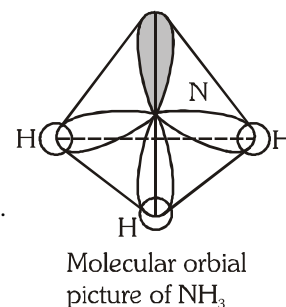
(b) Three sigma bonds & one lone pair of electron :

- (I) This condition is shown by following compounds & ions.



- (II) sp^3 hybridisation, pyramidal shape & bond angle will be less than $109^\circ 28'$.

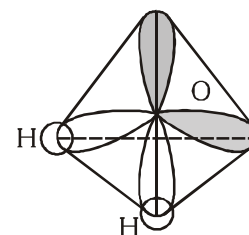
Due to the presence of one lone pair electron on nitrogen it repels bond pair electron more therefore bond angle reduced from $109^\circ 28'$ to 107° . The repulsion between lp - bp > bp - bp.



(c) Two sigma bonds & two lone pair of electrons :

- (I) This condition is shown by following compounds and ions.
 H_2O , OCl_2 , OBr_2 , OF_2 , OI_2 etc.
- (II) In all above examples, the central atom showing sp^3 hybridisation, angular shape and bond angle will be either less than $109^\circ 28'$ or more than $109^\circ 28'$.

In H_2O the hybridisation on O atom is sp^3 , but due to presence of two lone pair electrons they repel each other

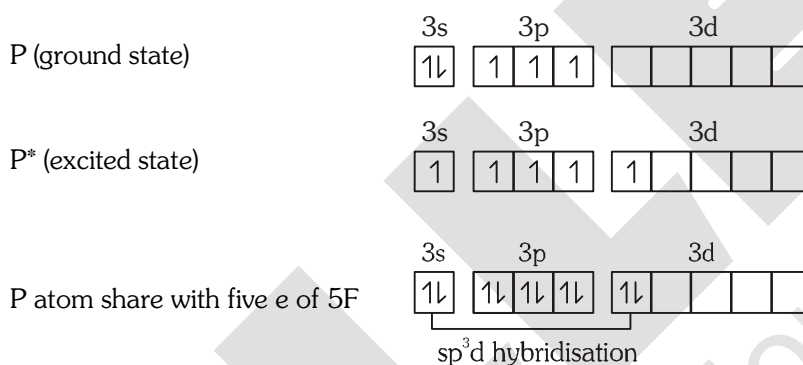


Molecular orbital picture of H_2O

(D) sp^3d Hybridisation :

- (I) In this hybridisation one s orbital, three p orbitals and one d orbital are mixed to give five new hybrid orbitals which are equivalent in shape and energy called as sp^3d hybrid orbitals.
- (II) Out of these five hybrid orbitals, three hybrid orbitals are at 120° angle and two hybrid orbitals are perpendicular to the plane of three hybrid orbitals that is trigonal planar, the shape of molecule becomes trigonal bipyramidal.

For example, PF_5 showing sp^3d hybridisation



- (III) In this hybridisation d_{z^2} orbital is hybridised with s and p orbitals.

In this way five sp^3d hybrid orbitals form five sigma bond with five F atoms and give a molecule of PF_5 , shape of this molecule is trigonal bipyramidal.

Two axial P-Cl bonds are longer than three equatorial P-Cl bonds due to repulsion between 3 equatorial bp of e^- and 2 axial b.p. of e^-

In above hybridisation, there are four conditions.

(a) Five sigma bonds and zero lone pair electron :

The following examples represent this conditions.

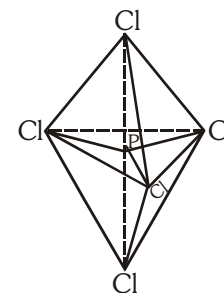
PCl_5 , PBr_5 , AsF_5 , AsCl_5 , SbCl_5 , SbF_5 etc.

The shape of all the above molecules is trigonal bipyramidal.

(b) Four sigma bonds and one lone pair of electron :

The following examples represent this condition.

SF_4 , SeF_4 , TeF_4 , PoF_4 , PF_4^- , SbF_4^- , SCl_4 , SeCl_4 , TeCl_4 etc.

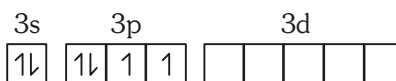


Structure of PCl_5

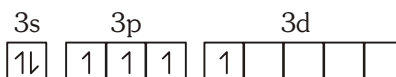
The shape of all above examples will be irregular tetrahedron or See-saw

Example SF_4

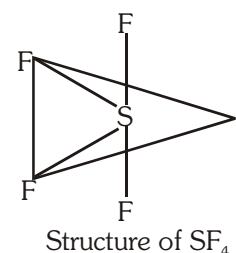
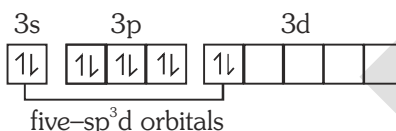
S ground state



S excited state



S atom share with four e^- from 4F atoms

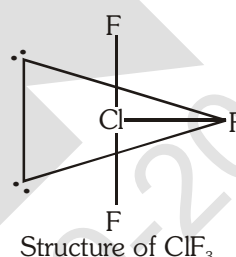


(c) Three sigma bonds & two lone pair of electrons :

The following examples represent this condition.

ClF_3 , BrF_3 , IF_3 , BrCl_3 , ICl_3 etc.

The shape of all above compounds is 'T' shape.



(d) Two sigma bonds & three lone pair of electrons :

The following examples represent this condition.

ICl_2^- , IBr_2^- , ClF_2^- , IF_2^- , BrF_2^- , XeF_2 , I_3^- , Br_3^-

The geometry of above examples will be linear shape.

(E) sp^3d^2 Hybridisation :

(I) In this hybridisation, one s-orbital, three p-orbitals & two d-orbitals (dz^2 , dx^2-y^2) are mixed to give six new hybrid orbitals known as sp^3d^2 hybrid orbitals.

(II) The geometry of molecule obtained from above six hybrid orbitals will be symmetrical octahedral or square bipyramidal.

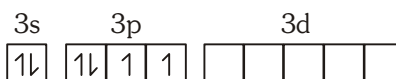
(III) The angle between all hybrid orbitals will be 90° .

Example : SF_6 , AlF_6^{3-} , PF_6^- , ICl_5 , XeF_4 , XeOF_4 , ICl_4^- ,

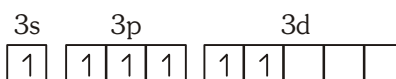
(IV) Two 'd' orbital participates in the hybridisation are $d_{x^2-y^2}$ and d_{z^2} .

SF_6

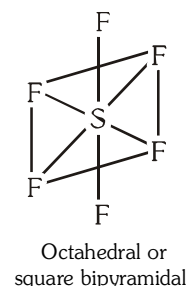
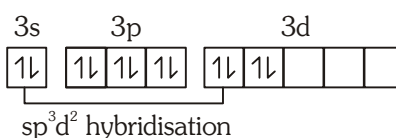
S (ground state)



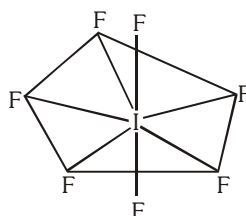
S (excited state)



S (after hybridisation)



- (I) In this hybridisation, one s-orbital, three p-orbitals & three d-orbitals are mixed to give seven new hybrid orbitals known as sp^3d^3 hybrid orbitals.
- (II) These seven sp^3d^3 orbitals are configured in pentagonal bipyramidal shape.
- (III) Five bond angles are of 72° and 10 bond angles of 90° .
- (IV) The following examples showing sp^3d^3 hybridisation – IF_7 & XeF_6 .



EXAMPLES ON sp^3d HYBRIDISATION

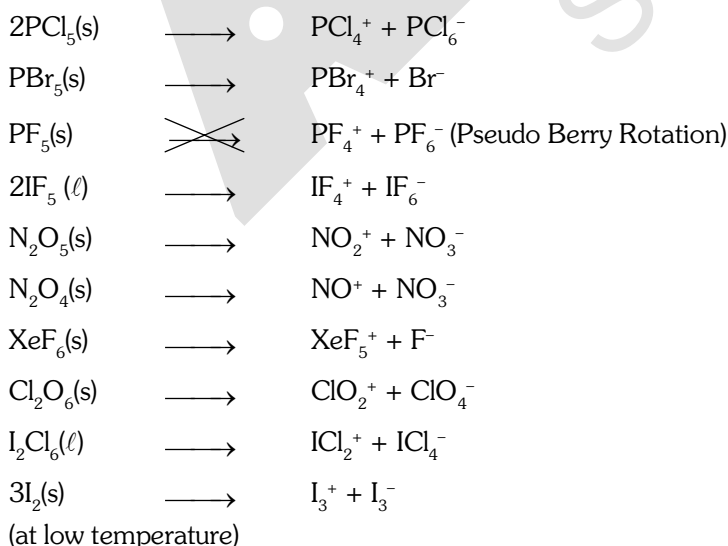
Example	σ bond	l.p.e.	Hybridisation	Bond angle	Shape
PCl_5	5	-	sp^3d	$120^\circ, 180^\circ \text{ \& } 90^\circ$	Trigonal bipyramidal
SF_4	4	1	sp^3d	$< 180^\circ, < 90^\circ, < 120^\circ$	Irregular tetrahedron
ClF_3	3	2	sp^3d	87.6°	T-shape
IF_3	3	2	sp^3d	87.6°	T-shape
ICl_3	3	2	sp^3d	87.6°	T-shape
Br_3^-	2	3	sp^3d	180°	Linear
ICl_2^-	2	3	sp^3d	180°	Linear
XeF_2	2	3	sp^3d	180°	Linear
PCl_4^+	4	-	sp^3		Tetrahedron
NH_4^+	4	-	sp^3		Tetrahedron
NF_3	3	1	sp^3		Pyramidal
H_3O^+	3	1	sp^3		Pyramidal
SO_3^{2-}	3	1	sp^3		Pyramidal
XeO_3	3	1	sp^3		Pyramidal
H_2O	2	2	sp^3		Angular (V)
NH_2^-	2	2	sp^3		Angular (V)
OF_2	2	2	sp^3		Angular (V)
Cl_2O	2	2	sp^3		Angular (V)
Diamond	4	-	sp^3		Tetrahedron
SiO_2	4	-	sp^3		Tetrahedron
SiC	4	-	sp^3		Tetrahedron
NO_3^-	3	-	sp^2	120°	Trigonal planar
SO_2	2	1	sp^2	$< 120^\circ$	Angular (V)
HCO_3^-	3	-	sp^2	120°	Trigonal planar
SnCl_2	2	1	sp^2	$< 120^\circ$	Angular (V)
NO_2^+	2	-	sp	180°	Linear
N_3^-	2	-	sp	180°	Linear

BEGINNER'S BOX-3

- Which of the following is incorrect about hybridization?
 - (1) The concept of hybridization is not applied to isolated atoms.
 - (2) Hybridization is the mixing of at least two non-equivalent atomic orbitals.
 - (3) The number of hybrid orbitals generated is more than the number of pure atomic orbitals that participate in the hybridization process.
 - (4) Hybridization requires an input of energy.
- The hybridization state of the central atom in HgCl_2 is—
 - (1) sp
 - (2) sp^2
 - (3) sp^3
 - (4) dsp^2
- The hybridization state of the central atom in AlI_3 is—
 - (1) dsp^2
 - (2) sp^3
 - (3) sp^2
 - (4) sp
- In C_3O_2 , the hybridization state of C is—
 - (1) sp^2
 - (2) sp
 - (3) sp^3
 - (4) dsp^2
- By hybridization, we mean the hybridization of—
 - (1) electrons
 - (2) atomic orbitals
 - (3) atoms
 - (4) protons
- The d- orbitals involved in sp^3d hybridisation is:-
 - (1) $d_{x^2-y^2}$
 - (2) d_{z^2}
 - (3) d_{xy}
 - (4) d_{xz}
- A sp^3 hybrid orbital contains:-
 - (1) $\frac{3}{4}$ s- character
 - (2) $\frac{1}{4}$ p - character
 - (3) $\frac{3}{4}$ p - character
 - (4) $\frac{1}{2}$ s - character
- In the protonation of NH_3 molecule, following statement is true:-
 - (1) A covalent bond is formed
 - (2) Hydrogen bond is formed
 - (3) Hybridisation state of N is changed
 - (4) Shape of NH_3 molecule is changed
- The shape of sulphate ion is :-
 - (1) Hexagonal
 - (2) Square planar
 - (3) Trigonal bipyramidal
 - (4) Tetrahedral
- In which following compound, central atom has four bond pair and one lone pair:-
 - (1) NH_4^+
 - (2) ICl_4^-
 - (3) SF_4
 - (4) XeF_4
- In which molecule s - p overlapping occurs ?
 - (1) CH_4
 - (2) NH_3
 - (3) H_2O
 - (4) None of these

Hybridisation in solid state :

- Compounds which change their structure in solid state/liquid state.



Hybridisation in radicals :

Radical	hybridisation
CH ₃	
CF ₃	
ClO ₃	
NO ₂	

Existence and Nonexistence :

Identify existing / non existing compounds / ions

- (a) PCl₆⁻ (b) NH₅ (c) PH₅ (d) (CCl₆)²⁻ (e) (SiF₆)²⁻ (f) (SiCl₆)²⁻ (g) ClBr₇
 (h) SH₆ (i) XeH₄ (j) XeOF₄ (k) FCl₃ (l) ClF₃ (m) BH₄⁻ (n) (PI₆)⁻

2.7 VALENCE SHELL ELECTRON PAIR REPULSION THEORY (VSEPT)

- (a) If the central atom possess only bonded pairs of electrons along with identical atoms then shape of the compound is symmetrical and according to Sidgwick & Powl.

eg.	BF ₃	—	120°	—	triangular
	CH ₄	—	109° 28'	—	tetrahedral
	CO ₂	—	180°	—	linear

- (b) If the central atom possess bonded pair of electrons as well as lone pair of electron, then shape of the molecule will be unsymmetrical ie. the original bond angle will disturbed due to repulsion between lone pair of electrons.

Similarly on having different type of side atoms, molecule becomes unsymmetrical due to unequal force of repulsion between e⁻. Order of repulsion is - [ℓp - ℓp > ℓp - bp > bp - bp]

$$\left[\text{Bond angle} \propto \frac{1}{\text{Number of lone pair of electrons}} \right]$$

- (c) By increasing one lone pair of electron, bond angle is decreased approx by 2.5°.

eg.:-	CH ₄	NH ₃	H ₂ O	→	sp ³
	109°	107°	105°		hybridisation

2.8 BOND PARAMETERS**Bond Length**

The average distance between the nucleus of two atoms is known as bond length, normally it is represented in Å.
 eg. A — B

It depends mainly on electronegativities of constituent atoms.

Case -I. Electronegativity difference is zero then -

$$\text{Bond length} = r_A + r_B \quad \text{or} \quad d_{A-B} = r_A + r_B$$

where

r_A = covalent radius of A

r_B = covalent radius of B

x_A = electronegativity of A

x_B = electronegativity of B

If $r_A = r_B$ then Bond length = $2r_A$ or $2r_B$

Example : - Cl - Cl

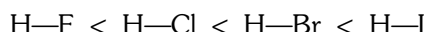
Case - II Electronegativity difference is not equal to zero then -

Bond length is given by Shomaker & Stevenson formula is - Bond length = $r_A + r_B - 0.09 (x_A - x_B) \text{Å}$

$(x_A - x_B)$ = Difference in electronegativities

Factors Affecting Bond Length

(a) **Δ electronegativity** :- Bond length $\propto \frac{1}{\Delta EN}$ (While B.E. $\propto \Delta EN$)



(b) **Bond order or number of bonds** :- Bond length $\propto \frac{1}{\text{Number of bonds or bond order}}$

Bond energy $\propto$ Number of bond

ex.	C—C,	C = C,	C $\equiv$ C	
Bond length	1.54 Å	1.34 Å	1.20 Å	← increasing
Bond energy	80	140	180-200 K.Cal.	increasing →
	C—O	C = O	C $\equiv$ O	
	1.43 Å	1.20 Å	1.13 Å	
			C $\equiv$ N	
	1.47 Å	1.28 Å	1.15 Å	

(c) **Resonance** :- Due to resonance bond length affected

Ex.1. Benzene

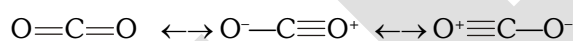
C—C bond length 1.54 Å

C=C bond length 1.34 Å



But bond length is between single and double bond is = 1.39 Å

Ex.2. Bond length of C—O in CO₂ is 1.15 Å Resonance occurs in CO₂ is as follows -



Bond length = 1.15 Å (Between double & triple bond)

(d) **Hybridization** :- Bond length $\propto \frac{1}{s \text{ character}}$

Example :-

s-character increases ↓

Compound	Hybridisation	Bond length
Ethane	sp ³ —sp ³	1.54 Å
	sp ³ —sp ²	1.51 Å
	sp ³ —sp	1.47 Å
C=C—C=C	sp ² —sp ²	1.46 Å
C=C—C≡C	sp ² —sp	1.42 Å
C≡C—C≡C	sp—sp	1.37 Å

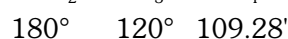
Bond Angle

The minimum angle between any two adjacent bonds is known as bond angle.

It is represented in degree (°), min (') and second (")

Factors affecting the bond angle -

Step I : Hybridisation or % 's' character : Bond angle $\propto$ % s character



Step II : Lone pair

When hybridisation is same, lone pair are different.

$$\text{Bond angle} \propto \frac{1}{\text{No. of lone pair}}$$

Example :-

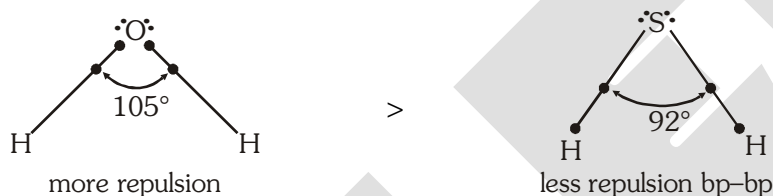
	CH₄		$\ddot{\text{N}}\text{H}_3$		H₂$\ddot{\text{O}}$:
Hybridisation	sp ³		sp ³		sp ³
Bond angle	109	>	107	>	105
	No l.p.		one l.p.		two l.p.

- In the different molecules if central atom have same number of lone pair of electron then bond angle will depend on electronegativities of A & B.

Step III : Central Atom

$$\text{Bond angle} \propto \text{Electronegativity of central atom}$$

In AB_x type of molecules if side atoms are same and EN of central atom increases the bond angle increases.



- Electronegativity of 'O' > Electronegativity of 'S'
- Bond angle of - NH₃ > PH₃ > AsH₃

Example :-

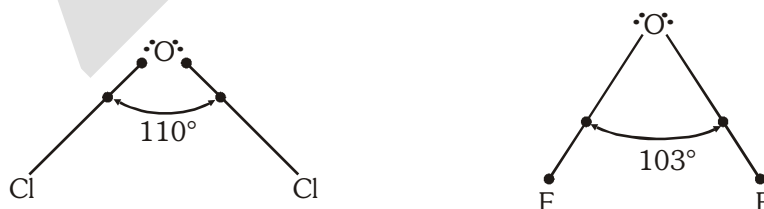
	$\ddot{\text{N}}\text{H}_3$	$\ddot{\text{P}}\text{H}_3$	$\ddot{\text{As}}\text{H}_3$
Bond angle	107°	93°	91°

-
- Electronegativity decreasing.
 - Bond angle will decrease

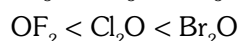
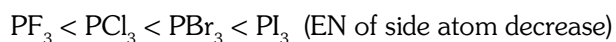
Step IV : Side atom

$$\text{Bond angle} \propto \frac{1}{\text{electronegativity of bonded atom}} \propto \text{size of side atom}$$

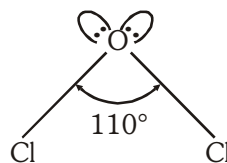
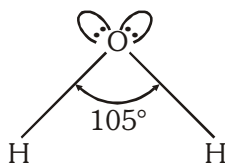
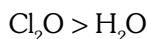
In AB_x type molecules, if central atoms are same and the EN of side atoms increases then bond angle decreases.



Electronegativity of Fluorine is greater than chlorine



Bond angle depends on size of side atom, On increasing size of side atom bond angle increases.



When hybridisation is same, lone pair are same, Central atom is same, bonded atoms are different.

sp^3	OF_2	103 - 105°	Electronegativity of bonded atom is decreasing
sp^3	Cl_2O	109 - 111°	
sp^3	Br_2O	116 - 118°	

Bond Energy (BE)

Bond energy may be defined as -

- Bond formation energy :- Energy released when any bond is formed is known as bond formation energy or bond energy.
- Bond dissociation energy :- Energy required to dissociate any bond is known as Bond dissociation energy.

Calculation of released energy is more difficult than the dissociation energy therefore dissociation energy of bond is calculated and is assumed as bond energy or bond formation energy.

Case-I In diatomic molecule :

Bond energy = bond dissociation energy

Example:- $\text{N}_2 > \text{O}_2 > \text{H}_2 > \text{F}_2$

Case-II For polyatomic molecule :-

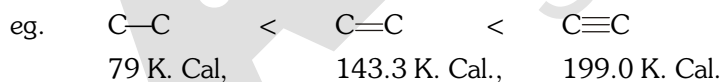
Bond energy $\approx$ Bond dissociation energy (D)

Factors Affecting The Bond Energy

- Electronegativity
- Bond order
- Atomic size
- Bond polarity
- Resonance
- Hybridisation
- Lone pair electron

(a) Δ Electronegativity :- Bond energy $\propto \Delta \text{EN}$ eg. $\text{HF} > \text{HCl} > \text{HBr} > \text{HI}$

(b) Bond order :- Bond energy $\propto$ Bond order.



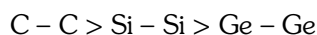
(c) Atomic size :- Bond energy $\propto \frac{1}{\text{Atomic size}}$ eg. $\text{C}\equiv\text{C} < \text{C}\equiv\text{N} < \text{N}\equiv\text{N}$

Exception :- In case of halogen group, order of bond energy is -

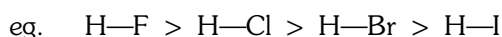


Because of higher electron density and small size of F atoms, repulsion between electrons of two F atoms, weakens the bond energy.

Other example $\text{S}-\text{S} > \text{O}-\text{O}$



(d) Bond Polarity :- Bond energy $\propto$ Bond polarity



9. The bond length does not affected by:-
 (1) Electron affinity (2) Bond order (3) Hybridisation (4) Resonance
10. In allene structure three carbon atoms are joined by :
 (1) Three sigma bonds and three pi bonds (2) Two sigma bonds and one pi bond
 (3) Two sigma bonds and two pi bonds (4) Three pi bonds only

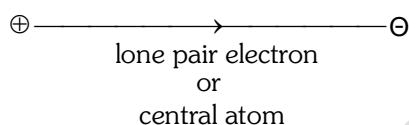
2.9 DIPOLE MOMENT

POLARITY OF BOND (IONIC NATURE IN COVALENT BOND)

- (a) Polarity of any polar covalent bond or molecule is measured in terms of dipole moment.
 (b) For measurement of extent of polarity, Pauling introduced the concept of dipole moment (μ).
 The product of positive or negative charge (q) and the distance (d) between two poles is called dipole moment.

Here - $\mu = q \times d$ (magnitude of charge $\times$ distance),

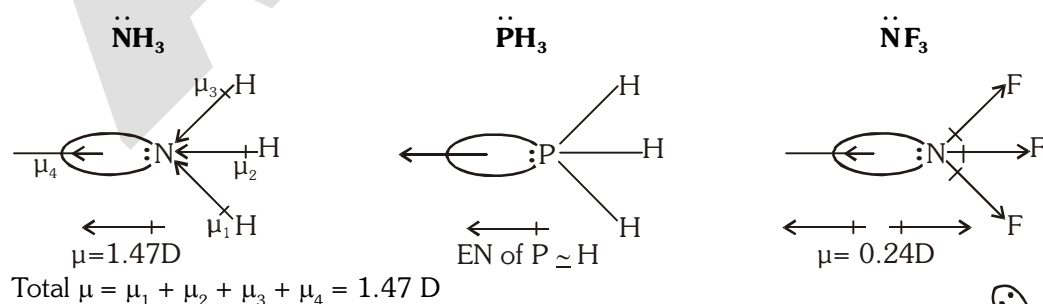
- (c) Dipole moment is a vector quantity i.e. it has both magnitude as well as direction.
 (d) Direction of dipole moment is represented by an arrow pointing from electro +ve to electro -ve element and from central atom to lone pair of electrons.



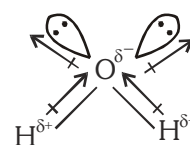
- (e) Unit of dipole moment is Debye
 $1 \text{ Debye} = 1 \times 10^{-18} \text{ e.s.u. cm} = 3.33 \times 10^{-30} \text{ coulomb metre}$
- (f) In the diatomic molecule μ depends upon difference of EN i.e. $\mu \propto \Delta \text{EN}$
 order of μ , $\text{H-F} > \text{H-Cl} > \text{H-Br} > \text{H-I}$
 $\mu = 0$ for H-H , F-F , Cl-Cl , Br-Br , O-O
- (g) For polyatomic molecules μ depends on the vector sum of dipole moments of all the covalent bonds.
 (h) For PCl_5 and SF_6 , etc. $\mu = 0$ due to their symmetrical geometry (According to charge).
 (i) Benzene, naphthalene, diphenyl have $\mu = 0$ due to planar structure.
 (j) If the vector sum is zero, than compound is non-polar compound or symmetrical compound (and it is not essential that individual μ of every bond should be zero).

Example - BX_3 , CCl_4 , SiCl_4 , CH_4 , CO_2 , CS_2 , PCl_5 , SiH_4 etc.

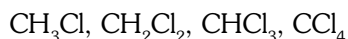
In these examples the bond B-F , C-Cl , C-H , C-O , P-Cl etc. are polar even though compounds are non-polar.



- (k) Dipole moment of H_2O is 1.85 D which is resultant μ of two O-H bonds.
 μ of $\text{H}_2\text{O} > \mu$ of H_2S because electronegativity of oxygen is higher than sulphur.
 (l) Angular structure of molecule have greater dipole moment.



Ex. Write the order of the dipole moment of following compounds ?



Sol. Right order is $\longrightarrow$ CCl_4 < CHCl_3 < CH_2Cl_2 < CH_3Cl
 $\mu = 0$ 1.02 1.55 1.93

Applications of Dipole Moment

(a) To determine polarity and geometry of molecule :-

If $\mu = 0$ compound is non polar and symmetrical

eg. $\text{CO}_2, \text{BF}_3, \text{CCl}_4, \text{CH}_4, \text{BeF}_2$ etc.

If $\mu \neq 0$ compound will be polar and unsymmetrical.

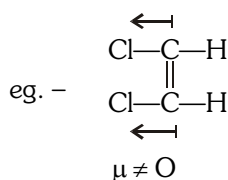
$\text{H}_2\text{O}, \text{SO}_2, \text{NH}_3, \text{Cl}_2\text{O}, \text{CH}_3\text{Cl}, \text{CHCl}_3$ etc.

(b) To calculate % ionic character :-

$$\% \text{ Ionic character} = \frac{\text{Experimental value of } \mu}{\text{Theoretical Value of } \mu} \times 100$$

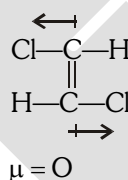
(c) To distinguish cis form or trans form :-

(I) Dipole moment of cis isomers is normally higher than trans isomers.



cis-form

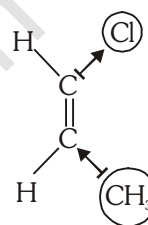
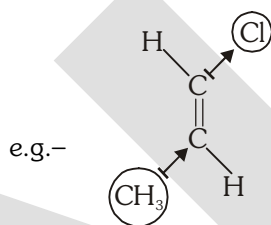
Polar ($\mu \neq 0$)



Trans-form

Non polar ($\mu = 0$)

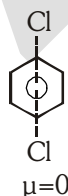
(II) If two groups have opposite inductive effect then trans-isomer will have greater dipole moment.



(d) To locate position of substituents in aromatic compounds.

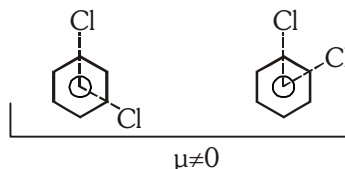
$$\mu \propto \frac{1}{\text{Bond angle}}$$

(I) If same substituents are present in the symmetrical position μ of benzene ring compounds will be zero.



Angle 180°

p-dichloro benzene



Angle 120°

m-dichloro benzene

Angle 60°

o-dichloro benzene

(II) As angle between vector decreases value of μ increases.

Illustrations

Illustration 1. Calculate the μ of HCl ? If bond distance is $1.34 \text{ \AA}$, charge = 4.8×10^{-10} esu and calculate % ionic character if experimental value of $\mu = 1.08 \text{ D}$?

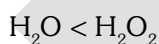
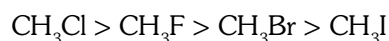
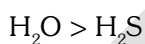
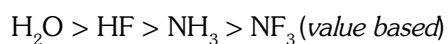
Solution $\mu = q \times d = 4.8 \times 10^{-10} \times 1.34 \times 10^{-8}$

$\mu = 6.4 \times 10^{-18} \text{ esu cm.}$

% Ionic character = $\frac{1.08 \times 10^{-18}}{6.4 \times 10^{-18}} \times 100 = 16.79 \%$

GOLDEN KEY POINTS

● **Some important orders of dipole moments :**



BEGINNER'S BOX-5

- Which of the following contains polar and nonpolar bonds?
 (1) H_2O_2 (2) CH_4 (3) HCN (4) NH_4Cl
- Carbon tetrachloride has no net dipole moment because of—
 (1) Similar electron affinity of C and Cl (2) its regular tetrahedral geometry
 (3) its planar geometry (4) similar sizes of C and Cl atoms
- Which of the following molecules is nonpolar?
 (i) PbCl_4 (ii) BF_3 (iii) SnCl_2 (iv) CS_2
 (1) (i), (ii), (iii) (2) (i), (ii), (iii), (iv) (3) (i), (ii), (iv) (4) (ii), (iii), (iv)
- Which of the following has the highest dipole moment ?
 (1) *o*-Dichlorobenzene (2) *m*-Dichlorobenzene (3) *p*-Dichlorobenzene (4) All have equal values
- Both CO_2 and H_2O contain polar covalent bonds but CO_2 is nonpolar while H_2O is polar because—
 (1) H atom is smaller than C atom
 (2) CO_2 is a linear molecule while H_2O is an angular molecule
 (3) O – H bond is more polar than C – H bond
 (4) CO_2 contains multiple bonds while H_2O has only single bonds

2.10 MOLECULAR ORBITAL THEORY (MOT)

MOT put forward by Hund & Mulliken, which can be applied to explain the properties, which the old VBT (Valence bond theory) was unable to explain. eg. Paramagnetic nature of O_2 molecule, as per VBT ($:O = O:$) it should be diamagnetic.

Definition : The atomic orbital lose their identity during molecule formation (overlapping) and form new orbitals termed as molecular orbitals.

Characteristic of Molecular Orbitals

- Molecular orbital formed by overlapping of atomic orbital of same energy.
- Number of molecular orbital formed = number of atomic orbitals involved in overlapping.
- Half of the molecular orbital have lower energy are called Bonding molecular orbital.
- Half are of higher energy - termed as Antibonding molecular orbital.
- Electronic configuration in various molecular orbital are governed by same three rules.
 - (1) Aufbau's rule
 - (2) Hund's rule
 - (3) Pauli's exclusion principle.
- Wave function for bonding molecular orbital is $\psi_b = \psi_A + \psi_B$
- Wave function for antibonding molecular orbital is $\psi_a = \psi_A - \psi_B$
 ψ_A = wave function of atom A
 ψ_B = wave function of atom B

Comparison of Bonding molecular orbital & Antibonding molecular orbital:

Bonding molecular orbital(BMO)	Antibonding molecular orbital (ABMO)
1. Bonding MO is the result of the linear combination of AO when their wave function are added $\psi_b = \psi_A + \psi_B$	1. ABMO is result of linear combination of AO when their wave function are subtracted $\psi_a = \psi_A - \psi_B$
2. Generally it does not have nodal plane.	2. It always have a nodal plane between two nuclei of bonded atom.
3. Electron density increases between two nuclei resulting attraction between two atoms.	3. Electron density decreases in between two nuclei, leads to repulsion between two atoms.
4. Energy of BMO is less, hence stable.	4. Energy of ABMO is high.
5. Electron placed in a BMO stabilises a molecule.	5. Electron placed in the ABMO destabilises the molecule.

Notation of molecular orbitals

As atomic orbitals are known by letters s, p, d and f depending on their shapes. Similarly for molecular orbital

For bonding molecular orbital - σ, π etc.

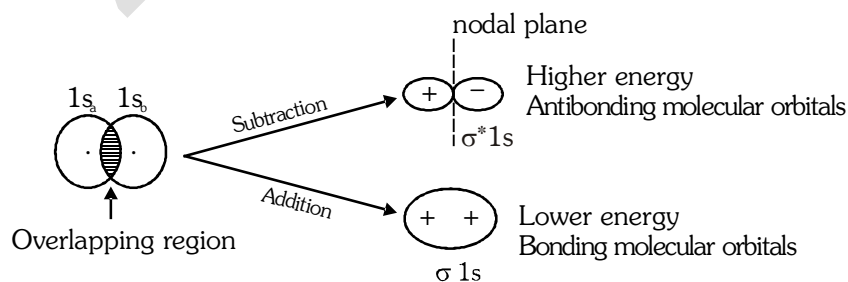
For antibonding molecular orbital- σ^*, π^* etc.

are used for different shapes of electron cloud.

Shapes of Molecular Orbitals (L.C.A.O. Method)

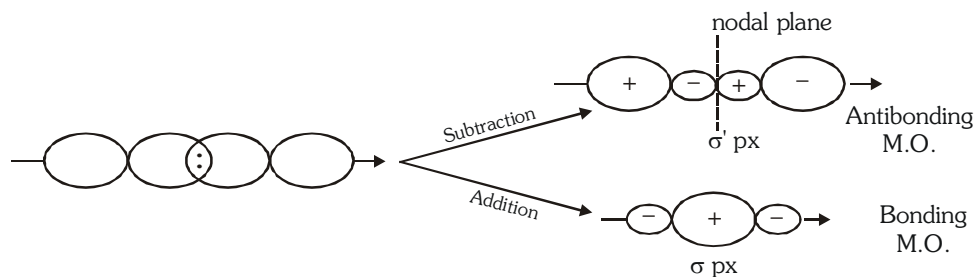
(A) (**σ molecular orbital**) :- It is formed by two ways -

- Combination of s-orbitals -



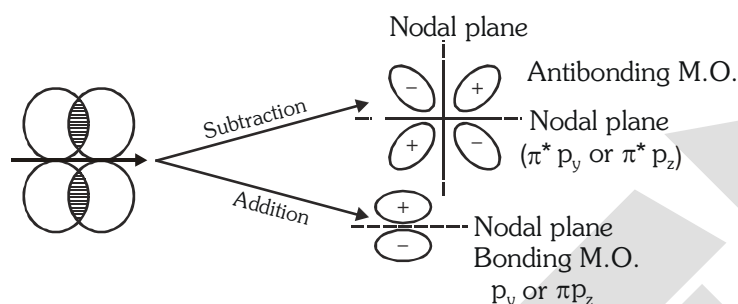
- $\sigma^* 1s$ have one nodal plane

(b) End on overlapping of p-orbitals - (Linearly) :-



● $\sigma^* p_x$ have one nodal plane

(B) π (pi) molecular orbitals :-



Positive sign, represent maximum probability finding of electrons.

$\pi^* p_y$ or $\pi^* p_z$ have two nodal plane

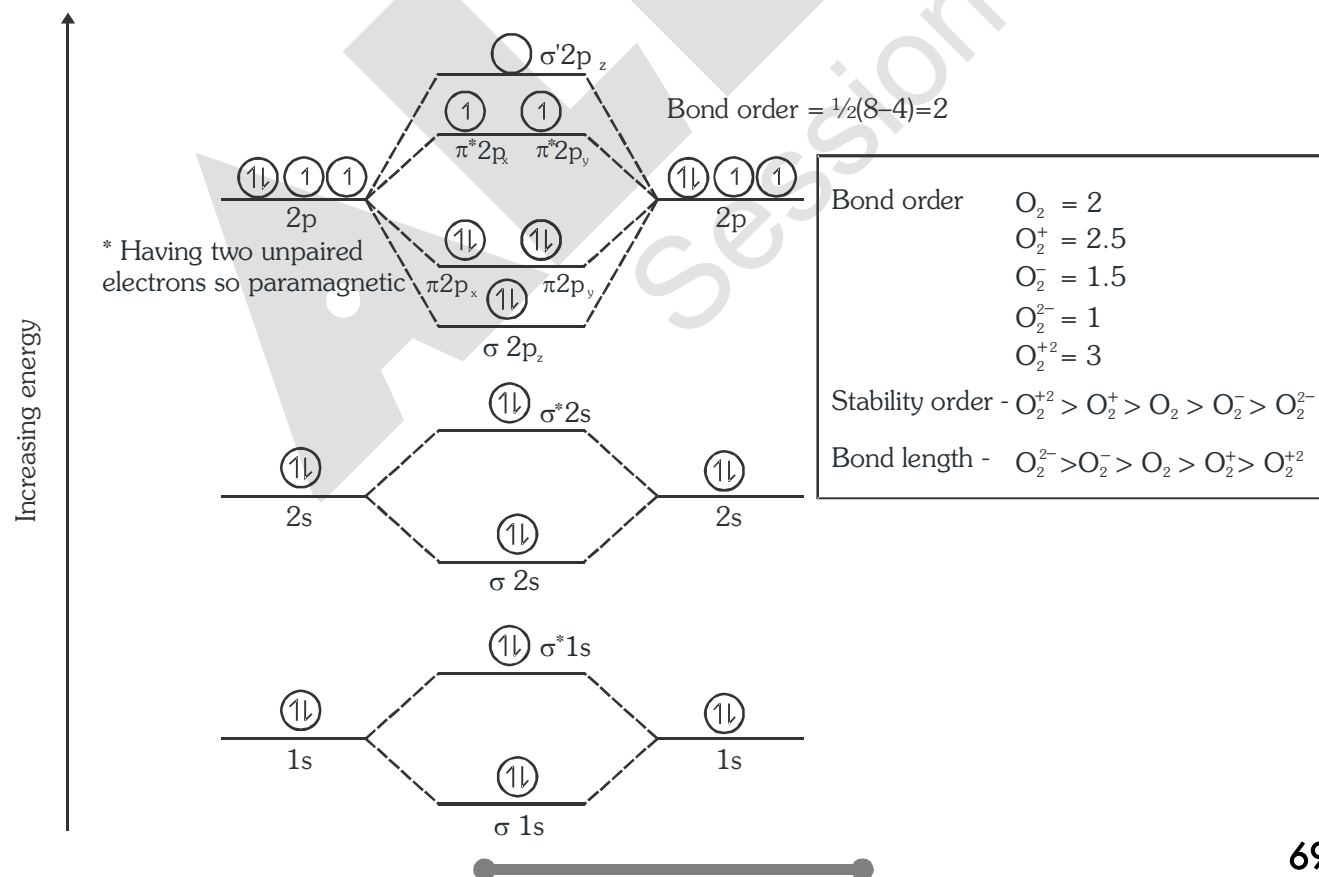
ENERGY LEVEL DIAGRAM OF MOLECULAR ORBITAL

(A) Energy level diagram for O_2, F_2, Ne_2 (Beyond N_2)

On the basis of Aufbau's rule - increasing order of energies of various molecular orbitals is -

$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_z < \pi 2p_x \equiv \pi 2p_y < \pi^* 2p_x \equiv \pi^* 2p_y < \sigma^* 2p_z$

For O_2 molecule



(B) Energy level diagram for B₂, C₂ and N₂ molecules (upto N₂)

$$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \pi 2p_x \equiv \pi 2p_y < \sigma 2p_z < \pi^* 2p_x \equiv \pi^* 2p_y < \sigma^* 2p_z$$

(a) Hydrogen molecule

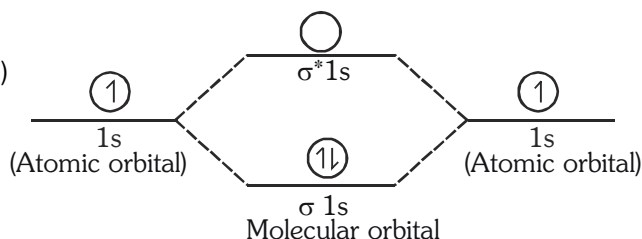
Having two H atoms with one electron each (1s')

M.O. configuration of $H_2 = (\sigma 1s)^2 (\sigma^* 1s)^0$

Bond order = $\frac{1}{2} [N_b - N_a]$
= $\frac{1}{2} [2 - 0] = 1$ ie. single bond.

Having paired electron so diamagnetic.

Stability $\rightarrow$ quite stable (having single bond)



(b) H_2^+ ion —

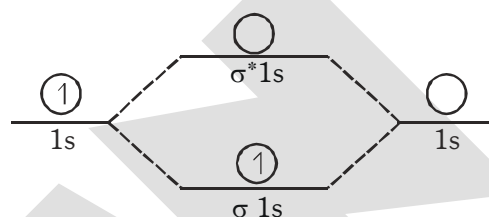
Configuration of $H_2^+ = (\sigma 1s)^1 (\sigma^* 1s)^0$

One electron in bonding molecular orbital.

So paramagnetic

Bond order = $\frac{1}{2} [1 - 0] = \frac{1}{2}$

Less stable. (Incomparision to H_2)



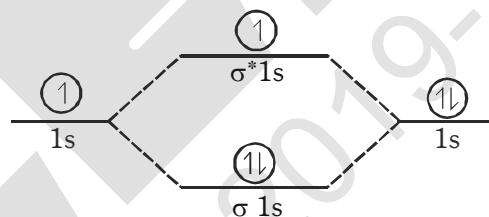
(c) H_2^- anion -

M.O. configuration - $(\sigma 1s)^2 (\sigma^* 1s)^1$

Paramagnetic

Bond order = $\frac{1}{2} [2 - 1] = \frac{1}{2}$

Stability is less than H_2^+ because H_2^- contain an ABMO electron



Stability order	H_2	H_2^+	H_2^-
Bond order	1.0	0.5	0.5

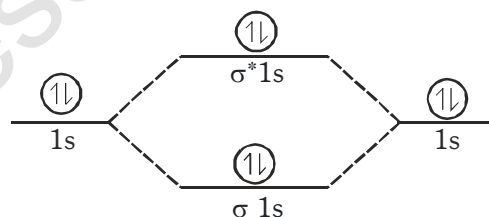
(d) Helium molecule (He_2)

M.O. configuration - $(\sigma 1s)^2 (\sigma^* 1s)^2$

Diamagnetic

Bond order = $\frac{1}{2} [2 - 2] = 0$ (zero)

- Bond order zero indicates no linkage between He atoms. Hence He_2 molecule does not exists.



Comparison between VBT and MOT

1. According to VBT electron moves around only one nucleus	1. According to MOT electron moves under influence of two or more nuclei
2. Orbitals are monocentric	2. Orbitals are polycentric
3. According to VBT O_2 is diamagnetic	3. According to MOT O_2 is paramagnetic

BEGINNER'S BOX-6

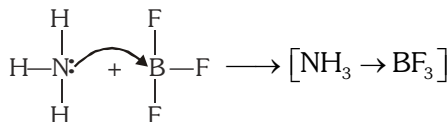
- Which of the following is incorrect regarding the MO theory ?
 - (1) The number of molecular orbitals formed is always equal to the number of atomic orbitals combined.
 - (2) The more stable the bonding molecular orbital, the less stable the corresponding antibonding molecular orbital.
 - (3) In a stable molecule, the number of electrons in bonding molecular orbitals is always equal to that in antibonding molecular orbitals.
 - (4) Like an atomic orbital, each molecular orbital can accommodate up to two electrons with opposite spins in accordance with the Pauli exclusion principle.
- If the z-axis is the molecular axis, then π MOs are formed by the overlap of—
 - (i) p_z and p_z
 - (ii) p_y and p_y
 - (iii) s_z and p_z
 - (iv) p_x and p_x
 - (1) (ii), (iv)
 - (2) (ii), (iii)
 - (3) (i), (ii)
 - (4) (i), (iii)
- If the z-axis is taken as the internuclear axis, then which of the following combinations of atomic orbitals is a nonbonding combination ?
 - (1) s and p_y
 - (2) p_x and d_{yz}
 - (3) p_x and p_y
 - (4) all of these
- Which of the following is the correct order of stability ?
 - (1) $H_2 > H_2^+ > He_2 > He_2^+$
 - (2) $H_2 > He_2^+ > H_2^+ > He_2$
 - (3) $H_2 > H_2^+ > He_2^+ > He_2$
 - (4) $H_2 > He_2 > He_2^+ > H_2^+$
- Bond order in C_2^+ is:-
 - (1) $\frac{1}{2}$
 - (2) $\frac{2}{3}$
 - (3) $\frac{3}{2}$
 - (4) 1
- In which of the following set, the value of bond order will be 2.5:-
 - (1) O_2^+ , NO, NO^{+2} , CN
 - (2) CN, NO^{+2} , CN^- , F_2
 - (3) O_2^+ , NO^{+2} , O_2^{+2} , CN^-
 - (4) O_2^{-2} , O_2^- , O_2^+ , O_2
- Of the following species which has the highest bond order and shortest bond length :
NO, NO^+ , NO^{2+} , NO^-
 - (1) NO only
 - (2) Bond order of NO is highest and bond length of NO^{2+} is shortest
 - (3) NO^+ only
 - (4) NO^{2+} only
- The diamagnetic molecule is
 - (1) Super oxide ion
 - (2) Oxygen molecule
 - (3) Carbon molecule
 - (4) Unipositive ion of nitrogen molecule
- On the basis of molecular orbital theory which molecules does not exist
 - (1) H_2
 - (2) He_2
 - (3) HeH
 - (4) Li_2
- Maximum bond energy will be shown by the species
 - (1) O_2^+
 - (2) O_2
 - (3) O_2^-
 - (4) O_2^{-2}

2.11 CO-ORDINATE BOND

- (1) It is a covalent bond in which the shared e^- pair come from one atom is called coordinate bond.
- (2) Necessary conditions for the formation of coordinate bond are -
 - (a) Octet of donor atom should be complete and should have atleast one lone pair of electron.
 - (b) Acceptor atom should have deficiency of at least one pair of electron.

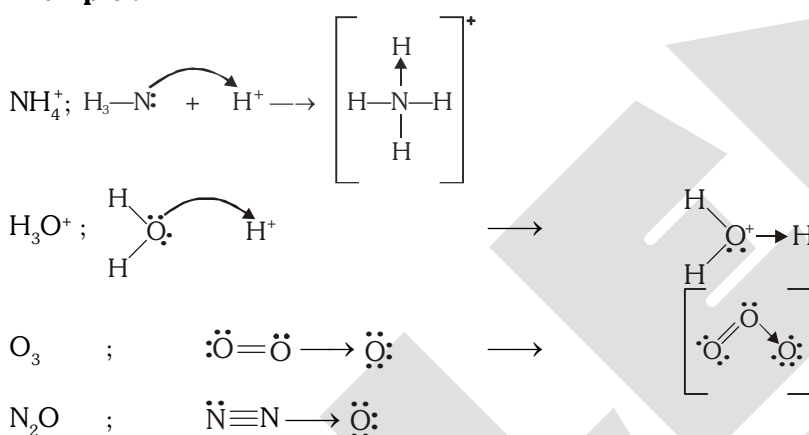


- (3) Atom which provide electron pair for sharing is called donor.
- (4) Other atom which accepts electron pair is called acceptor. That is why it is called donor-acceptor or dative bond.



BF_3 is electron deficient compound.

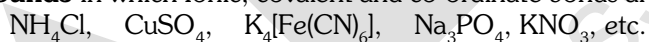
Example :



Metal co-ordinate compounds -



Compounds in which Ionic, covalent and co-ordinate bonds are present, are as follows -



2.12 FORMAL CHARGE

The difference between the valence electrons in an isolated atom and the number of valence electrons assigned to that atom in a given Lewis structure is called that atom's formal charge. The formal charge, abbreviated FC, on an atom in a Lewis structure is given by the following relationship:

Formal charge on a atom in a Lewis structure

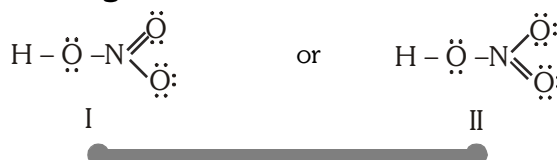
$$= (\text{total number of valence electrons in the isolated atom}) - (\text{Total number of nonbonding electrons}) - \frac{1}{2} (\text{total number of bonding electrons}) \dots (i)$$

$$\text{or} \quad \text{FC} = (\text{Valence electrons}) - (\text{Nonbonding electrons}) - \frac{1}{2} (\text{bonding electrons})$$

$$= (\text{Valence electrons or group number}) - [(\text{Number of unshared } e^-s) + (\text{Number of bonds})]$$

To find the formal charge, we count the bonding electrons as though they are equally shared between the two bonded atoms.

Q. Calculate the formal charges on the various atoms of nitric acid (HNO_3) molecule which has been described by the following Lewis structures:



The H atom : It has one valence electron, zero lone pair (or zero nonbonding electrons), and forms one bond (two bonding electrons).

Substituting in Eq. (i), we write $FC = (1) - (0) - \frac{1}{2}(2) = 0$.

The O atom bonded to H: It has six valence electrons, two lone pairs (or four nonbonding electrons), and from two bonds (or four bonding electrons).

Hence, we write $FC = (6) - (4) - \frac{1}{2}(4) = 0$.

The N atom: It has five valence electrons, zero lone pair (or zero nonbonding electrons), and forms four bonds (or has eight bonding electrons).

Thus, we write $FC = (5) - (0) - \frac{1}{2}(8) = +1$

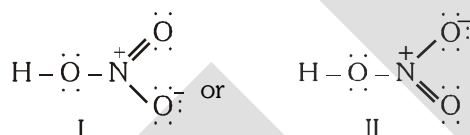
The end O atom in N = O: It has six valence electrons in the free state *but* in the Lewis structure (I or II), it has two lone pairs (or four nonbonding electrons) and forms two bonds (or has four bonding electrons).

Thus, we can write $FC = (6) - (4) - \frac{1}{2}(4) = 0$

The end O atom in N - O : It has six valence electrons in the free state but in the Lewis structure (I or II), it has three lone pairs or six nonbonding electrons and forms one bond (or has two bonding electrons).

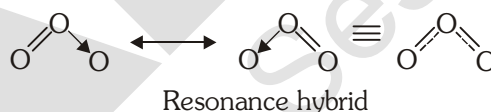
thus, we write $FC = (6) - (6) - \frac{1}{2}(2) = -1$

We can now write the Lewis structures (I and II) for nitric acid molecule including the formal charges as



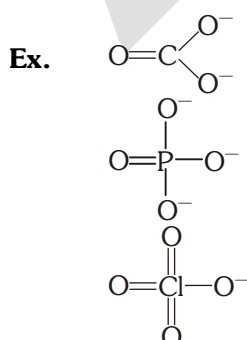
2.13 RESONANCE

- (a) It has been found that the observed properties of certain compounds cannot be satisfactorily explained by writing a single Lewis structure. The molecule is then supposed to have many structures, each of which can explain most of the properties of the molecule but none can explain all the properties of the molecules. The actual structure is in between of all these contributing structures and is called resonance hybrid and the different individual structures are called resonating structures or canonical forms. This phenomenon is called resonance.
- (b) Let us discuss resonance in ozone, according to its resonance structure it should have one single bond ($O-O = 1.48\text{\AA}$) but experiments show that both the bonds are same which can be proved by its resonance hybrid as shown below.

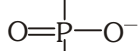


Note : To calculate bond order in the polyatomic molecule or ion use following formula :

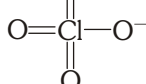
$$\text{Bond order} = \frac{\text{Total number of bonds in a molecule}}{\text{Resonating Structures}}$$



$$\text{Bond order} = \frac{4}{3} = 1.33$$



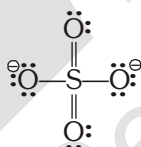
$$\text{Bond order} = \frac{5}{4} = 1.25$$



$$\text{Bond order} = \frac{7}{4} = 1.75$$

BEGINNER'S BOX-7

- Which of the following ions has resonating structures ?
 (1) SO_4^{2-} (2) PO_4^{3-} (3) SO_3^{2-} (4) All of these
- How many resonating structures can be drawn for NO_2 ?
 (1) Six (2) Four (3) Five (4) Two
- Which of the following is true for nitrate anion
 (1) Formal charge on N is zero (2) Bond order of NO bond is $\frac{4}{3}$
 (3) Average formal charge on oxygen is $-\frac{1}{3}$ (4) There are 2 π -bonds in the ion
- Which of the following contains Co-ordinate and covalent bonds:-
 (a) N_2H_5^+ (b) H_3O^+ (c) HCl (d) H_2O
 Correct answer is :-
 (1) a & d (2) a & b (3) c & d (4) Only a
- The correct statement for the reaction -
 $\text{NH}_3 + \text{H}^+ \rightarrow \text{NH}_4^+$:-
 (1) Hybridisation state is changed (2) Bond angle increases
 (3) NH_3 act as a Lewis acid (4) Regular geometry is changed
- The number of coordinate bonds presents in SO_3 molecule are
 (1) 1 (2) 2 (3) 3 (4) 4
- One of the resonating structure of SO_4^{2-} is



Which set of formal charge on oxygen and bond order is correct

- (1) 0.5 and 1.5 (2) 1.5 and 3 (3) 2 and 3 (4) 1.5 and 1.5
- Resonance is not shown by -
 (1) C_6H_6 (2) CO_2 (3) CO_3^{2-} (4) SiO_2
- Bond length of C - O is minimum in -
 (1) CO (2) CO_2 (3) CO_3^{2-} (4) HCOO^-

Subjective Type Questions

- Discuss resonance and formal charge in N_3^- and N_2O ?
- Give the average formal charge and average bond order of XO bond in the following oxy compounds?
 (a) SO_4^{2-} (b) SO_3^{2-} (c) NO_2^- (d) ClO_2^- (e) ClO_3^-
 (f) ClO_4^- (g) HCO_3^- (h) CO_3^{2-} (i) HSO_3^- (j) PO_4^{3-}

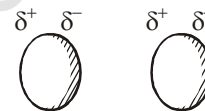
2.14 METALLIC BOND

- (1) The constituent particles of metallic solids are metal atoms which are held together by metallic bond.
- (2) In order to explain the nature of metallic bond Lorentz proposed a simple theory known as electron gas model or electron sea model.
- (3) A metal atom is supposed to consist of two parts, valence electrons and the remaining part (the nucleus and the inner shells) called kernel.
- (4) The kernels of metal atoms occupy the lattice sites while the space between the kernel is occupied by valence electrons.
- (5) Due to small ionisation energy the valence electrons or metal atoms are not held by the nucleus firmly. Therefore, the electrons leave the field of influence of one kernel and come under the influence of another kernel. Thus the electrons are not localised but are mobile.
- (6) The simultaneous attraction between the kernels and the mobile electrons which hold the kernel together is known as metallic bond.

WEAKER FORCES

2.15 Vander Waal's Forces

- (a) These are weak; non directional, non valence force of attraction among neutral species.
- (b) These are electrical in nature, due to induced polarity caused by temporary displacement of electrons towards one end of the inert atoms, becoming a temporary dipole.
- (c) This temporary dipole in one molecule can induce opposite dipoles in surrounding molecule due to displacement of electrons, one end becomes -ve and other +ve. These partially charged ends, induce surrounding molecules accordingly.
- (d) Strength of vander waal force depends on ease of distortion of electron cloud.



van der Waal's force $\propto$ size of atom or molecule $\propto$ atomic wt. or molecular weight.

- (e) Therefore m.p. and b.p. of noble gas elements (inert gas atom) and halogens increases down the group.

Types of van der Waal's Forces

- (a) **Keesom force or dipole dipole force** - one dipole molecule orient the other dipole molecule, to bring opposite ends close to each other for attraction. This is called orientation effect. Example - HCl, H₂O, NH₃ etc.
- (b) **Debye force or dipole induce dipole force** :- Forces which results from the interaction of a permanent dipole and induced dipole are called Debye force.
eg. When non-polar substance like benzene come in contact with polar molecules like NH₃, induced dipole moment in benzene appears (Induction effect).
- (c) **London dispersion force or Instantaneous dipole instantaneous induce dipole attraction** :-
Due to vibration or moment to atom electron cloud and nuclei shifts temporarily towards opposite ends, leads to attraction between them, eg. In atoms of noble gases and halogens.

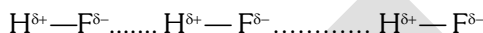
2.16 HYDROGEN BONDING

Definition

- (a) It is an electrostatic attractive force between covalently bonded hydrogen atom of one molecule and an electronegative atom.



- (b) It is not formed in ionic compounds.
 (c) H-bond forms in polar covalent compounds, (not in non-polar).
 (d) It is very weak bond but stronger than vander waal's force.
 (e) It is also known as dipole-dipole attraction.



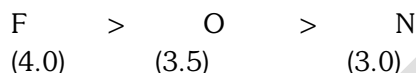
Main condition for H-bonding

- (a) H— should be covalently bonded with high electro -ve element like F, O, N.
 (b) Atomic size of electro -ve element should be small.

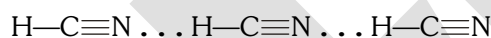
Order of atomic size is -



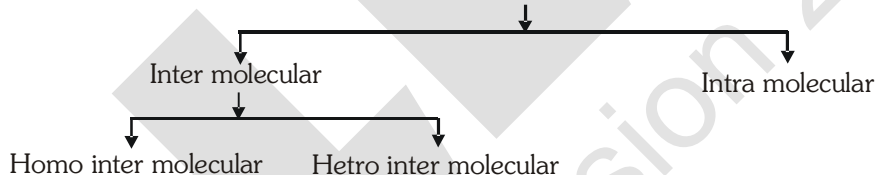
Order of electronegativity is -



- (c) Strength of H-bond $\propto$ Electronegativity of Z (element) $\propto \frac{1}{\text{atomic size of Z}}$
 (d) Hydrogen bonding occurs in HCN, due to $(-C \equiv N)$ triple bond (sp hybridisation), electronegativities of carbon and nitrogen increases.



TYPES OF HYDROGEN BONDING

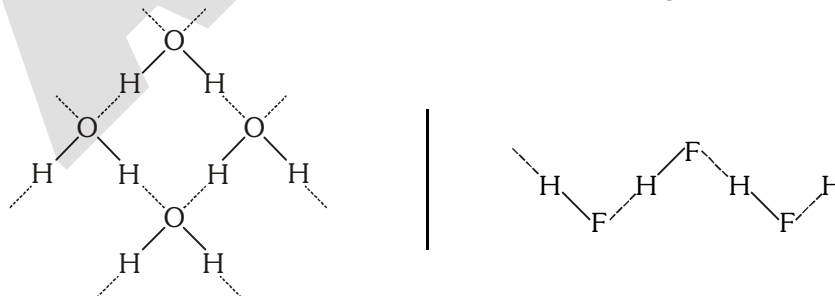


Intermolecular H-Bond

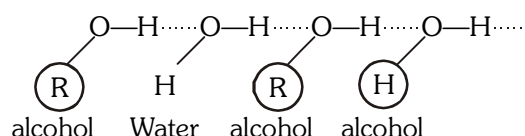
H-bond formation between two or more molecules of either the same or different compounds known as **Inter molecular H-bonding**.

These are of two types :-

- (i) **Homo intermolecular** :- H-bond between molecules of same compound.



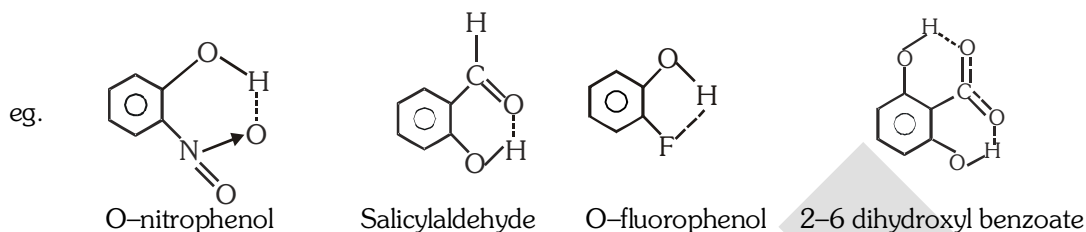
- (ii) **Hetro intermolecular** :- H-bond between molecules of different compounds
Eg. between alcohol and water



Intramolecular H-bond

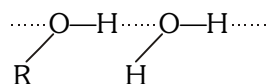
It takes place within the molecule.

- (a) H-bonded with electronegative element of a functional group, form H-bond with another electronegative element present on nearest position on the same molecule.
- (b) This type of H-bond is mostly occurred in organic compounds (Aromatic)
- (c) It results in ring formation (Chelation).

**Effect of H-bond on Physical Properties****(A) Solubility****(a) Inter molecular H-bonding**

- (I) Few organic compounds (Non-polar) are soluble in water (Polar solvent) due to H-bonding.

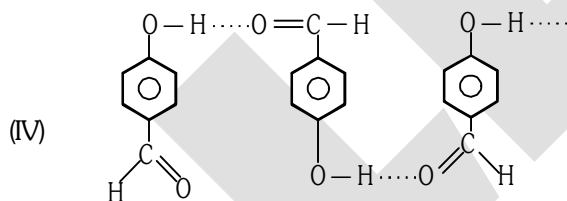
Ex. - Alcohol in water



Other examples - Glucose, Fructose etc. dissolve in water.

- (II) Ketone, ether, alkane etc. are insoluble (no H-bond). Dimethyl ether is soluble in water while diethyl ether is partially soluble, due to bulky ethyl groups H-bonding interrupts.

- (III) Solubility order- $\text{CH}_3\text{OCH}_3 < \text{CH}_3\text{OH}$

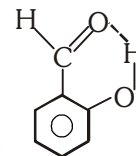


p-hydroxy benzaldehyde.

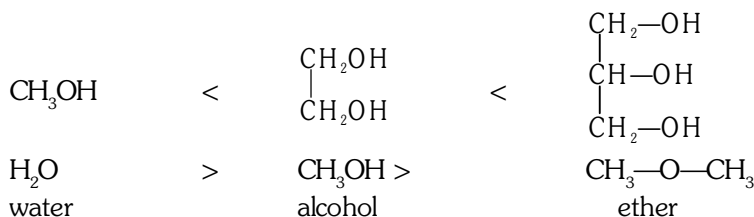
It can form H-bond with water molecule so it can dissolve.

(b) Intra molecular H-bonding

- (I) It decreases solubility as it form chelate by H-bonding, so H- is not free for other molecule.
- (II) It can not form H-bond with water molecule so it can not dissolve.

**(B) Viscosity**

H-bond associates molecules together so viscosity increases.

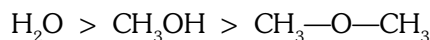


(C) **Surface Tension**

Surface tension of a liquid $\propto$ extent of H-bonding.

(D) **Melting point and boiling point**

(I) Due to intermolecular H-bond M.P. & B.P. of compounds increases.



(II) Boiling points of VA, VIA, VIIA hydrides are as shown below :



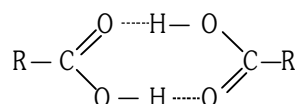
(III) But sudden increase in boiling point of NH_3 , H_2O and HF is due to H-bonding.

(IV) $\text{H}_2\text{O} > \text{HF} > \text{NH}_3$ (BP comparison due to combined effect of strength and number of H bond)

Intramolecular H-bonding gives rise to ring formation, so the force of attraction among these molecules are vander waal force. So M.P. and B.P. are low.

(E) **Molecular weight**

Molecular wt. of CH_3COOH is double of its molecular formula, due to dimer formation occur by H-bonding.



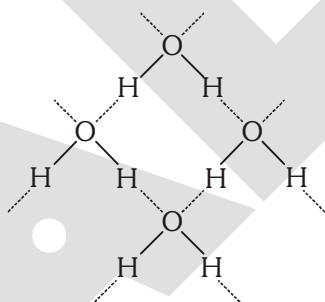
(F) **Physical states**

H_2O is liquid while H_2S is gas.

Water and Ice :- Both have H-bonding even then density of ice is less than water.

Volume of ice is more because of open cage like crystal structure, form by association of water molecules with the help of H-bond.

H_2O becomes solid due to four hydrogen bond among water molecule are formed in tetrahedral manner.

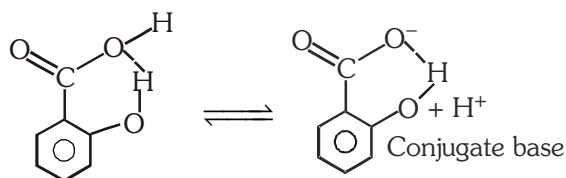


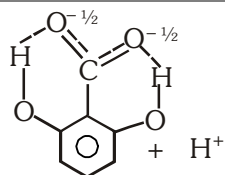
GOLDEN KEY POINTS

Effect of intramolecular H-bonding

Strength of acid

(I) The formation of intramolecular H-bonding in the conjugate base of an acid gives extra stability to conjugate base and hence acid strength increases eg. Salicylic acid is stronger than benzoic acid
2, 6 - dihydroxy benzoic acid > salicylic acid > benzoic acid.





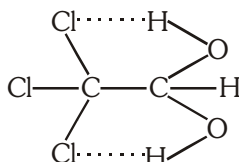
2, 6-dihydroxy benzoate ion.

(II) $\text{C}_2\text{H}_5\text{SH}$ is more acidic than $\text{C}_2\text{H}_5\text{OH}$. In $\text{C}_2\text{H}_5\text{OH}$, H-bond forms so H^+ is not free.

(III) HF is weaker acid than HI , due to H-bond in $\text{H}-\text{F}$, H^+ is not free

● **Stability of chloral hydrate**

If two or more OH group on the same atom are present it will be unstable, but chloral hydrate is stable (due to H-bonding).



Chloral hydrate

BEGINNER'S BOX-8

- Two ice cubes are pressed over each other until they unite to form one block. The force mainly responsible for holding them together is—
 - van der Waals force
 - dipole-dipole interaction
 - H bonding
 - covalent bonding
- The vapor pressure of *o*-nitrophenol at any given temperature is predicted to be—
 - higher than that of *p*-nitrophenol
 - lower than that of *p*-nitrophenol
 - same as that of *p*-nitrophenol
 - higher or lower depending upon the size of the vessel
- The hydrogen bond is strongest in:—
 - $\text{O}-\text{H} \cdots \text{S}$
 - $\text{S}-\text{H} \cdots \text{O}$
 - $\text{F}-\text{H} \cdots \text{F}$
 - $\text{O}-\text{H} \cdots \text{O}$
- H_2O boils at higher temperature than H_2S , because it is capable of forming:—
 - Ionic bonds
 - Covalent bonds
 - Hydrogen bonds
 - Metallic bonds
- Maximum number of H-bonding is shown by
 - H_2O
 - H_2Se
 - H_2S
 - HF
- Which is the weakest among the following types of bonds?
 - Debye force
 - Metallic bond
 - Dipole-dipole bond
 - Hydrogen bond
- The boiling point of *p*-nitrophenol is higher than that of *o*-nitrophenol because :
 - NO_2 group at *p*-position behaves in a different way from that at *o*-position
 - intramolecular hydrogen bonding exists in *p*-nitrophenol
 - there is intermolecular hydrogen bonding in *p*-nitrophenol
 - p*-nitrophenol has a higher molecular weight than *o*-nitrophenol
- In which molecule the Vander Waals force (dispersion force) is likely to be the most important in determining the m.pt. and b.pt. :
 - Br_2
 - CO
 - H_2S
 - HCl
- Covalent-molecules are usually held in a crystal structure by
 - Dipole-dipole attraction
 - Electrostatic attraction
 - Hydrogen bond
 - Van-der waal's attraction
- In solid argon the atoms are held together
 - by ionic bonds
 - by hydrogen bonds
 - By vander-waals forces
 - By hydrophobic bonds

2.17 IONIC OR ELECTROVALENT BOND

The chemical bond formed between two or more atoms as a result of the complete transfer of one or more electrons from one atom to another is called Ionic or electrovalent bond.

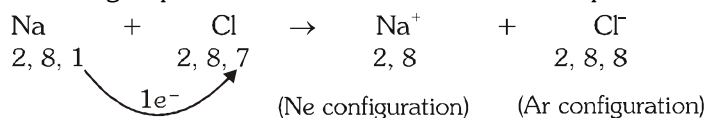
Electro +ve atom loses electron (group IA to IIIA)

Electro -ve atom gains electron (group VA to VIIA)

Electrostatic force of attraction between cation and anion is called ionic bond or electrovalent bond.

Nature of ionic bond $\propto$ Electronegativity difference.

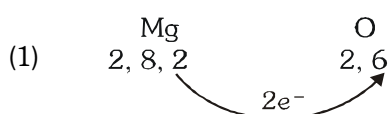
e.g. IA and VIIA group elements form maximum ionic compound.



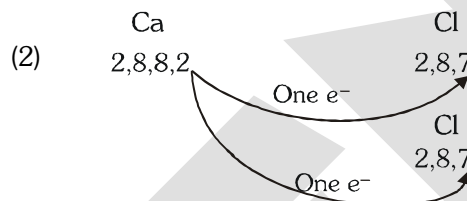
More the distance between two elements in periodic table more will be ionic character of bond.

Total number of electron lose or gained is called electrovalency.

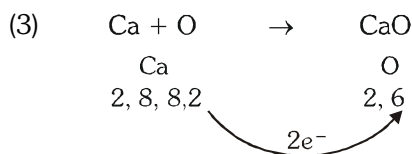
Example –



electrovalency of Mg = 2
electrovalency of O = 2



electrovalency of Ca = 2
electrovalency of Cl = 1



electrovalency of Ca = 2
electrovalency of O = 2

The force of attraction is equal in all direction so ionic bond is non-directional. A definite three dimensional structure is formed called crystal lattice. Energy released during the formation of one mole crystal lattice is called lattice energy.

Ionic compound do not have molecular formula. It has only empirical formula.

e.g. NaCl is empirical formula of sodium chloride

Conditions for Forming Ionic Bonds

Formation of Ionic bond depends upon these three factors –

(a) Ionisation energy :

Amount of energy required to remove an electron from the outermost orbit of an isolated gaseous atom to form the +ve ion or cation. (energy absorbed)

Lesser Ionisation energy $\rightarrow$ Greater tendency to form cation.



(b) Electron affinity :

Amount of energy released when an electron is added to an isolated gaseous atom to form -ve ion (anion) energy released.

Higher electron affinity $\rightarrow$ Greater tendency to form anion



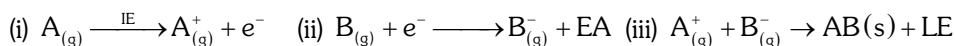
- (c) **Lattice energy** - (Energy released) Amount of energy released when one mole of crystal lattice is formed

Higher lattice energy $\rightarrow$ Greater will be the stability or strength of ionic compound.
or Amount of energy required to break the crystal.

- (d) **Overall lowering of energy :**

Energy must be released during bond formation.

Energy changes are involved in the following steps –



This concludes that for lower value of IE and higher value of EA there is more ease of formation of the ionic compound which is summarised as Born Haber Cycle.

GOLDEN KEY POINTS

● Factors affecting lattice energy

- (1) Magnitude of charge $\rightarrow U \propto z^+ z^-$ (Ionic charge)

Lattice energy $\propto$ Magnitude of charge



- $\longrightarrow$
- Lattice energy increases
 - Charge of cation increases

- (2) Size of Cation :- Lattice energy $\propto \frac{1}{r^+ + r^-}$



- $\longrightarrow$
- Size of cation increasing
 - Size of anion is constant
 - Lattice energy decreases.

● Representation of formula of compounds :

- (1) Write the symbols of the ions side by side in such a way that positive ion is at the left and negative ion is at the right as A^+B^-
- (2) Write their electrovalencies in figure at the top of each symbol as $A^x B^y$

- (3) Now apply criss cross rule as



Examples : Calcium chloride

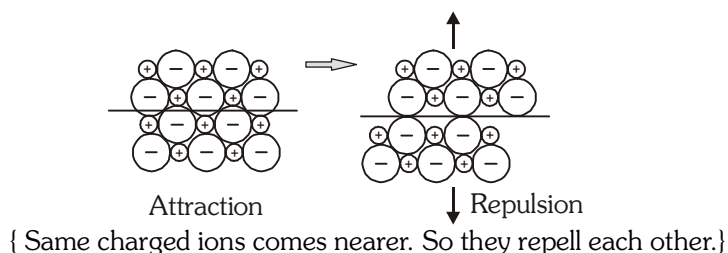


Properties of Ionic Compound

- (a) **Physical state** –

Ionic compounds are hard, crystalline and brittle due to strong electrostatic force of attraction.

Brittleness $\rightarrow$



(b) **Isomorphism –**

- (1) Two compounds are said to be isomorphous if they have similar no. of electrons i.e. similar configuration of their cation and anion.
- (2) They have similar crystal structure.

Example –

	Na ⁺	F ⁻	Mg ⁺²	O ⁻²
Valency	+1,	-1	+2,	-2
electronic configuration	2, 8	2, 8	2, 8	2, 8
similarly	Ca ⁺²	2Cl ⁻¹	2K ⁺¹	S ⁻²
	2, 8, 8	2, 8, 8 } 2, 8, 8 }	2, 8, 8 } 2, 8, 8 }	2, 8, 8

(c) **Boiling point and melting point –**

Ionic compounds have high boiling point and melting point due to strong electrostatic force of attraction among oppositely charged ions.

(d) **Conductivity –**

It depends on ionic mobility. In solid state - No free ions - Bad conductor of electricity.

In fused state or aqueous solution Due to free ions - Good conductor of electricity.

Conductivity order Solid state < fused state < Aqueous solution

(e) **Solubility –**

Highly soluble in Polar solvents like water.

Less soluble in non polar solvents like benzene.

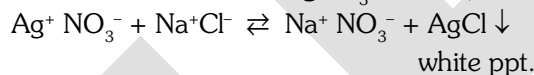
Ex. NaCl form a true solution in water but is colloid in benzene

(f) **Ionic reaction –**

(a) Ionic compounds shows ionic reaction and covalent compounds shows - molecular reaction.

(b) Ionic reactions are faster than molecular reaction because of free ions.

e.g. When NaCl is added in AgNO₃ solution, white ppt of AgCl is formed at once.



(g) Ionic bond non-directional and does not show stereo isomerism

2.18 Polarisation (Fajan's Rule) (Covalent Nature in Ionic Bond)

When a cation approaches an anion closely the positive charge of a cation attract the electron cloud of the anion towards itself, due to the electrostatic force of attraction between them.

At the same time cation also repel the positively charge nucleus of anion.

Due to this combined effect, cloud of anion is bulged or elongated towards the cation. This is called distortion, deformation or Polarisation of the anion by the cation and anion is called Polarised.

Polarisation Power

The ability of cation to polarised a nearby anion is called Polarisation power of cation.



Polarizability

Ability of an anion to get polarised by the cation.

Polarisation of anion causes some sharing of electron between the ions so ionic bond acquires certain covalent character.

[Polarisation $\propto$ Covalent character]

Magnitude of polarisation depends upon a no. of factors, suggested by Fajan and are known as Fajan's rule.

Fajan's Rule (Factors Affecting Polarisation)**(a) Size of cation** : Polarisation of the anion increases as the size of cation decreases.

$$\text{Polarisation} \propto \frac{1}{\text{size of cation}}$$

eg.

In a group



- Size of cation increases
- Covalent character decreases
- Ionic character increases

Greatest polarising power of Be²⁺, shows its maximum covalent characterIn a period — Na⁺, Mg⁺², Al⁺³, Si⁺⁴

- Cation size decreases
- Covalent character increases

(b) Size of anion :- If the size of the anion increases for a given cation, the covalent character increases.
Polarisation $\propto$ size of anion.

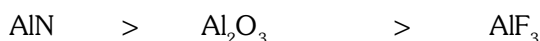
- size of anion increases
- Covalent character increases
- Ionic character decreases

(c) Charge on cation and anion :-Polarisation $\propto$ charge on cation or anion**(i) Polarisation (covalent character) $\propto$ Charge on cation**

- Charge on cation increases
- Covalent character increases
- Ionic character decreases (M.P. decreases)

Ex. Write the increasing order of M.P. & B.P. of following compounds.**Ans.** (1) Sn⁺⁴ < Sn⁺² (2) Fe⁺³ < Fe⁺² (3) Pb⁺⁴ < Pb⁺²(Charge on cation $\propto$ polarisation power $\propto$ covalent character $\propto \frac{1}{\text{M.P.}}$)**(ii) Polarisation $\propto$ Charge of anion**

- Charge increases
- Covalent character increases



Charge on anion $\propto$ polarisation $\propto$ covalent nature $\propto \frac{1}{\text{M.P.}}$

(d) **Electronic configuration of cation : -**

Order of Polarisation power : $8e^- < (18+2)e^- < 18e^-$

$\text{CuCl} \longrightarrow \text{Cu}^+ \quad 2, 8, 18 \quad (\text{Covalent})$

$\text{NaCl} \longrightarrow \text{Na}^+ \quad 2, 8 \quad (\text{Ionic})$

Cu^+ and Na^+ both the cation (Pseudo inert & inert respectively) have same charge and size but polarising power of Cu^+ is more than Na^+ because -

$$\begin{array}{ccc} z_{\text{eff}} \text{ of } ns^2p^6 \text{ (inert)} & < & z_{\text{eff}} \text{ of } ns^2p^6d^{10} \text{ (pseudo inert)} \\ \text{Na}^+ & < & \text{Cu}^+ \\ (\text{Ionic}) & & (\text{Covalent}) \end{array}$$

So CuCl has more covalent character than NaCl .

Order of covalent character

- $\text{LiF} < \text{LiCl} < \text{LiBr} < \text{LiI}$
- $\text{CaCl}_2 < \text{FeCl}_2 < \text{FeCl}_3$
- $\text{SrCl}_2 < \text{SnCl}_2 < \text{CdCl}_2$
- $\text{SF}_2 < \text{SF}_4 < \text{SF}_6$
- $\text{Hg}_2\text{Cl}_2 < \text{HgCl}_2$
- $\text{ZnCl}_2 < \text{CdCl}_2 < \text{HgCl}_2$

Note :

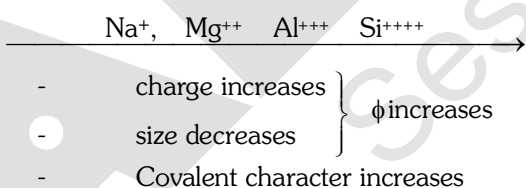
Polarisation power of a cation is usually called ionic potential or charge density.

$$\text{Ionic potential } \phi \text{ (phi)} = \frac{\text{Charge on cation}}{\text{Size of cation}}$$

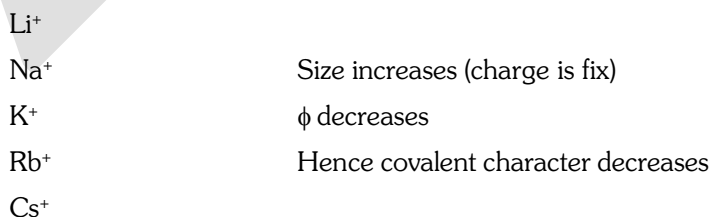
To determine covalent and ionic character of molecule

$$\phi \propto \frac{\text{Covalent Character}}{\text{Ionic character}}$$

From left (larger size) to right (smaller size) in a period ϕ increases so covalent character increases.



From top to bottom in a group ϕ decreases so covalent, character decreases.



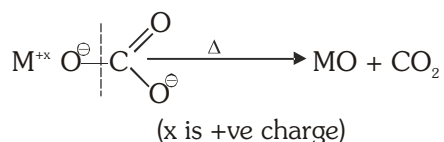
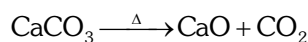
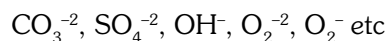
BEGINNER'S BOX-9

- The electrovalency of the element is equal to the—
 - (1) number of electrons lost
 - (2) number of electrons gained
 - (3) number of electrons transferred
 - (4) number of electrons lost or gained by the atom of the element during the formation of ions of ionic compound
- Which of the following polar solvents has the highest dielectric constant?
 - (1) H_2O
 - (2) D_2O
 - (3) CH_3OH
 - (4) $\text{C}_2\text{H}_5\text{OH}$
- Which of the following cations possess neither noble gas nor pseudo noble gas configurations?
 - (i) Bi^{3+}
 - (ii) Pb^{2+}
 - (iii) Sn^{2+}
 - (iv) Tl^+
 - (1) (ii), (iii)
 - (2) (i), (iv)
 - (3) (i), (ii), (iii)
 - (4) (i), (ii), (iii), (iv)
- Ionic bond formation involves :
 - (1) Elimination of protons
 - (2) Sharing of electrons
 - (3) Overlapping of orbitals
 - (4) Formation of octets
- The hydration of ionic compounds involves —
 - (1) Evolution of heat
 - (2) Weakening of attractive forces
 - (3) Dissociation into ions
 - (4) All
- The hydration energy of Mg^{+2} is greater than the hydration energy of
 - (1) Al^{+3}
 - (2) Mg^{+3}
 - (3) Na^+
 - (4) Be^{+2}
- Among the following which compounds will show the highest lattice energy ?
 - (1) KF
 - (2) NaF
 - (3) CsF
 - (4) RbF
- The lattice energy of the lithium is in the following order :
 - (1) $\text{LiF} > \text{LiCl} > \text{LiBr} > \text{LiI}$
 - (2) $\text{LiCl} > \text{LiF} > \text{LiBr} > \text{LiI}$
 - (3) $\text{LiBr} > \text{LiCl} > \text{LiF} > \text{LiI}$
 - (4) $\text{LiI} > \text{LiBr} > \text{LiCl} > \text{LiF}$
- Among LiCl , BeCl_2 , BCl_3 and CCl_4 , the covalent bond character follows the order :
 - (1) $\text{LiCl} < \text{BeCl}_2 > \text{BCl}_3 > \text{CCl}_4$
 - (2) $\text{LiCl} > \text{BeCl}_2 < \text{BCl}_3 < \text{CCl}_4$
 - (3) $\text{LiCl} < \text{BeCl}_2 < \text{BCl}_3 < \text{CCl}_4$
 - (4) $\text{LiCl} > \text{BeCl}_2 > \text{BCl}_3 > \text{CCl}_4$

2.19 THERMAL DECOMPOSITION

Thermal stability of metal carbonates

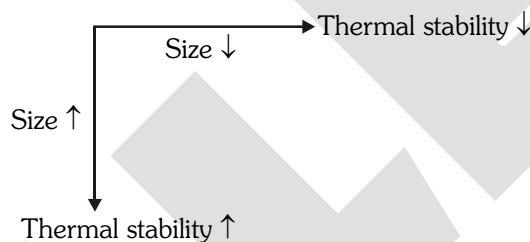
Thermal stability of compound having poly atomic anion



Polarising Power (↑) Thermal Stability (↓)

$$\text{Thermal stability} \propto \frac{1}{\text{Pol. power}} \propto \frac{\text{size of cation}}{\text{charge of cation}}$$

Compounds having poly atomic anions :



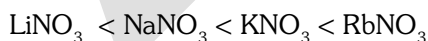
For fluoride, hydride & normal oxide

$\left[\text{Thermal stability} \propto \frac{1}{\text{size}} \right]$ for a group

[Thermal stability $\propto \Delta E_N$] for a period

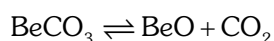
Thermal Stability order

For Example



Note :

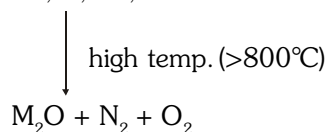
- (i) LiHCO_3 and IIA group bicarbonate does not exist in solid state.
- (ii) Carbonate, Sulphates & hydroxide of Na, K, Rb & Cs do not decompose at high temperature only melt.
- (iii) BeCO_3 is kept in CO_2 atmosphere due to less thermal stability.



Heating Effect

- (a) Metal carbonate $\xrightarrow{\Delta}$ metal oxide + CO_2
- (b) Metal hydroxide $\xrightarrow{\Delta}$ metal oxide + H_2O
- (c) Metal bicarbonate $\xrightleftharpoons{\Delta}$ metal carbonate + CO_2 + H_2O
- (d) Ammonium salts having CO_3^{-2} , PO_4^{-3} , SO_4^{-2} , X-anion (non oxidising or weak oxidising) gives NH_3 gas on decomposition.
Ammonium salt having $\text{Cr}_2\text{O}_7^{-2}$, ClO_3^{-1} , NO_2^{-1} , NO_3^{-1} (strong oxidising anion) gives N_2 or N_2O gas on decomposition.
- (e) Metal nitrate $\xrightarrow{\Delta}$ metal oxide + NO_2 + O_2

Except : Na, K, Rb, Cs nitrate $\xrightarrow{\text{low temp.}}$ MNO_2 + $\frac{1}{2} \text{O}_2$

**Note :**

- (i) Some less stable metal oxide like Ag_2O & HgO further decompose into metal & oxygen.
- (ii) Metal salts having high percentage of oxygen like KMnO_4 , $\text{K}_2\text{Cr}_2\text{O}_7$ & KClO_3 give O_2 gas on decomposition.

Complete following reactions :

$\text{PbCO}_3 \xrightarrow{\Delta}$	$\text{FeCO}_3 \xrightarrow{\Delta}$
$\text{ZnCO}_3 \xrightarrow{\Delta}$	$\text{MgCO}_3, \text{CaCO}_3 \xrightarrow{\Delta}$
$\text{CuCO}_3, \text{Cu(OH)}_2 \xrightarrow{\Delta}$	$\text{NaCO}_3, 10\text{H}_2\text{O} \xrightarrow{\quad} \text{X} \xrightarrow{\quad} \xrightarrow{\Delta}$
$2\text{NaHCO}_3 \xrightarrow{\Delta}$	$\text{NH}_4\text{NO}_2 \xrightarrow{\Delta}$
$\text{NH}_4\text{NO}_3 \xrightarrow{\Delta}$	$(\text{NH}_4)_2\text{SO}_4 \xrightarrow{\Delta}$
$(\text{NH}_4)_2\text{CO}_3 \xrightarrow{\Delta}$	$\text{Ba(N}_3)_2 \text{ or } \text{NaN}_3 \xrightarrow{\Delta}$
$\text{Pb(NO}_3)_2 \xrightarrow{\Delta}$	$\text{Zn(NO}_3)_2 \xrightarrow{\Delta}$
$\text{Ca(NO}_3)_2 \xrightarrow{\Delta}$	$\text{LiNO}_3 \xrightarrow{\Delta}$
$\text{NaNO}_3 \xrightarrow{400^\circ\text{C}}$	$\text{PbCl}_4 \xrightarrow{\Delta}$
$\text{FeCl}_3 \xrightarrow{\Delta}$	$\text{AuCl}_3 \xrightarrow{\Delta}$
$\text{CuSO}_4 \cdot 5\text{H}_2\text{O} \xrightarrow{<200^\circ\text{C}} \text{A} \xrightarrow{>200^\circ\text{C}} \text{B} \xrightarrow{800^\circ\text{C}}$	$\text{BaCl}_2 \cdot 2\text{H}_2\text{O} \xrightarrow{\Delta}$
$\text{ZnSO}_4 \cdot 7\text{H}_2\text{O} \xrightarrow{<200^\circ\text{C}} \text{A} \xrightarrow{>200^\circ\text{C}} \text{B} \xrightarrow{800^\circ\text{C}}$	$\text{AlCl}_3 \cdot 6\text{H}_2\text{O} \xrightarrow{\Delta}$
$\text{FeSO}_4 \cdot 7\text{H}_2\text{O} \xrightarrow{<200^\circ\text{C}} \text{A} \xrightarrow{>200^\circ\text{C}} \text{B} \xrightarrow{750^\circ\text{C}}$	$\text{CaCl}_2 \cdot 6\text{H}_2\text{O} \xrightarrow{\Delta}$
$\text{Fe}_2(\text{SO}_4)_3 \xrightarrow{\Delta}$	$\text{FeCl}_3 \cdot 6\text{H}_2\text{O} \xrightarrow{\Delta}$
$\text{CaSO}_4 \cdot 2\text{H}_2\text{O} \xrightarrow{<200^\circ\text{C}} \text{A} \xrightarrow{>200^\circ\text{C}} \text{B} \xrightarrow{\Delta}$	$\text{MgCl}_2 \cdot \text{H}_2\text{O} \xrightarrow{\Delta}$
$2\text{KMnO}_4 \xrightarrow{\Delta}$	$\text{K}_2\text{Cr}_2\text{O}_7 \xrightarrow{\Delta}$
$2\text{KClO}_3 \xrightarrow{\Delta}$	$(\text{NH}_4)_2\text{Cr}_2\text{O}_7 \xrightarrow{\Delta}$

2.20 SOLUBILITY OF IONIC COMPOUNDS

(a) Solubility of ionic compounds :

Soluble in polar solvents like water which have high dielectric constant

Factors affecting solubility of ionic compounds :

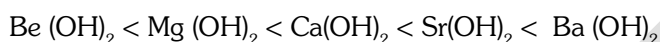
- (i) Dielectric constant of the solvent increases the solubility of compound increases



- (ii) If heat of hydration of ions exceeds the lattice energy (L.E.) of ionic compounds, the ionic compounds will be soluble in water.
- Lattice energy as well as hydration energy depend on the size of ions. Both lattice energy and hydration energy decrease with increase in ionic size.

Two general rule, regarding the solubility is that -

(a) If the anion and the cation are of comparable size, the cationic radius will influence the lattice energy. Since lattice energy decreases much more than the hydration energy with increasing ionic size, solubility will increase as we go down the group.



Solubility increases

(b) If the anion is large compared to the cation, (i.e. compound contain ions with widely different radii) the lattice energy will remain almost constant i.e. change is very small within a particular group. Since the hydration energies decrease down a group, solubility will decrease.



Solubility decreases

Note :

- If common ion is small like Na^+ , Li^+ , O^{2-} , F^- , OH^- , IIA cation then lattice energy dominates.
- If common ion is large like Cs^+ , Rb^+ , Br^- , I^- , polyatomic anion like CO_3^{2-} , SO_4^{2-} then hydration energy dominates.

Some important solubility orders:

Examples

- $\text{BeSO}_4 > \text{MgSO}_4 > \text{CaSO}_4 > \text{SrSO}_4 > \text{BaSO}_4$ (SO_4^{2-} larger)
- $\text{CsF} > \text{CsCl} > \text{CsBr} > \text{CsI}$ [Cs^+ (larger)]
- $\text{BeS} > \text{MgS} > \text{CaS} > \text{SrS} > \text{BaS}$ (S^{2-} larger)

*Important facts about solubility

- All metal chlorides are soluble except Ag, Pb, Hg
- All metal sulphides are insoluble except : IA and ammonium
- Highly insoluble sulphides are of Hg^{2+} , Pb^{2+} , Cu^{2+} , Cd^{2+} , Bi^{3+}
- All hydroxides are insoluble except IA, lower IIA (Ca^{+2} , Sr^{+2} , Ba^{+2}) and ammonium
- Most insoluble hydroxides are of Al^{3+} , Fe^{3+} , Cr^{3+}
- All metal nitrate, Acetate, perchlorate are soluble in water
- Generally metal sulphates are soluble in water except Ag, Hg, Pb, Ca, Sr, Ba
- All alkali metal salts are soluble in water except . LiF , Li_2CO_3 , Li_3PO_4 , $\text{Li}_2\text{C}_2\text{O}_4$

2.21 MELTING POINT

Melting point of ionic compounds

The two factors which mainly influence the melting point of ionic compound are

- (i) Lattice enthalpy (ii) Polarisation

M.P. $\propto$ LE when $\left\{ \begin{array}{l} \text{Cation} = \text{Na}^+, \text{K}^+, \text{Rb}^+, \text{Cs}^+ \\ \text{or} \\ \text{Anion} = \text{F}^-, \text{O}^{2-}, \text{N}^{3-}, \text{C}^{4-}, \text{H}^- \end{array} \right.$

Melting point of ionic compound > covalent compound

Except Giant molecules

↓
Diamond, Carborundum (SiC)
Norbide (B₄C), Silica (SiO₂), Borazone (BN)_x

Order of Hardness : Diamond > Norbide > SiC > Al₂O₃

Some important melting point orders :

Examples



Melting Point and Boiling Point of non-metallic molecules

- (i) $\text{CH}_3\text{—O—CH}_3 < \text{C}_2\text{H}_5\text{OH}$ (boiling point)
- (ii) $\text{CH}_4 < \text{SiH}_4 < \text{GeH}_4 < \text{SnH}_4 < \text{PbH}_4$ (boiling point)
- (iii) $\text{I-Cl} > \text{Br}_2$ (boiling point)
- (iv) $\text{H}_2 < \text{O}_2 < \text{H}_2\text{O}$ (boiling point)
- (v) $\text{CH}_3\text{-OH} < \text{H}_2\text{O}$ (surface tension)
- (vi) $\text{HCl} < \text{HNO}_3 < \text{H}_2\text{SO}_4 < \text{H}_3\text{PO}_4$ (melting point)
- (vii) $\text{H}_2\text{O} < \text{D}_2\text{O}$ (boiling point)
- (viii) O-nitro phenol < p-nitro phenol (boiling point)
- (ix) $\text{H}_2\text{O} \gg \text{H}_2\text{S} < \text{H}_2\text{Se} < \text{H}_2\text{Te}$ (melting point)
- (x) $\text{NH}_3 \gg \text{PH}_3 < \text{AsH}_3 < \text{SbH}_3$ (melting point)
- (xi) $\text{HCl} < \text{HBr} < \text{HI} < \text{HF}$ (boiling point)
- (xii) $\text{HCl} < \text{HBr} < \text{HF} < \text{HI}$ (melting point)

Melting point & boiling point of elements

- (i) $\text{Li} > \text{Na} > \text{K} > \text{Rb} > \text{Cs}$ (melting point)
 - (ii) $\text{Be} > \text{Ca} > \text{Sr} > \text{Ba} > \text{Mg}$ (melting point)
 - (iii) **3d series**
 Melting point $\propto$ number of unpaired electrons
 $\therefore$ In 3d series (MP) maximum = Cr (chromium)
 In 4d series (MP) maximum = Mo
 In 5d series (MP) maximum = W
 - (iv) $3d < 4d < 5d$ (melting point)
 - (v) $\text{Zn} > \text{Cd} > \text{Hg}$ (Melting point)
- $\text{He} < \text{Ne} < \text{Ar} < \text{Kr} < \text{Xe}$ (boiling point)
 $\text{F}_2 < \text{Cl}_2 < \text{Br}_2 < \text{I}_2$ (boiling point)
 $\text{N}_2 < \text{P}_4 < \text{As}_4$ (boiling point)
 $\text{O}_2 < \text{S}_8 < \text{Se}_8 < \text{Te}_8$ (boiling point)
 $\text{H}_o > \text{He}$ (melting point)

Note : Maximum melting point in Boron family $\rightarrow$ Boron

Minimum melting point in Boron family $\rightarrow$ Gallium

Maximum melting point in all elements $\rightarrow$ Carbon

Minimum melting point in all elements $\rightarrow$ Helium

Maximum melting point in Metals → Tungsten

Minimum melting point in Metals $\rightarrow$ Mercury

COLOUR OF COMPOUNDS

More the covalent character, more will be the colour intensity.

Colour intensity $\propto \phi$ (Covalent character)

	AgF	AgCl	AgBr	AgI
Example :	(Colourless)	(White)	(Light yellow)	(Dark yellow)

BEGINNER'S BOX-10

Arrange in order of thermal stability :

1. BeCO_3 , MgCO_3 , CaCO_3 , SrCO_3 , BaCO_3
 KO_2 , RbO_2 , CsO_2
 NH_3 , PH_3 , AsH_3 , SbH_3 , BiH_3
 H_2O , H_2S , H_2Se , H_2Te
 BeO , MgO , CaO , SrO , BaO
 NH_3 , H_2O , HF
 NaF , MgF_2 , AlF_3
 Na_2SO_4 , MgSO_4 , $\text{Al}_2(\text{SO}_4)_3$
 CCl_4 , SiCl_4 , GeCl_4 , SnCl_4 , PbCl_4

2. Give solubility order for :-

- | | |
|-------------------------------------|------------------------------|
| (i) alkali metal hydroxide | (ii) Sodium halides |
| (iii) alkali metal per chlorates | (iv) alkali metal oxides |
| (v) alkaline earth metal carbonates | (vi) alkali metal carbonates |
| (vii) alkali metal bicarbonates | (viii) silver halides |
| (ix) lead (II) halides | (x) mercury (II) halides |

3. Give the order of melting points

- Na_2O , MgO , Al_2O_3
- CaCl_2 , FeCl_2 , FeCl_3
- NaF , MgO , ScN , TiC
- SnCl_2 , SnCl_4
- KCl , CuCl , CuCl_2

ANSWER KEY

BEGINNER'S BOX-1	Que.	1	2	3	4						
	Ans.	3	4	2	1						

BEGINNER'S BOX-2	Que.	1	2	3	4	5	6	7	8		
	Ans.	3	1	4	3	1	3	1	2		

BEGINNER'S BOX-3	Que.	1	2	3	4	5	6	7	8	9	10	11
	Ans.	3	1	3	2	2	2	3	4	4	3	4

BEGINNER'S BOX-4	Que.	1	2	3	4	5	6	7	8	9	10
	Ans.	1	3	2	3	3	2	1	2	1	3

BEGINNER'S BOX-5	Que.	1	2	3	4	5				
	Ans.	1	2	3	1	2				

BEGINNER'S BOX-6	Que.	1	2	3	4	5	6	7	8	9	10
	Ans.	3	1	4	3	3	1	3	3	2	1

BEGINNER'S BOX-7	Que.	1	2	3	4	5	6	7	8	9	
	Ans.	4	4	2	2	2	2	1	4	1	

BEGINNER'S BOX-8	Que.	1	2	3	4	5	6	7	8	9	10
	Ans.	3	1	3	3	1	1	3	1	4	3

BEGINNER'S BOX-9	Que.	1	2	3	4	5	6	7	8	9		
	Ans.	4	1	4	4	4	3	2	1	3		

EXERCISE-I (Conceptual Questions)

Build Up Your Understanding

OCTET RULE :

- Among the following element group number is not equals to valence electrons -
 (1) Na (2) N
 (3) Ne (4) None
- Which of the following does not act as lewis acid ?
 (1) BF_3 (2) SnCl_4
 (3) CCl_4 (4) SF_4

COVALENT BOND

- The strength of bonds by 2s - 2s, 2p - 2p and 2p - 2s overlapping has the order :-
 (1) $s - s > p - p > s - p$
 (2) $s - s > p - s > p - p$
 (3) $p - p > s - p > s - s$
 (4) $p - p > s - s > p - s$
- In which of the excitation state of chlorine ClF_3 is formed:-
 (1) In ground state
 (2) In third excitation state
 (3) In first excitation state
 (4) In second excitation state
- A sigma bond is formed by the overlapping of :-
 (1) s-s orbital alone
 (2) s and p orbitals alone
 (3) s-s, s-p or p-p orbitals along internuclear axis
 (4) p-p orbital along the sides
- Which overlapping is involved in HCl molecule :-
 (1) s-s overlap (2) p-p overlap
 (3) s-d overlap (4) s-p overlap
- Which is not characteristic of π -bond:-
 (1) π - bond is formed when a sigma bond already formed
 (2) π - bond are formed from hybrid orbitals
 (3) π - bond may be formed by the overlapping of p-orbitals
 (4) π -bond results from lateral overlap of atomic orbitals
- π bond is formed :-
 (1) By overlapping of hybridised orbitals
 (2) Overlapping of s - s orbitals
 (3) Head on overlapping of p -p orbitals
 (4) By p - p collateral overlapping
- Which of the following overlapping is strongest ?
 (1) 1s - 1s (2) 2p - 2p
 (3) 2s - 2p (4) 1s - 2p
- p-p overlapping will be observed in the molecules of:
 (1) Hydrogen (2) Hydrogen bromide
 (3) Hydrogen chloride (4) Chlorine

- Which compound of xenon is not possible
 (1) XeF_2 (2) XeF_4 (3) XeF_5 (4) XeF_6
- Similarity of fluorine and oxygen may not be attributed to-
 (1) Their atomic and ionic radii are closely similar
 (2) The atom of both elements attain an octet of electrons in their valence shell
 (3) Both of them are highly electronegative elements
 (4) Both form stable $p\pi - p\pi$ multiple bonds with themselves
- Higher is the bond order, greater is -
 (1) Bond dissociation energy
 (2) Covalent character
 (3) Bond length
 (4) Paramagnetism

HYBRIDISATION

- In the protonation of H_2O , change occurs in
 (1) Hybridisation state of oxygen
 (2) Shape of molecule
 (3) Hybridisation and shape both
 (4) None
- In the compound

$$\overset{1}{\text{CH}_2}=\overset{2}{\text{CH}}-\overset{3}{\text{CH}_2}-\overset{4}{\text{CH}_2}-\overset{5}{\text{C}}\equiv\overset{6}{\text{CH}}$$
 the $\text{C}^2 - \text{C}^3$ bond is formed by the overlapping of :-
 (1) $sp - sp^2$ (2) $sp^3 - sp^3$
 (3) $sp - sp^3$ (4) $sp^2 - sp^3$
- Among the following species identify the isostructural pairs:-
 NF_3 , NO_3^- , BF_3 , H_3O^+ , HN_3
 (1) $[\text{NF}_3, \text{NO}_3^-]$ and $[\text{BF}_3, \text{H}_3\text{O}^+]$
 (2) $[\text{NF}_3, \text{HN}_3]$ and $[\text{NO}_3^-, \text{BF}_3]$
 (3) $[\text{NF}_3, \text{H}_3\text{O}^+]$ and $[\text{NO}_3^-, \text{BF}_3]$
 (4) $[\text{NF}_3, \text{H}_3\text{O}^+]$ and $[\text{HN}_3, \text{BF}_3]$
- Which of the set of species have same hybridisation state but different shapes:-
 (1) NO_2^+ , NO_2 , NO_2^-
 (2) ClO_4^- , SF_4 , XeF_4
 (3) NH_4^+ , H_3O^+ , OF_2
 (4) SO_4^{2-} , PO_4^{3-} , ClO_4^-

- 18.** Which of the following elements can not exhibit sp^3d hybridisation state:-
 (a) C (b) P (c) Cl (d) B
 Correct answer is:-
 (1) a, c (2) a, d (3) b, c (4) b, d
- 19.** Which of the following species are expected to be planar:-
 (a) NH_3 (b) NH_3^{2+} (c) CH_3^+ (d) PCl_3
 The correct answer is:-
 (1) b and c (2) c and d
 (3) b and d (4) a and d
- 20.** In which following set of compound/ion has linear shape
 (1) CH_4 , NH_4^+ , BH_4^- (2) CO_3^{2-} , NO_3^- , BF_3
 (3) NO_2^+ , CO_2 , XeF_2 (4) $BeCl_2$, BCl_3 , CH_4
- 21.** Which order of decreasing bond angle is correct:-
 (1) $CCl_4 > BF_3 > NO_2^+$
 (2) $NH_3 > NCl_3 > NBr_3$
 (3) $Br_2O > Cl_2O > OF_2$
 (4) $PCl_3 > PBr_3 > PI_3$
- 22.** Which of the following set is not correct:-
 (1) SO_3 , O_3 , NH_4^+ all have coordinate bonds
 (2) H_2O , NO_2 , ClO_2^- , all are 'V' shape molecules
 (3) I_3^- , ICl_2^- , NO_2^+ ; all are linear molecules
 (4) SF_4 , SiF_4 , XeF_4 are tetrahedral in shape
- 23.** Shape of a molecule having 4 bond pairs and two lone pairs of electrons, will be :-
 (1) Square planar (2) Tetrahedral
 (3) Linear (4) Octahedral
- 24.** The shape of IF_4^+ will be :-
 (1) Square planar
 (2) Tetrahedral
 (3) Pentagonal bipyramidal
 (4) Distorted tetrahedral
- 25.** Which of the following has pyramidal shape ?
 (1) BF_3 (2) H_3O^+
 (3) NO_3^- (4) CO_3^{2-}
- 26.** A σ bonded molecule MX_3 is T-shaped. The number of non-bonding pairs of electrons is
 (1) 0
 (2) 2
 (3) 1
 (4) Can be predicted if atomic number of M is known.
- 27.** Amongst CO_3^{2-} , AsO_3^{3-} , XeO_3 , ClO_3^- , BO_3^{3-} and SO_3^{2-} the non-planar species are :-
 (1) XeO_3 , ClO_3^- , SO_3^{2-} , AsO_3^{3-}
 (2) AsO_3^{3-} , XeO_3 , CO_3^{2-}
 (3) BO_3^{3-} , CO_3^{2-} , SO_3^{2-}
 (4) AsO_3^{3-} , BO_3^{3-} , CO_3^{2-}
- 28.** The type of hybrid orbitals used by chlorine atom in ClO^- , ClO_2^- , ClO_3^- and ClO_4^- is/are :-
 (1) sp , sp^2 , sp^3 and sp^3d (2) sp and sp^3
 (3) Only sp^3 (4) only sp
- 29.** On the basis of hybridization of one s & one p orbitals they are arranged at :-
 (1) Two orbitals mutually at 90° angle
 (2) two orbitals mutually at 180° angle
 (3) Two orbitals mutually at 120° angle
 (4) Two orbitals mutually at 150° angle
- 30.** Which of the following having a square planar structure is
 (1) NH_4^+ (2) BF_4^- (3) XeF_4 (4) CCl_4
- 31.** When p-character of hybridised orbital (formed by s and p orbitals) increases. Then the bond angle
 (1) Decreases (2) Increases
 (3) Becomes twice (4) Remains unaltered
- 32.** Which orbitals overlap to form bond in OF_2
 (1) $sp^3 - 2p$ (2) $sp^2 - 2p$
 (3) $sp - 2p$ (4) $p - p$
- 33.** In compounds X, all the bond angles are exactly $109^\circ 28'$. X is :
 (1) Chloromethane
 (2) Carbon tetrachloride
 (3) Iodoform
 (4) Chloroform
- 34.** Among the following orbitals/bonds, the angle is minimum between :
 (1) sp^3 bonds
 (2) p_x and p_y orbitals
 (3) $H-O-H$ bond in water
 (4) sp bonds
- 35.** The structure and hybridization of $Si(CH_3)_4$ is :
 (1) bent, sp
 (2) trigonal, sp^2
 (3) octahedral, sp^3d
 (4) tetrahedral, sp^3
- 36.** The AsF_5 molecule is trigonal bipyramidal. The hybrid orbitals used by the As atoms for bonding are:
 (1) $d_{x^2-y^2}$, d_{z^2} , s , p_x , p_y (2) d_{xy} , s , p_x , p_y , p_z
 (3) s , p_x , p_y , p_z , d_{z^2} (4) $d_{x^2-y^2}$, s , p_x , p_y

37. When the hybridization state of carbon atom changes from sp^3 , sp^2 and sp , the angle between the hybridized orbitals.
 (1) decrease considerably
 (2) increase progressively
 (3) decrease gradually
 (4) all of these
38. The hybridization states of the central atoms of the ions I_3^- , ICl_4^- and ICl_2^- are respectively :
 (1) sp^2 , dsp^2 , sp^3
 (2) sp^3d , sp^3d^2 and sp^3d
 (3) sp^3d , sp^3d , dsp^2
 (4) sp , sp , dsp^2
39. Molecular shapes of SF_4 , CF_4 and XeF_4 are :-
 (1) The same, with 2, 0 and 1 lone pairs of electrons respectively
 (2) The same, with 1, 1 and 1 lone pairs of electrons respectively
 (3) Different, with 0, 1 and 2 lone pairs of electrons respectively
 (4) Different, with 1, 0 and 2 lone pairs of electrons respectively
40. Which of the following two are isostructural :-
 (1) XeF_2 , IF_2^- (2) NH_3 , BF_3
 (3) CO_3^{2-} , SO_3^{2-} (4) PCl_5 , ICl_5
41. Select the correct matching :

List I		List II	
A : XeF_4	1. Pyramidal		
B : XeF_6	2. T-shape		
C : XeO_3	3. Distorted octahedral		
D : $XeOF_2$	4. Square planar		

A	B	C	D
(1) 4	3	1	2
(2) 1	2	3	4
(3) 2	1	3	4
(4) 4	1	3	2
42. Which one of the following pair is a correct with respect to molecular formula of xenon compound and hybridization state of xenon in it :
 (1) XeF_4 , sp^3 (2) XeF_2 , sp
 (3) XeF_2 , sp^3d (4) XeF_4 , sp^2
43. The xenon compound(s) that are iso-structural with IBr_2^- and BrO_3^- respectively are :
 (1) Linear XeF_2 and pyramidal XeO_3
 (2) Bent XeF_2 and pyramidal XeO_3
 (3) Bent XeF_2 and planar XeO_3
 (4) Linear XeF_2 and tetrahedral XeO_3

DIPOLE MOMENT

44. Which statement is correct:-
 (1) All the compounds having polar bonds, have dipole moment
 (2) SO_2 is non-polar
 (3) H_2O molecule is non polar, having polar bonds
 (4) PH_3 is polar molecule having non polar bonds
45. BeF_2 has zero dipole moment where as H_2O has a dipole moment because :-
 (1) Water is linear
 (2) H_2O is bent
 (3) F is more electronegative than O
 (4) Hydrogen bonding is present in H_2O
46. Which of the following species are symmetrical :-
 (a) XeF_4 (b) XeF_6
 (c) SO_2 (d) NH_3^{+2}
 Correct answer is:-
 (1) a and b (2) b and c
 (3) c and d (4) a and d
47. Which of the following molecule have zero dipole moment:-
 (1) BF_3 (2) CH_2Cl_2
 (3) NF_3 (4) SO_2
48. The dipole moment of NH_3 is:-
 (1) Less than dipole moment of NCl_3
 (2) Higher than dipole moment of NCl_3
 (3) Equal to the dipole moment of NCl_3
 (4) None of these
49. Which of the following order of polarities of molecules is correct:-
 (1) $HF > NH_3 > PH_3$
 (2) $CH_4 > NH_3 > H_2O$
 (3) $CH_3Cl < CH_2Cl_2 < CHCl_3$
 (4) $BF_3 > BeF_2 > F_2$
50. Which set of molecules is polar :-
 (1) XeF_4 , IF_7 , SO_3 (2) PCl_5 , C_6H_6 , SF_6
 (3) $SnCl_2$, SO_2 , NO_2 (4) CO_2 , CS_2 , C_2H_6
51. Which of the following has symmetrical structure :
 (1) PCl_3 (2) CH_2Cl_2
 (3) $CHCl_3$ (4) CCl_4
52. Species having zero dipole moment :-
 (1) XeF_4
 (2) 1,2,4 trichloro benzene
 (3) SF_4
 (4) CH_2Cl_2

53. What conclusion can be drawn from the fact that BF_3 has no dipole moment but PF_3 does
 (1) BF_3 is not symmetrical but PF_3 is
 (2) BF_3 molecule must be linear
 (3) Atomic radius of P is larger than that of B
 (4) BF_3 molecule must be planar triangular
54. PCl_5 is non polar because :-
 (1) P – Cl bond is non-polar
 (2) Its dipole moment is zero
 (3) P – Cl bond is polar
 (4) P & Cl have equal electronegativity
55. Dipole moment of CO_2 is zero which implies that :
 (1) Carbon and oxygen have equal electronegativities
 (2) Carbon has no polar bond
 (3) CO_2 is a linear molecule
 (4) Carbon has bond moments of zero value
56. The correct order of dipole moment is :
 (1) $\text{CH}_4 < \text{NF}_3 < \text{NH}_3 < \text{H}_2\text{O}$
 (2) $\text{NF}_3 < \text{CH}_4 < \text{NH}_3 < \text{H}_2\text{O}$
 (3) $\text{NH}_3 < \text{NF}_3 < \text{CH}_4 < \text{H}_2\text{O}$
 (4) $\text{H}_2\text{O} < \text{NH}_3 < \text{NF}_3 < \text{CH}_4$
57. Which of the following molecule does not possess a permanent dipole moment :
 (1) H_2S (2) SO_2 (3) SO_3^{2-} (4) CS_2
58. Which of the following has the highest value of dipole moment :
 (1) HCl (2) HF (3) HI (4) HBr

MOLECULAR ORBITAL THEORY

59. The ion that is isoelectronic with CO and having same bond order is :-
 (1) CN^- (2) O_2^+ (3) O_2^- (4) N_2^+
60. Which of the following is paramagnetic:-
 (1) O_2^- (2) CN^- (3) CO (4) NO^+
61. In the following which of the two are paramagnetic
 (a) N_2 (b) CO (c) B_2 (d) NO_2
 Correct answer is :-
 (1) a and c (2) b and c
 (3) c and d (4) b and d
62. The bond order of CO molecule on the basis of molecular orbital theory is
 (1) Zero (2) 2 (3) 3 (4) 1
63. The energy of σ_{2s} orbital is greater, than σ_{1s}^* orbital because
 (1) σ_{2s} orbital is bigger than σ_{1s}^* orbital
 (2) σ_{2s} orbital is a bonding orbital where as σ_{1s}^* is an antibonding orbital
 (3) σ_{2s} orbital has a greater value of n than σ_{1s}^* orbital
 (4) None
64. The no. of antibonding electron pair in O_2^- is
 (1) 4 (2) 3
 (3) 8 (4) 10
65. Which of the following species will have the minimum bond energy
 (1) N_2 (2) N_2^-
 (3) N_2^+ (4) N_2^{2-}
66. Which of the following ion has not bond order of 2.5 ?
 (1) O_2^- (2) O_2^+
 (3) N_2^+ (4) N_2^-
67. In a homonuclear molecule which of the following set of orbitals are degenerate ?
 (1) σ_{2s} and σ_{1s} (2) π_{2p_x} and π_{2p_y}
 (3) π_{2p_x} and σ_{2p_z} (4) σ_{2p_z} and $\pi_{2p_x}^*$

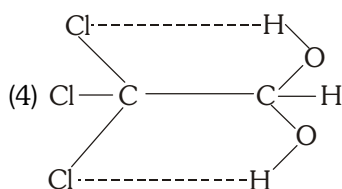
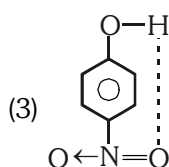
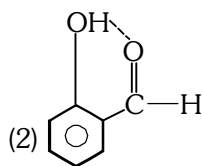
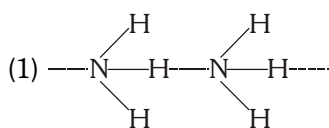
COORDINATE BOND

68. In Co-ordinate bond, the acceptor atoms must essentially contain in its valency shell an orbital:-
 (1) With paired electron
 (2) With single electron
 (3) With no electron
 (4) With three electron
69. The bonds present in N_2O_5 are :-
 (1) Only ionic
 (2) Covalent & coordinate
 (3) Only covalent
 (4) Covalent & ionic
70. Dative bond is present in
 (1) SO_3 (2) NH_3
 (3) K_2CO_3 (4) BF_3
71. Which of the following has no coordinate bond ?
 (1) PH_3 (2) $\text{P}_2\text{H}_6^{+2}$
 (3) P_2H_5^+ (4) PH_4^+
72. The compound containing co-ordinate bond is :
 (1) H_2SO_4 (2) O_3
 (3) SO_3 (4) All of these

WEAK FORCES

73. In dry ice the bond present between two molecules is
 (1) Ionic bond (2) Covalent bond
 (3) Hydrogen bond (4) Vander Waal
74. Intermolecular hydrogen bonds are not present in:-
 (1) $\text{CH}_3\text{CH}_2\text{OH}$ (2) CH_3COOH
 (3) $\text{C}_2\text{H}_5\text{NH}_2$ (4) CH_3OCH_3

75. In which of the following molecule, the shown hydrogen bond is not possible:-



76. Correct order of volatility is:-

- (1) $\text{HF} > \text{HCl} > \text{HBr} > \text{HI}$
- (2) $\text{HCl} > \text{HBr} > \text{HI} > \text{HF}$
- (3) $\text{HI} > \text{HBr} > \text{HCl} > \text{HF}$
- (4) $\text{HBr} < \text{HCl} < \text{HI} < \text{HF}$

77. The correct order of volatility is:-

- (1) $\text{NH}_3 < \text{H}_2\text{O}$
- (2) p-nitro phenol < o-nitro phenol
- (3) $\text{CH}_3\text{OH} > \text{CH}_3 - \text{O} - \text{CH}_3$
- (4) $\text{HF} > \text{HCl}$

78. The incorrect order of decreasing boiling points is

- (1) $\text{NH}_3 > \text{AsH}_3 > \text{PH}_3$
- (2) $\text{H}_2\text{O} > \text{H}_2\text{Se} > \text{H}_2\text{S}$
- (3) $\text{Br}_2 > \text{Cl}_2 > \text{F}_2$
- (4) $\text{CH}_4 > \text{GeH}_4 > \text{SiH}_4$

79. Acetic acid exists as dimer in benzene due to:-

- (1) Condensation reaction
- (2) Hydrogen bonding
- (3) Presence of carboxyl group
- (4) None of the above

80. Maximum no. of hydrogen bonds formed by a water molecule in ice is

- (1) 4
- (2) 3
- (3) 2
- (4) 1

81. Strongest hydrogen bond is shown by :

- (1) Water
- (2) Ammonia
- (3) Hydrogen fluoride
- (4) Hydrogen sulphide

82. Density of ice is less than that of water because of

- (1) presence hydrogen bonding
- (2) crystal modification of ice
- (3) open porous structure of ice due to hydrogen bonding
- (4) different physical states of these

83. NH_3 has abnormally high boiling point because it has :

- (1) Alkaline nature
- (2) Distorted shape
- (3) sp^3 - Hybridisation
- (4) Hydrogen bonding

84. Which of the following is soluble in water ?

- (1) CS_2
- (2) $\text{C}_2\text{H}_5\text{OH}$
- (3) CCl_4
- (4) CHCl_3

85. KF combines with HF to form KHF_2 . The compound contains the species :

- (1) K^+ , F^- and H^+
- (2) K^+ , F^- and HF
- (3) K^+ and $[\text{HF}_2]^-$
- (4) $[\text{KHF}]^+$ and F_2

86. Which of the following compounds show intramolecular hydrogen bonding :

- (A) o-nitrophenol
- (B) p-nitrophenol
- (C) phenol
- (D) salicylaldehyde
- (1) A & B
- (2) A & C
- (3) A & D
- (4) B & C

87. The pair of molecules forming strongest hydrogen bonds are :

- (1) SiH_4 and SiF_4
- (2) $\text{CH}_3 - \text{C}(=\text{O}) - \text{CH}_3$ and CHCl_3
- (3) $\text{H} - \text{C}(=\text{O}) - \text{OH}$ and $\text{CH}_3 - \text{C}(=\text{O}) - \text{OH}$
- (4) H_2O and H_2

IONIC BOND

88. Which one is the correct statement with reference to solubility of MgSO_4 in water:

- (1) Hydration energy of MgSO_4 is higher in comparison to its lattice energy
- (2) Ionic potential of Mg^{2+} is very low
- (3) SO_4^{2-} ion mainly contributes towards hydration energy
- (4) Size of Mg^{2+} and SO_4^{2-} are similar

- 89.** Conditions for ionic bond formation is/are :
 (a) Small cation, large anion
 (b) Low IP of cation, high electron affinity of anion
 (c) Large cation, small anion and less charge
 (d) Less lattice energy
 Correct answer is:
 (1) a, d (2) b, c and d
 (3) b and c (4) a, b
- 90.** Capacity of solvent to neutralise charge on ionic compound is called:-
 (1) Solvation energy (2) Dielectric constant
 (3) Dipole moment (4) Solubility
- 91.** The force responsible for dissolution of ionic compound in water is –
 (1) Dipole – dipole forces
 (2) Ion – dipole force
 (3) Ion – ion force
 (4) Hydrogen bond
- 92.** Born Haber cycle is mainly used to determine
 (1) Lattice energy (2) Electron affinity
 (3) Ionisation energy (4) Electronegativity
- 93.** An ionic compound $A^+ B^-$ is most likely to be formed when –
 (1) Ionization energy of A is low
 (2) Electron affinity of B is high
 (3) Electron affinity of B is low
 (4) Both (1) and (2)
- 94.** The pair of elements which on combination are most likely to form an ionic compound is
 (1) Na and Ca (2) K and O
 (3) O and Cl (4) Al and I
- 95.** Choose the compound of each of the following pairs that has the more solubility in water :-
 (a) $BeSO_4$, $BaSO_4$ (b) $NaCl$, $MgCl_2$
 (c) $AgCl$, AgI
 (1) $BeSO_4$, $MgCl_2$, AgI (2) $BeSO_4$, $NaCl$, $AgCl$
 (3) $BaSO_4$, $NaCl$, $AgCl$ (4) $BaSO_4$, $MgCl_2$, AgI
- 96.** Highest melting point would be of
 (1) $AlCl_3$ (2) $LiCl$
 (3) $NaCl$ (4) $BeCl_2$
- 97.** Which of the following substance will have highest b.p.t. ?
 (1) He (2) CsF
 (3) NH_3 (4) $CHCl_3$
- 98.** As compared to covalent compounds electrovalent compounds generally possess
 (1) High m.p. and high b.p.
 (2) Low m.p. and low b.p.
 (3) Low m.p. and high b.p.
 (4) high m.p. and low b.p.
- 99.** The electronic configuration of metal M is $1s^2 2s^2 2p^6 3s^1$. The formula of its oxide will be :
 (1) MO (2) M_2O
 (3) M_2O_3 (4) MO_2
- 100.** Which of the following pairs will form the most stable ionic bond ?
 (1) Na and Cl (2) Mg and F
 (3) Li and F (4) Na and F
- 101.** For two ionic solids, CaO and KI . Identify the wrong statement among the following :
 (1) Lattice energy of CaO is much larger than that of KI
 (2) KI is soluble in benzene
 (3) CaO has higher melting point
 (4) KI has lower melting point
- 102.** Which of the following does not show electrical conduction ?
 (1) diamond
 (2) graphite
 (3) sodium chloride (fused)
 (4) potassium
- 103.** The most covalent halide is:-
 (1) AlF_3 (2) $AlCl_3$
 (3) $AlBr_3$ (4) AlI_3
- 104.** Ionic potential (ϕ) of electropositive element will be highest in which of the following compound:-
 (1) $CsCl$ (2) $MgCl_2$
 (3) AlF_3 (4) SF_6
- 105.** $LiCl$ is soluble in organic solvent while $NaCl$ is not because :-
 (1) Lattice energy of $NaCl$ is less than that of $LiCl$
 (2) Ionisation potential of Li is more than that of Na
 (3) Li^+ has more hydration energy than Na^+ ion
 (4) $LiCl$ is more covalent compound than that $NaCl$
- 106.** The most stable carbonate is
 (1) Li_2CO_3 (2) $BeCO_3$
 (3) $CaCO_3$ (4) $BaCO_3$
- 107.** Correct order of covalent character of alkaline earth metal chloride in
 (1) $BeCl_2 < MgCl_2 < CaCl_2 < SrCl_2$
 (2) $BeCl_2 < CaCl_2 < SrCl_2 < MgCl_2$
 (3) $BeCl_2 > MgCl_2 > CaCl_2 > SrCl_2$
 (4) $SrCl_2 > BeCl_2 > CaCl_2 > MgCl_2$
- 108.** Which of the compound is least soluble in water
 (1) AgF (2) $AgCl$
 (3) $AgBr$ (4) AgI

109. CCl_4 is more covalent than LiCl because :

- (1) There is more polarization of Cl in CCl_4
- (2) There is more polarization of Cl in LiCl
- (3) CCl_4 has more weight
- (4) None of above

110. The M.P. of SnCl_4 is less than that of SnCl_2 , the suitable reason for the observed fact is

- (1) There is more charge on Sn^{+4}
- (2) The size of Sn^{+4} is small
- (3) Ionic potential of Sn^{+4} is high
- (4) The shape of SnCl_4 is tetrahedral

111. The correct order of decreasing polarisable ions is:

- (1) Cl^- , Br^- , I^- , F^-
- (2) F^- , I^- , Br^- , Cl^-
- (3) F^- , Cl^- , Br^- , I^-
- (4) I^- , Br^- , Cl^- , F^-

112. Ionic conductances of hydrated M^+ ions are in the order –

- (1) $\text{Li}^+(\text{aq}) > \text{Na}^+(\text{aq}) > \text{K}^+(\text{aq}) > \text{Rb}^+(\text{aq}) > \text{Cs}^+(\text{aq})$
- (2) $\text{Li}^+(\text{aq}) > \text{Na}^+(\text{aq}) < \text{K}^+(\text{aq}) < \text{Rb}^+(\text{aq}) < \text{Cs}^+(\text{aq})$
- (3) $\text{Li}^+(\text{aq}) > \text{Na}^+(\text{aq}) > \text{K}^+(\text{aq}) > \text{Rb}^+(\text{aq}) < \text{Cs}^+(\text{aq})$
- (4) $\text{Li}^+(\text{aq}) < \text{Na}^+(\text{aq}) < \text{K}^+(\text{aq}) < \text{Rb}^+(\text{aq}) < \text{Cs}^+(\text{aq})$

113. Which of the following halides has the highest melting point –

- (1) NaCl
- (2) KCl
- (3) NaBr
- (4) NaF

114. Which of the following does not give an oxide on heating –

- (1) MgCO_3
- (2) Li_2CO_3
- (3) ZnCO_3
- (4) K_2CO_3

115. Which decomposes on heating –

- (1) NaOH
- (2) KOH
- (3) LiOH
- (4) RbOH

116. Which of the following forms metal oxide on heating

- (1) Na_2CO_3
- (2) Li_2CO_3
- (3) K_2SO_4
- (4) NaHCO_3

117. Increasing order of stability of –

- I. K_2CO_3 II. MgCO_3 III. Na_2CO_3
- (1) $\text{I} < \text{II} < \text{III}$
- (2) $\text{II} < \text{III} < \text{I}$
- (3) $\text{II} < \text{I} < \text{III}$
- (4) $\text{I} < \text{III} < \text{II}$

118. Which of the following carbonate will not decompose on heating :-

- (1) BaCO_3
- (2) ZnCO_3
- (3) Na_2CO_3
- (4) Li_2CO_3

119. The decomposition temperature is maximum for:-

- (1) BeCO_3
- (2) CaCO_3
- (3) K_2CO_3
- (4) Li_2CO_3

EXERCISE-I (Conceptual Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	3	3	3	3	4	2	4	1	4	3	4	1	2	4
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	3	3	2	1	3	3	4	1	4	2	2	1	3	2	3
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	1	1	2	2	4	3	2	2	4	1	1	3	1	4	2
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	4	1	2	1	3	4	1	4	2	3	1	4	2	1	1
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	3	3	3	2	4	1	2	3	2	1	1	4	4	4	3
Que.	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Ans.	2	2	4	2	1	3	3	4	2	3	3	3	1	3	2
Que.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Ans.	2	1	4	2	2	3	2	1	2	2	2	1	4	4	4
Que.	106	107	108	109	110	111	112	113	114	115	116	117	118	119	
Ans.	4	3	4	1	3	4	4	4	4	3	2	2	3	3	

EXERCISE-II (Previous Year Questions)

AIPMT/NEET & AIIMS (2006-2018)

AIPMT-2006

- The number of unpaired electrons in a paramagnetic diatomic molecule of an element with atomic number 16 is
(1) 1 (2) 2 (3) 3 (4) 4
- Which of the following species has a linear shape:
(1) O_3 (2) NO_2^-
(3) SO_2 (4) NO_2^+
- Which of the following is not isostructural with $SiCl_4$?
(1) NH_4^+ (2) SCl_4
(3) SO_4^{2-} (4) PO_4^{3-}
- The electronegativity difference between N and F is greater than that between N and H yet the dipole moment of NH_3 (1.5 D) is larger than that of NF_3 (0.2 D). This is because
(1) in NH_3 the atomic dipole and bond dipole are in the opposite directions whereas in NF_3 these are in the same direction
(2) in NH_3 as well as in NF_3 the atomic dipole and bond dipole are in the same direction
(3) in NH_3 the atomic dipole and bond dipole are in the same direction whereas in NF_3 these are in opposite directions
(4) in NH_3 as well as in NF_3 the atomic dipole and bond dipole are in opposite directions
- In which of the following molecules are all the bonds **not** equal :-
(1) NF_3 (2) ClF_3
(3) BF_3 (4) AlF_3

AIPMT-2007

- The correct order of increasing thermal stability of K_2CO_3 , $MgCO_3$, $CaCO_3$ and $BeCO_3$ is
(1) $BeCO_3 < MgCO_3 < CaCO_3 < K_2CO_3$
(2) $MgCO_3 < BeCO_3 < CaCO_3 < K_2CO_3$
(3) $K_2CO_3 < MgCO_3 < CaCO_3 < BeCO_3$
(4) $BeCO_3 < MgCO_3 < K_2CO_3 < CaCO_3$
- In which of the following pairs the two species are iso-structural
(1) SO_3^{2-} and NO_3^-
(2) BF_3 and NF_3
(3) BrO_3^- and XeO_3
(4) SF_4 and XeF_4

- The correct order of C-O bond length among CO , CO_3^{2-} , CO_2 is
(1) $CO < CO_3^{2-} < CO_2$ (2) $CO_3^{2-} < CO_2 < CO$
(3) $CO < CO_2 < CO_3^{2-}$ (4) $CO_2 < CO < CO_3^{2-}$
- In which of the following hydration energy is higher than lattice energy
(1) $MgSO_4$ (2) $CaSO_4$
(3) $BaSO_4$ (4) $SrSO_4$
- Which one of the following orders correctly represents the increasing acid strengths of the given acids :
(1) $HOCIO_3 < HOCIO_2 < HOCIO < HOCl$
(2) $HOCl < HOCIO < HOCIO_2 < HOCIO_3$
(3) $HOCIO < HOCl < HOCIO_2 < HOCIO_3$
(4) $HOCIO_2 < HOCIO_3 < HOCIO < HOCl$

AIPMT-2008

- Four diatomic species are listed below in different sequences. Which of these presents the correct order of their increasing bond order :
(1) $C_2^{2-} < He_2^+ < NO < O_2^-$
(2) $He_2^+ < O_2^- < NO < C_2^{2-}$
(3) $O_2^- < NO < C_2^{2-} < He_2^+$
(4) $NO < C_2^{2-} < O_2^- < He_2^+$
- The angular shape of ozone molecule (O_3) consists of
(1) 1 sigma and 1 pi bonds
(2) 2 sigma and 1 pi bonds
(3) 1 sigma and 2 pi bonds
(4) 2 sigma and 2 pi bonds
- The correct order of increasing bond angles in the following triatomic species is:-
(1) $NO_2^+ < NO_2 < NO_2^-$ (2) $NO_2^+ < NO_2^- < NO_2$
(3) $NO_2^- < NO_2^+ < NO_2$ (4) $NO_2^- < NO_2 < NO_2^+$

AIPMT-2009

- In which of the following molecules/ions BF_3 , NO_2^- , NH_2^- and H_2O , the central atom is sp^2 hybridized :
(1) BF_3 and NO_2^- (2) NO_2^- and NH_2^-
(3) NH_2^- and H_2O (4) NO_2^- and H_2O

15. According to MO theory which of the following lists ranks the nitrogen species in terms of increasing bond order :
- (1) $N_2^- < N_2^{2-} < N_2$ (2) $N_2^- < N_2 < N_2^{2-}$
 (3) $N_2^{2-} < N_2^- < N_2$ (4) $N_2 < N_2^{2-} < N_2^-$
16. In the case of alkali metals, the covalent character decreases in the order :
- (1) $MI > MBr > MCl > MF$
 (2) $MCl > MI > MBr > MF$
 (3) $MF > MCl > MBr > MI$
 (4) $MF > MCl > MI > MBr$
17. What is the dominant intermolecular force or bond that must be overcome in converting liquid CH_3OH to a gas :
- (1) London or dispersion force
 (2) Hydrogen bonding
 (3) Dipole-dipole interaction
 (4) Covalent bonds

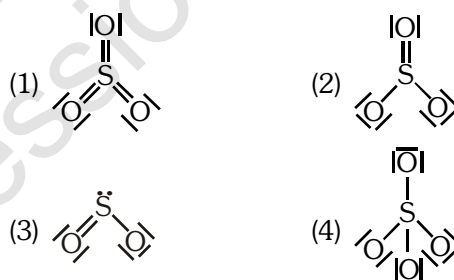
AIPMT-2010

18. Some of the properties of the two species, NO_3^- and H_3O^+ are described below. Which one of them is correct :-
- (1) Isostructural with same hybridization for the central atom.
 (2) Isostructural with different hybridization for the central atom.
 (3) Similar in hybridization for the central atom with different structures.
 (4) Dissimilar in hybridization for the central atom with different structures.
19. In which of the following molecules the central atom does not have sp^3 hybridization :-
- (1) SF_4 (2) BF_4^-
 (3) NH_4^+ (4) CH_4
20. Which one of the following species does not exist under normal conditions :
- (1) Li_2 (2) Be_2^+
 (3) Be_2 (4) B_2
21. In which of the following pairs of molecules/ions, the central atoms have sp^2 hybridization :
- (1) BF_3 and NH_2^- (2) NO_2^- and NH_3
 (3) BF_3 and NO_2 (4) NH_2^- and H_2O
22. Which of the following alkaline earth metal sulphates has hydration enthalpy higher than the lattice enthalpy :
- (1) $SrSO_4$ (2) $CaSO_4$
 (3) $BeSO_4$ (4) $BaSO_4$
23. In which one of the following species the central atom has the type of hybridisation which is not the same as that present in the other three :
- (1) PCl_5 (2) SF_4
 (3) I_3^- (4) $SbCl_5^{2-}$

24. Property of the alkaline earth metals that increases with their atomic number :-
- (1) Electronegativity
 (2) Solubility of their hydroxides in water
 (3) Solubility of their sulphates in water
 (4) Ionization energy

AIPMT Pre-2011

25. Considering the state of hybridization of carbon atoms, find out the molecule among the following which is linear :
- (1) $CH_3-CH=CH-CH_3$
 (2) $CH_3-C \equiv C-CH_3$
 (3) $CH_2=CH-CH_2-C \equiv CH$
 (4) $CH_3-CH_2-CH_2-CH_3$
26. Which of the following has the minimum bond length :
- (1) O_2^+ (2) O_2^- (3) O_2^{2-} (4) O_2
27. Which of the two ions from the list given below that have the geometry that is explained by the same hybridization of orbitals, NO_2^- , NO_3^- , NH_2^- , NH_4^+ , SCN^- :
- (1) NO_2^- and NO_3^- (2) NH_4^+ and NO_3^-
 (3) SCN^- and NH_2^- (4) NO_2^- and NH_2^-
28. Which of the following compounds has the lowest melting point :
- (1) $CaCl_2$ (2) $CaBr_2$
 (3) CaI_2 (4) CaF_2
29. Which of the following structures is the most preferred and hence of lowest energy for SO_3 :



AIIMS-2011

30. In solid ice, oxygen atom is surrounded:
- (1) Tetrahedrally by 4 hydrogen atoms
 (2) Octahedrally by 2 oxygen and 4 hydrogens atoms
 (3) Tetrahedrally 2 hydrogen and 2 oxygens atoms
 (4) Octahedrally by 6 hydrogens atoms
31. Decreasing order of bond angle is :
- (1) $BeCl_2 > NO_2 > SO_2$
 (2) $BeCl_2 > SO_2 > NO_2$
 (3) $SO_2 > BeCl_2 > NO_2$
 (4) $SO_2 > NO_2 > BeCl_2$

AIPMT Pre-2012

32. Which one of the following pairs is isostructural (i.e. having the same shape and hybridization) :
 (1) NF_3 and BF_3 (2) $[\text{BF}_4^-]$ and $[\text{NH}_4^+]$
 (3) $[\text{BCl}_3]$ and $[\text{BrCl}_3]$ (4) $[\text{NH}_3]$ and $[\text{NO}_3^-]$
33. Which of the following species contains three bond pairs and one lone pair around the central atom:
 (1) NH_2^- (2) PCl_3 (3) H_2O (4) BF_3
34. The pair of species with the same bond order is:
 (1) NO , CO (2) N_2 , O_2
 (3) O_2^{2-} , B_2 (4) O_2^+ , NO^+
35. Bond order of 1.5 is shown by:
 (1) O_2^{2-} (2) O_2 (3) O_2^+ (4) O_2^-

AIPMT Mains-2012

36. During change of O_2 to O_2^- ion, the electron adds on which one of the following orbitals :
 (1) σ^* orbital (2) σ orbital
 (3) π^* orbital (4) π orbital
37. Four diatomic species are listed below. Identify the correct order in which the bond order is increasing in them :
 (1) $\text{C}_2^{2-} < \text{He}_2^+ < \text{O}_2^- < \text{NO}$
 (2) $\text{He}_2^+ < \text{O}_2^- < \text{NO} < \text{C}_2^{2-}$
 (3) $\text{NO} < \text{O}_2^- < \text{C}_2^{2-} < \text{He}_2^+$
 (4) $\text{O}_2^- < \text{NO} < \text{C}_2^{2-} < \text{He}_2^+$

NEET-UG-2013

38. Which one of the following molecules contains no π bond :
 (1) NO_2 (2) CO_2 (3) H_2O (4) SO_2
39. XeF_2 is isostructural with :
 (1) BaCl_2 (2) TeF_2 (3) $[\text{ICl}_2^-]$ (4) SbCl_3
40. Dipole induced dipole interactions are present in which of the following pairs :
 (1) SiF_4 and He atoms (2) H_2O and alcohol
 (3) Cl_2 and CCl_4 (4) HCl and He atoms
41. Which of the following is a polar molecule :
 (1) XeF_4 (2) BF_3 (3) SF_4 (4) SiF_4
42. Which of the following is paramagnetic :
 (1) NO^+ (2) CO (3) O_2^- (4) CN^-
43. Identify the correct order of solubility in aqueous medium :
 (1) $\text{Na}_2\text{S} > \text{ZnS} > \text{CuS}$ (2) $\text{CuS} > \text{ZnS} > \text{Na}_2\text{S}$
 (3) $\text{ZnS} > \text{Na}_2\text{S} > \text{CuS}$ (4) $\text{Na}_2\text{S} < \text{CuS} > \text{ZnS}$

AIIMS-2013

44. Total no. of Antibonding electrons present in O_2 will be :-
 (1) 6 (2) 8 (3) 4 (4) 2

AIPMT-2014

45. Which of the following molecules has the maximum dipole moment :
 (1) CO_2 (2) CH_4 (3) NH_3 (4) NF_3
46. Which one of the following species has plane triangular shape :
 (1) N_3^- (2) NO_3^- (3) NO_2^- (4) CO_2

AIIMS-2014

47. Heat of sublimation of Na = x J
 Ionization energy of Na = y J
 Bond dissociation energy of Cl_2 = a J
 e^- gain enthalpy of Cl = $-b$ J
 Enthalpy of formation of NaCl = $-c$ J
 Then what will be lattice energy of $\text{NaCl}_{(s)}$
 (1) $b + \left(x + y + c - \frac{a}{2}\right)$
 (2) $b - \left(x + y + c - \frac{a}{2}\right)$
 (3) $b - \left(\frac{a}{2} + x + y + c\right)$
 (4) $c - \frac{a}{2}x + y + c$

AIPMT-2015

48. The correct bond order in the following species is:-
 (1) $\text{O}_2^{2+} < \text{O}_2^- < \text{O}_2^+$ (2) $\text{O}_2^+ < \text{O}_2^- < \text{O}_2^{2+}$
 (3) $\text{O}_2^- < \text{O}_2^+ < \text{O}_2^{2+}$ (4) $\text{O}_2^{2+} < \text{O}_2^+ < \text{O}_2^-$
49. Which of the following pairs of ions are isoelectronic and isostructural :
 (1) $[\text{ClO}_3^-]$, $[\text{CO}_3^{2-}]$ (2) $[\text{SO}_3^{2-}]$, $[\text{NO}_3^-]$
 (3) $[\text{ClO}_3^-]$, $[\text{SO}_3^{2-}]$ (4) $[\text{CO}_3^{2-}]$, $[\text{SO}_3^{2-}]$
50. Which of the following options represents the correct bond order :
 (1) $\text{O}_2^- < \text{O}_2 < \text{O}_2^+$ (2) $\text{O}_2^- > \text{O}_2 < \text{O}_2^+$
 (3) $\text{O}_2^- < \text{O}_2 > \text{O}_2^+$ (4) $\text{O}_2^- > \text{O}_2 > \text{O}_2^+$

51. Solubility of the alkaline earth's metal sulphates in water decreases in the sequence :-
 (1) $\text{Ca} > \text{Sr} > \text{Ba} > \text{Mg}$ (2) $\text{Sr} > \text{Ca} > \text{Mg} > \text{Ba}$
 (3) $\text{Ba} > \text{Mg} > \text{Sr} > \text{Ca}$ (4) $\text{Mg} > \text{Ca} > \text{Sr} > \text{Ba}$
52. Maximum bond angle at nitrogen is present in which of the following :
 (1) NO_2^- (2) NO_2^+
 (3) NO_3^- (4) NO_2

RE-AIPMT-2015

53. On heating which of the following releases CO_2 most easily :
 (1) MgCO_3 (2) CaCO_3
 (3) K_2CO_3 (4) Na_2CO_3
54. Decreasing order of stability of O_2 , O_2^- , O_2^+ and O_2^{2-} is :-
 (1) $\text{O}_2 > \text{O}_2^+ > \text{O}_2^{2-} > \text{O}_2^-$
 (2) $\text{O}_2^- > \text{O}_2^{2-} > \text{O}_2^+ > \text{O}_2$
 (3) $\text{O}_2^+ > \text{O}_2 > \text{O}_2^- > \text{O}_2^{2-}$
 (4) $\text{O}_2^{2-} > \text{O}_2^- > \text{O}_2 > \text{O}_2^+$
55. In which of the following pairs, both the species are not isostructural :
 (1) NH_3 , PH_3
 (2) XeF_4 , XeO_4
 (3) SiCl_4 , PCl_4
 (4) Diamond, silicon carbide
56. The variation of the boiling points of the hydrogen halides is in the order $\text{HF} > \text{HI} > \text{HBr} > \text{HCl}$. What explains the higher boiling point of hydrogen fluoride ?
 (1) The bond energy of HF molecules is greater than in other hydrogen halides
 (2) The effect of nuclear shielding is much reduced in fluorine which polarises the HF molecule
 (3) The electronegativity of fluorine is much higher than for other elements in the group.
 (4) There is strong hydrogen bonding between HF molecules

AIIMS-2015

57. SF_4 & XeF_2 shape respectively are
 (1) Linear and distorted tetrahedral
 (2) See-saw and linear
 (3) T-shape and trigonal bipyramidal
 (4) Tetrahedral and linear
58. In ClF_3 lone pair are present at approx
 (1) 180° (2) 120°
 (3) 90° (4) 60°

59. In solid phase of XeF_6 , it consists of
 (1) $\text{XeF}_5^+ + \text{F}^-$ (2) $\text{XeF}_4 + \text{F}_2$
 (3) $\text{XeF}_5^+ + \text{XeF}_2^-$ (4) $\text{XeF}_2 + 2\text{F}_2$
60. Correct order of bond length
 (1) $\text{CO} \approx \text{CO}_3^{2-} > \text{CO}_2$
 (2) $\text{CO}_2 \approx \text{CO}_3^{2-} > \text{CO}$
 (3) $\text{CO} < \text{CO}_2 < \text{CO}_3^{2-}$
 (4) $\text{CO}_3^{2-} < \text{CO}_2 < \text{CO}$
61. Correct order of bond energy
 (1) $\text{I}-\text{Cl} < \text{I}_2$
 (2) $\text{Br}-\text{F} < \text{F}-\text{F}$
 (3) $\text{Br}-\text{F} < \text{I}-\text{I}$
 (4) $\text{F}-\text{F} < \text{Cl}-\text{Cl}$

NEET-I 2016

62. Consider the molecules CH_4 , NH_3 and H_2O . Which of the given statements is false ?
 (1) The H-C-H bond angle in CH_4 , the H-N-H bond angle in NH_3 , and the H-O-H bond angle in H_2O are all greater than 90°
 (2) The H-O-H bond angle in H_2O is larger than the H-C-H bond angle in CH_4 .
 (3) The H-O-H bond angle in H_2O is smaller than the H-N-H bond angle in NH_3 .
 (4) The H-C-H bond angle in CH_4 is larger than the H-N-H bond angle in NH_3 .
63. Which one of the following orders is correct for the bond dissociation enthalpy of halogen molecules?
 (1) $\text{I}_2 > \text{Br}_2 > \text{Cl}_2 > \text{F}_2$
 (2) $\text{Cl}_2 > \text{Br}_2 > \text{F}_2 > \text{I}_2$
 (3) $\text{Br}_2 > \text{I}_2 > \text{F}_2 > \text{Cl}_2$
 (4) $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$
64. Predict the correct order among the following :-
 (1) lone pair - lone pair > lone pair - bond pair > bond pair - bond pair
 (2) lone pair - lone pair > bond pair - bond pair > lone pair - bond pair
 (3) bond pair - bond pair > lone pair - bond pair > lone pair - lone pair
 (4) lone pair - bond pair > bond pair - bond pair > lone pair - lone pair

65. Match the compounds given in column I with the hybridisation and shape given in column II and mark the **correct** option.

Column-I		Column-II	
(a)	XeF_6	(i)	Distorted octahedral
(b)	XeO_3	(ii)	Square planar
(c)	XeOF_4	(iii)	pyramidal
(d)	XeF_4	(iv)	Square pyramidal

Code :-

(a)	(b)	(c)	(d)
(1) (i)	(iii)	(iv)	(ii)
(2) (i)	(ii)	(iv)	(iii)
(3) (iv)	(iii)	(i)	(ii)
(4) (iv)	(i)	(ii)	(iii)

NEET-II 2016

66. The correct geometry and hybridization for XeF_4 are:
- (1) Planar triangle, sp^3d^3
 - (2) square planar, sp^3d^2
 - (3) octahedral, sp^3d^2
 - (4) trigonal bipyramidal, sp^3d
67. Among the following which one is a wrong statement?
- (1) SeF_4 and CH_4 have same shape
 - (2) I_3^+ has bent geometry
 - (3) PH_5 and BiCl_5 do not exist
 - (4) $\pi\text{-}\pi$ bonds are present in SO_2
68. The hybridizations of atomic orbitals of nitrogen in NO_2^+ , NO_3^- and NH_4^+ respectively are
- (1) sp , sp^2 and sp^3
 - (2) sp^2 , sp and sp^3
 - (3) sp , sp^3 and sp^2
 - (4) sp^2 , sp^3 and sp
69. Which of the following fluoro-compounds is most likely to behave as a Lewis base ?
- (1) CF_4
 - (2) SiF_4
 - (3) BF_3
 - (4) PF_3
70. Which of the following pairs of ions is isoelectronic and isostructural ?
- (1) SO_3^{2-} , NO_3^-
 - (2) ClO_3^- , SO_3^{2-}
 - (3) CO_3^{2-} , NO_3^-
 - (4) ClO_3^- , CO_3^{2-}

AIIMS 2016

71. Minimum lone pair on central atom will be present in :-
- (1) XeO_3
 - (2) XeF_4
 - (3) ClF_3
 - (4) HOCl
72. Which of the following has paramagnetic character in gaseous state :-
- (1) S^{-2}
 - (2) S_2
 - (3) S_8
 - (4) S_6
73. In which of the following molecule dipole moment is more than zero but less than one ?
- (1) NH_3
 - (2) NF_3
 - (3) BeF_2
 - (4) BF_3
74. Select the incorrect statement :-
- (1) hybrid orbitals are maximum repelled to each other
 - (2) only half filled atomic orbital are participate in hybridisation.
 - (3) Bond energy of hybrid orbitals are more than unhybrid atomic orbital.
 - (4) Without hybridisation, geometry of molecule is uncertain.
75. Which of the following has tetrahedral structure :-
- (1) XeF_4
 - (2) H_3PO_4
 - (3) SF_4
 - (4) ClF_3
76. Which of the following molecule has more than one lone pair on central atom :-
- (1) SO_2
 - (2) XeF_2
 - (3) PCl_5
 - (4) IF_7
77. In ice crystal nearest and second nearest atoms to water molecule respectively are :-
- (1) 4 & 4
 - (2) 4 & 6
 - (3) 6 & 6
 - (4) 6 & 4
78. Which of the following statement is true regarding H-bond :-
- (1) it is a directional bond
 - (2) bond strength is similar to O-H bond
 - (3) it is attraction between hydrogen and electronegative atom like as F, O, N
 - (4) it does not require minimum one lone pair of electron at electronegative atom

NEET(UG) 2017

79. Ionic mobility of which of the following alkali metal ions is lowest when aqueous solution of their salts are put under an electric field ?
 (1) K (2) Rb (3) Li (4) Na
80. Match the interhalogen compounds of column-I with the geometry in column II and assign the correct code.

Column-I		Column-II	
(a)	XX'	(i)	T-shape
(b)	XX' ₃	(ii)	Pentagonal bipyramidal
(c)	XX' ₅	(iii)	Linear
(d)	XX' ₇	(iv)	Square-pyramidal
		(v)	Tetrahedral

Code :

- | | | | |
|-----------|-------|-------|------|
| (a) | (b) | (c) | (d) |
| (1) (iii) | (i) | (iv) | (ii) |
| (2) (v) | (iv) | (iii) | (ii) |
| (3) (iv) | (iii) | (ii) | (i) |
| (4) (iii) | (iv) | (i) | (ii) |
81. Which of the following pairs of compounds is isoelectronic and isostructural ?
 (1) TeI₂, XeF₂ (2) IBr₂⁻, XeF₂
 (3) IF₃, XeF₂ (4) BeCl₂, XeF₂
82. The species, having bond angles of 120° is :-
 (1) ClF₃ (2) NCl₃ (3) BCl₃ (4) PH₃
83. Which of the following pairs of species have the same bond order ?
 (1) O₂, NO⁺ (2) CN⁻, CO
 (3) N₂, O₂⁻ (4) CO, NO

AIIMS 2017

84. Which of the following has longest bond length
 (1) C = O (2) N = O
 (3) C = C (4) C ≡ N
85. Which of the following has hydrogen bonded water
 (1) BaCl₂·2H₂O (2) CrCl₃·5H₂O
 (3) CuSO₄·5H₂O (4) CaCl₂·6H₂O

86. Which of the following is diamagnetic ?
 (1) NO (2) Cl₂⁺
 (3) S₂²⁻ (4) ClO
87. Number of lone pair in SO₂ on S & O are respectively
 (1) 0, 6 (2) 2, 4
 (3) 2, 5 (4) 1, 5

NEET(UG) 2018

88. Among CaH₂, BeH₂, BaH₂, the order of ionic character is
 (1) BeH₂ < CaH₂ < BaH₂
 (2) CaH₂ < BeH₂ < BaH₂
 (3) BeH₂ < BaH₂ < CaH₂
 (4) BaH₂ < BeH₂ < CaH₂
89. Magnesium reacts with an element (X) to form an ionic compound. If the ground state electronic configuration of (X) is 1s² 2s² 2p³, the simplest formula for this compound is
 (1) Mg₂X₃ (2) MgX₂ (3) Mg₂X (4) Mg₃X₂
90. Consider the following species:
 CN⁺, CN⁻, NO and CN
 Which one of these will have the highest bond order?
 (1) NO (2) CN⁻
 (3) CN⁺ (4) CN
91. Which one of the following elements is unable to form MF₆³⁻ ion ?
 (1) Ga (2) Al
 (3) B (4) In
92. In the structure of ClF₃, the number of lone pairs of electrons on central atom 'Cl' is
 (1) one (2) two
 (3) four (4) three

AIIMS 2018

93. Which statement is correct regarding N₂O₅ :-
 (1) It is anhydride of nitrous acid
 (2) It is brown color gas
 (3) It is paramagnetic
 (4) It exist as NO₂⁺ & NO₃⁻ in solid state
94. In s-p overlapping wrong statement is :-
 (1) Nodal plane formed in ABMO
 (2) Some orbital may have higher energy than combining orbitals
 (3) Combining orbitals should have same symmetry
 (4) Nodal plane is found in BMO

- 95.** Which pair not shows same shape :-
 (1) I_3^{-1} , $BeCl_2$ (2) BF_3 , ClO_3^{-1}
 (3) SO_2 , H_2O (4) XeF_4 , BrF_4^{-}
- 96.** In which at least one σ -bond of np (σ_{np}) orbital is present :-
 (1) B_2 (2) O_2
 (3) C_2 (4) Li_2
- 97.** Which is correct for dipole moment :-
 (1) $NH_3 < NF_3$ (2) $SO_2 < SO_3$
 (3) $NF_3 > BF_3$ (4) $SO_2 > H_2O$
- 98.** Which of the following is a molecular solid :-
 (1) solid SO_2 (2) SiC
 (3) C. (graphite) (4) ZnS
- 99.** The geometry and No. of lone pair on ClF_2^{-} and ClF_4^{-} are respectively :-
 (1) Square planar, 2 & Linear, 3
 (2) Square Planar, 3 & Linear, 2
 (3) Linear, 3 & Square planar, 2
 (4) Tetrahedral, 2 & linear, 3
- 100.** Wrong statement for N_3^{-} is :-
 (1) Isoelectronic with CO_2
 (2) Structure is linear
 (3) Both N-N bond lengths are unequal in hydrogen azide
 (4) Oxidation state for Nitrogen is same as in NH_2OH
- 101.** Bond order of which of following pair is different:-
 (1) F_2^{-} & O_2^{-2} (2) N_2^{-} & O_2^{+}
 (3) C_2 & B_2^{-2} (4) N_2 & O_2^{2+}
- 102.** Correct order for Bond angle is
 (1) $SO_2 > NH_3$ (2) $NH_3 < H_2O$
 (3) $SO_2 < H_2O$ (4) $H_2S > NH_3$
- 103.** Which of the following two species have same shape but different hybridisation
 (1) XeF_2 and ICl_2^{-} (2) NO_2^{+} and SO_2
 (3) $BeCl_2$ and NO_2^{+} (4) SO_2 and NH_2^{-}
- 104.** Which of the following has maximum %s character in N-H bond ?
 (1) N_2H_2 (2) NH_4^{+} (3) NH_3 (4) N_2H_4

EXERCISE-II (Previous Year Questions)

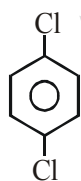
ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	2	4	2	3	2	1	3	3	1	2	2	2	4	1	3
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	1	2	4	1	3	3	3	4	2	2	1	1	3	1	1
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	1	2	2	3	4	3	2	3	3	4	3	3	1	1	3
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	2	3	3	3	1	4	2	1	3	2	4	2	2	1	3
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	4	2	2	1	1	2	1	1	4	2,3	1	2	2	2	2
Que.	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Ans.	2	2	3	3	1	2	3	2	3	3	3	4	1	4	2
Que.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	
Ans.	3	2	4	4	2	2	3	1	3	4	1	1	4	1	

EXERCISE-III (Analytical Questions)

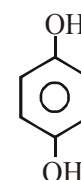
Check Your Understanding

- The thermal stability of alkaline earth metal carbonates MgCO_3 , CaCO_3 , BaCO_3 and SrCO_3 decreases as :
 (1) $\text{CaCO}_3 > \text{SrCO}_3 > \text{MgCO}_3 > \text{BaCO}_3$
 (2) $\text{BaCO}_3 > \text{SrCO}_3 > \text{MgCO}_3 > \text{CaCO}_3$
 (3) $\text{BaCO}_3 > \text{SrCO}_3 > \text{CaCO}_3 > \text{MgCO}_3$
 (4) $\text{MgCO}_3 > \text{CaCO}_3 > \text{SrCO}_3 > \text{BaCO}_3$
- Which of the following are isoelectronic and isostructural NO_3^- , CO_3^{2-} , ClO_3^- , SO_3
 (1) NO_3^- , CO_3^{2-} (2) SO_3 , NO_3^-
 (3) ClO_3^- , CO_3^{2-} (4) CO_3^{2-} , SO_3
- According to molecular orbital theory which of the following statement about the magnetic character and bond order is correct regarding O_2^+ ?
 (1) paramagnetic and bond order $< \text{O}_2$
 (2) paramagnetic and bond order $> \text{O}_2$
 (3) diamagnetic and bond order $< \text{O}_2$
 (4) diamagnetic and bond order $> \text{O}_2$
- The bond order in NO is 2.5 while that in NO^+ is 3. Which of the following statements is true for these two species ?
 (1) Bond length in NO^+ is equal to that in NO
 (2) Bond length in NO is greater than in NO^+
 (3) Bond length in NO^+ is greater than in NO
 (4) Bond length is unpredictable
- Number of σ and π bond in SO_4^{2-} are :-
 (1) 4, 2 (2) 4, 3
 (3) 4, 4 (4) 2, 3
- The paramagnetic property of the oxygen molecule is due to the presence of unpaired electrons present in :-
 (1) $(\sigma 2p_x)^1$ and $(\pi^* 2p_x)^1$
 (2) $(\sigma 2p_x)^1$ and $(\pi 2p_y)^1$
 (3) $(\pi^* 2p_y)^1$ and $(\pi^* 2p_x)^1$
 (4) $(\pi^* 2p_y)^1$ and $(\pi 2p_y)^1$
- The boiling point of methanol is greater than that of methylthiol because :-
 (1) There is intramolecular hydrogen bonding in methanol and intermolecular hydrogen bonding in methylthiol
 (2) There is intermolecular hydrogen bonding in methanol and no hydrogen bonding in methylthiol
 (3) There is no hydrogen bonding in methanol and intermolecular hydrogen bonding in methylthiol
 (4) There is intramolecular hydrogen bonding in methanol and no hydrogen bonding in methylthiol
- In silicon dioxide –
 (1) each silicon atom is surrounded by four oxygen atoms and each oxygen atom is bonded to two silicon atoms
 (2) each silicon atom is surrounded by two oxygen atoms and each oxygen atom is bonded to two silicon atoms
 (3) silicon atom is bonded to two oxygen atoms
 (4) there are double bond between silicon and oxygen atoms
- The number and type of bond between two carbon atom in calcium carbide are :
 (1) One sigma, one pi
 (2) One sigma, two pi
 (3) Two sigma, one pi
 (4) Two sigma, two pi
- What is the hybridization of oxygen atom in an alcohol molecule?
 (1) sp^3 (2) sp (3) sp^2 (4) sp^3d
- The decreasing values of bond angles from NH_3 (106°) to SbH_3 (91°) down group-15 of the periodic table is due to
 (1) decreasing lp – bp repulsion
 (2) increasing electronegativity
 (3) increasing bp – bp repulsion
 (4) increasing p-orbital character in sp^3
- In which of the following ionisation processes, the bond order has increased and the magnetic behaviour has changed
 (1) $\text{NO} \rightarrow \text{NO}^+$ (2) $\text{O}_2 \rightarrow \text{O}_2^+$
 (3) $\text{N}_2 \rightarrow \text{N}_2^+$ (4) $\text{C}_2 \rightarrow \text{C}_2^+$
- Which of the following hydrogen bonds is the strongest
 (1) $\text{F}-\text{H} \cdots \text{F}$ (2) $\text{O}-\text{H} \cdots \text{O}$
 (3) $\text{O}-\text{H} \cdots \text{F}$ (4) $\text{O}-\text{H} \cdots \text{N}$
- Which of the following species exhibits the diamagnetic behaviour
 (1) O_2^+ (2) O_2 (3) NO (4) O_2^{2-}
- The charge/size ratio of a cation determines its polarizing power. Which one of the following sequences represents the increasing order of the polarizing power of the cationic species, K^+ , Ca^{+2} , Mg^{+2} , Be^{+2}
 (1) $\text{Be}^{+2} < \text{K}^+ < \text{Ca}^{+2} < \text{Mg}^{+2}$
 (2) $\text{K}^+ < \text{Ca}^{+2} < \text{Mg}^{+2} < \text{Be}^{+2}$
 (3) $\text{Ca}^{+2} < \text{Mg}^{+2} < \text{Be}^{+2} < \text{K}^+$
 (4) $\text{Mg}^{+2} < \text{Be}^{+2} < \text{K}^+ < \text{Ca}^{+2}$

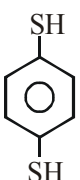
- 16.** Which one of the following constitutes a group of the isoelectronic species ?
 (1) C_2^{2-} , O_2^- , CO, NO
 (2) NO^+ , C_2^{2-} , CN^- , N_2
 (3) CN^- , N_2 , O_2^{2-} , C_2^{2-}
 (4) N_2 , O_2^- , NO^+ , CO
- 17.** Which of the following pairs of species have the same bond order ?
 (1) CN^- and NO^+ (2) CN^- and CN^+
 (3) O_2^- and CN^- (4) NO^+ and CN^+
- 18.** Among the following, the molecule with highest dipole moment is :-
 (1) CH_3Cl (2) CH_2Cl_2
 (3) $CHCl_3$ (4) CCl_4
- 19.** For a diatomic molecule the dipole moment is 1.2D and bond length is 1.0 Å then charge fraction on each atom is :-
 (1) 12% (2) 18%
 (3) 25% (4) 29%
- 20.** Which one of the following molecules is expected to exhibit diamagnetic behaviour ?
 (1) C_2 (2) N_2^-
 (3) O_2 (4) S_2
- 21.** For which of the following molecule significant $\mu \neq 0$
- 

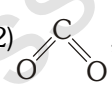
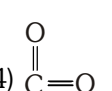
(a)



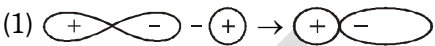
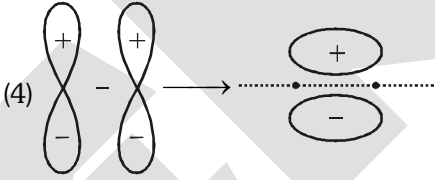
(b)
- 

(c)



(d)
- (1) Only (c) (2) (c) and (d)
 (3) Only (a) (4) (a) and (b)
- 22.** Which one of the following order is correct for the bond energies of halogen molecules :
 (1) $I_2 > Cl_2 > Br_2$ (2) $Br_2 > Cl_2 > I_2$
 (3) $I_2 > Br_2 > Cl_2$ (4) $Cl_2 > Br_2 > I_2$
- 23.** The boiling point of ICl is nearly 40°C higher than that of Br_2 although the two substances have the same relative molecular mass. This is because :-
 (1) ICl is ionic compound
 (2) I-Cl bond is stronger than Br - Br bond
 (3) ICl is polar covalent molecule while Br_2 is non polar
 (4) IP of Iodine is less than that of Br.
- 24.** Nitrogen does not form NF_5 because
 (1) Nitrogen is member of V group
 (2) It contains no empty d-orbital
 (3) The bond energy of $N \equiv N$ is very high
 (4) Inert pair effect exists in the molecule
- 25.** CO_2 is a gas, while SiO_2 is a solid but both are-
 (1) Covalent containing π -bond
 (2) Molecules having $p\pi - d\pi$ bonding
 (3) Acidic
 (4) Discrete molecules
- 26.** Which trihalide ion is unknown due to absence of vacant d-orbital?
 (1) F_3^- (2) Br_3^- (3) I_3^- (4) Cl_3^-
- 27.** Dipole moment is shown by
 (1) 1, 4 - dichlorobenzene
 (2) Cis 1, 2-dichloro ethene
 (3) Trans-1, 2- dichloro ethene
 (4) benzene
- 28.** Experiment shows that H_2O has a dipole moment where as CO_2 has not. Point out the structures which best illustrate these facts -
 (1) $O=C=O$, $H-O-H$
 (2) , $H-O-H$
 (3) $O=C=O$, 
 (4) , 
- 29.** In which of the species, bond order increases on removing one electron:-
 (a) NO (b) CN^- (c) O_2 (d) CO
 Correct answer is -
 (1) b and d (2) a and c
 (3) b, d and c (4) b and c
- 30.** Increasing order of bond length in NO, NO^+ and NO^- is :-
 (1) $NO > NO^- > NO^+$ (2) $NO^+ < NO < NO^-$
 (3) $NO < NO^+ < NO^-$ (4) $NO < NO^+ = NO^-$

31. N_2 and O_2 are converted into monoanions, N_2^- and O_2^- respectively. Which of the following statements is wrong ?
 (1) In N_2^- , N-N bond weakens
 (2) In O_2^- , O-O bond order increases
 (3) In O_2^- , O-O bond order decreases
 (4) N_2^- becomes paramagnetic
32. N_2 and O_2 are converted into monocations, N_2^+ and O_2^+ respectively. Which of the following is wrong ?
 (1) In N_2^+ , N-N bond weakens
 (2) In O_2^+ , the O-O bond order increases
 (3) In O_2^+ , paramagnetism decreases
 (4) N_2^+ becomes diamagnetic
33. The pair of compounds which can form a co-ordinate bond is :-
 (1) $(C_2H_5)_3B$ and $(CH_3)_3N$
 (2) HCl and HBr
 (3) BF_3 and NH_3
 (4) (1) & (3) both
34. In the neutralization process of NH_3 and $AlCl_3$ the compound formed will have the bonding
 (1) Ionic (2) Covalent
 (3) Coordinate (4) Hydrogen
35. Glycerol is more viscous than glycol the reason is :-
 (1) Higher molecular wt.
 (2) More covalent
 (3) More extent of hydrogen bonding
 (4) Complex structure
36. Incorrect order of viscosity :-
 (1) $H_2SO_4 > HNO_3$
 (2) $H_2O > CH_3OH$
 (3) o-nitro phenol > p-nitro phenol
 (4) Glycol > ether
37. Which of the following can form H-Bond with water
 (1) HCN (2) NH_4F
 (3) $NaHCO_3$ (4) All
38. Which of the following does not form a hydrogen bond with water
 (1) $(CH_3)_2CO$ (2) CH_3CN
 (3) CH_3OH (4) C_2H_6
39. A metal M readily forms its sulphate MSO_4 , which is water soluble. It forms an insoluble hydroxide $M(OH)_2$ which is soluble in $NaOH$ solution, then M is :-
 (1) Mg (2) Ca (3) Be (4) Ba

40. Which is most ionic :
 (1) P_2O_5 (2) MnO
 (3) CrO_3 (4) Mn_2O_7
41. Lithium chloride is highly soluble in -
 (1) C_6H_6 (2) H_2O
 (3) D_2O (4) All
42. Which of the following linear combinations of atomic orbitals is incorrectly depicted ?
 (1) 
 (2) 
 (3) 
 (4) 
43. Which of the following molecule contains net π -bond only
 (1) B_2 (2) C_2
 (3) C_2^{-2} (4) Both 1 & 2
44. Pick out the incorrect statement :-
 (1) sp^3d hybridisation involves $d_{x^2-y^2}$ orbital
 (2) Hybridised orbital form σ -bond when overlaps with other orbitals.
 (3) SF_2 molecule is more polar than CS_2
 (4) o-nitrophenol is more volatile than p-nitrophenol.
45. The group of substances in which at melting point covalent bond becomes weak :-
 (1) $NaCl$, KCl , $CaCl_2$
 (2) $I_{2(s)}$, $CH_{4(s)}$, dry ice
 (3) B_4C , diamond, SiC
 (4) All of the above
46. The incorrect statement is :-
 (1) π^* p have two nodal planes
 (2) Bond order of HeH^+ is 0.5
 (3) In NCO^- , C is sp hybridised
 (4) O_3 is polar while O_2 is non polar

47. Solid CH_4 is
 (1) Molecular solid
 (2) Ionic solid
 (3) Covalent solid
 (4) None of these
48. Which compound has the weakest bond
 (1) Diamond
 (2) Neon(solid)
 (3) KCl
 (4) water (ice)
49. Which of the following order is incorrect ?
 (1) $\text{Na}_2\text{O} < \text{K}_2\text{O} < \text{Rb}_2\text{O}$ (basic nature)
 (2) $\text{CH}_4 > \text{SiH}_4 > \text{GeH}_4 > \text{SnH}_4$ (Stability of hydride)
 (3) $\text{NH}_3 < \text{PH}_3 < \text{AsH}_3$ (basic nature)
 (4) $\text{N}_2\text{O}_5 > \text{P}_2\text{O}_5 > \text{As}_2\text{O}_5$ (acidic nature)
50. Which order are correct ?
 (a) Thermal stability : $\text{BeSO}_4 < \text{MgSO}_4 < \text{CaSO}_4 < \text{SrSO}_4 < \text{BaSO}_4$
 (b) Basic Nature : $\text{ZnO} > \text{BeO} > \text{MgO} > \text{CaO}$
 (c) Solubility in water : $\text{LiOH} > \text{NaOH} > \text{KOH} > \text{RbOH} > \text{CsOH}$
 (d) Melting point : $\text{NaCl} > \text{KCl} > \text{RbCl} > \text{CsCl} > \text{LiCl}$
 (1) a, b, d (2) b, c
 (3) a, d (4) All correct
51. Which of the following order is not correct ?
 (1) $\text{SF}_2 > \text{SF}_4 > \text{SF}_6$ (ionic character)
 (2) $\text{AlF}_3 < \text{Al}_2\text{O}_3 < \text{AlN}$ (covalent character)
 (3) $\text{CaCl}_2 < \text{SnCl}_2 < \text{CdCl}_2$ (covalent character)
 (4) $\text{ZnCl}_2 < \text{CdCl}_2 < \text{HgCl}_2$ (ionic character)
52. Match the column :-
 (A) $\text{NO}_3^- + \text{HCl}$ (p) Kessome attraction
 (B) $\text{Xe} + \text{H}_2\text{O}$ (q) Debye attraction
 (C) $\text{CH}_3-\overset{\text{O}}{\underset{\text{O}}{\text{C}}}-\text{CH}_3 + \text{CH}_3-\text{C}\equiv\text{N}$ (r) London force
 (D) $\text{CO}_2 + \text{CS}_2$ (s) Ion-dipole attraction
 (1) A-s, B-q, C-p, D-r (2) A-q, B-s, C-p, D-r
 (3) A-s, B-p, C-q, D-r (4) A-s, B-q, C-r, D-p
53. Correct order is :-
 (1) $\text{MgO} < \text{NiO} < \text{K}_2\text{O} < \text{Cs}_2\text{O}$ (basic strength)
 (2) $\text{Cs}^+ < \text{K}^+ < \text{Mg}^{2+} < \text{Be}^{2+}$ (ionic mobility in aqueous solution)
 (3) $\text{Li}_2\text{CO}_3 < \text{Na}_2\text{CO}_3 < \text{K}_2\text{CO}_3$ (solubility order)
 (4) $\text{NH}_3 < \text{PH}_3 < \text{AsH}_3$ (thermal stability)
54. Which of the following statement is not correct ?
 (1) NO has one unpaired electron in ABMO.
 (2) N-H bond length is higher in N_2H_4 as compared to N_2H_2
 (3) Both PF_5 and IF_5 have identical shape
 (4) Both SO_2 and BrF_3 are polar and planar
55. In which of the following process, hybridisation of central atom remains same ?
 (1) Solidification of PCl_5
 (2) Dimerisation of NO_2
 (3) Trimerisation of SO_3
 (4) BF_3 forms adduct with lewis base

EXERCISE-III (Analytical Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	1	2	2	1	3	2	1	2	1	4	1	1	4	2
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	2	1	1	3	1	2	4	3	2	3	1	2	3	2	2
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	2	4	4	3	3	3	4	4	3	2	1	4	4	1	3
Que.	46	47	48	49	50	51	52	53	54	55					
Ans.	2	1	2	3	3	4	1	3	3	2					

EXERCISE-IV (Assertion & Reason)

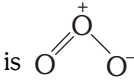
Target AIIMS

Directions for Assertion & Reason questions

These questions consist of two statements each, printed as Assertion and Reason. While answering these Questions you are required to choose any one of the following four responses.

- (A) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (B) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (C) If Assertion is True but the Reason is False.
 (D) If both Assertion & Reason are false.

1. **Assertion** : Ionic bonds are formed between metal and nonmetals
Reason : In ionic bonds electrons are shared
 (1) A (2) B (3) C (4) D
2. **Assertion** : D_2O is better solvent for ionic compounds, than H_2O .
Reason : dielectric constant of D_2O is higher than H_2O .
 (1) A (2) B (3) C (4) D
3. **Assertion** : Hydrated ionic radius of Li^+ is maximum in IA group elements.
Reason : Lithium is a metal
 (1) A (2) B (3) C (4) D
4. **Assertion** : $LiCl$ exhibits covalent character.
Reason : Lithium is lightest metal.
 (1) A (2) B (3) C (4) D
5. **Assertion** : K_2CO_3 do not gives CO_2 gas on heating.
Reason : Value of $\sqrt{\phi}$ for K^+ is high.
 (1) A (2) B (3) C (4) D
6. **Assertion** : CO_2 molecule is non-polar while SO_2 is polar.
Reason : Carbon atom is smaller than sulphur.
 (1) A (2) B (3) C (4) D
7. **Assertion** : CH_3OH is soluble in water
Reason : CH_3OH is ionic in nature
 (1) A (2) B (3) C (4) D
8. **Assertion** : Boiling point of H_2O is greater than C_2H_5OH .
Reason : Molecular wt. of H_2O is higher than C_2H_5OH .
 (1) A (2) B (3) C (4) D
9. **Assertion** : NO^+ is more stable than NO^- .
Reason : NO^+ do not have electron in antibonding orbitals.
 (1) A (2) B (3) C (4) D
10. **Assertion** : In MgO electrovalency of Mg is 2.
Reason : Mg shares two electrons with oxygen.
 (1) A (2) B (3) C (4) D
11. **Assertion** : Ionic reactions are faster than molecular reactions.
Reason : Ionic bonds are weaker than covalent bonds.
 (1) A (2) B (3) C (4) D
12. **Assertion** : LiI is more soluble in water than LiF .
Reason : LiI has more ionic character.
 (1) A (2) B (3) C (4) D
13. **Assertion** : Bond energy of $H-H$ bond is greater than $Cl-Cl$ bond.
Reason : H_2 is more covalent than Cl_2 .
 (1) A (2) B (3) C (4) D
14. **Assertion** : Ionic compounds tend to be non-volatile.
Reason : Inter ionic forces in these compounds are strong.
 (1) A (2) B (3) C (4) D
15. **Assertion** : NCl_3 has pyramidal shape
Reason : In NCl_3 central atom is sp^3 hybridised.
 (1) A (2) B (3) C (4) D
16. **Assertion** : BF_3 is planar while NF_3 is non-planar molecule.
Reason : $B-F$ bond is more polar than $N-F$ bond.
 (1) A (2) B (3) C (4) D
17. **Assertion** : Density of ice is greater than water
Reason : In ice H_2O molecules are closely packed.
 (1) A (2) B (3) C (4) D
18. **Assertion** : Bond order of O_2 and BN is same.
Reason : O_2 and BN are isoelectronic
 (1) A (2) B (3) C (4) D

- 19. Assertion** : p-nitrophenol is more viscous than o-nitrophenol.
Reason : In p-nitrophenol, intermolecular H-bonding occurs.
 (1) A (2) B (3) C (4) D
- 20. Assertion** : H_2O_2 is not used as solvent for ionic compounds
Reason : Dielectric constant of H_2O_2 is low
 (1) A (2) B (3) C (4) D
- 21. Assertion** : Ionic compounds exhibits electrical conductivity in solution state.
Reason : In solution state electrons of ionic compounds are free.
 (1) A (2) B (3) C (4) D
- 22. Assertion** : p-nitrophenol is more volatile than o-nitrophenol.
Reason : Molecular wt. of p-nitrophenol is higher than o-nitrophenol.
 (1) A (2) B (3) C (4) D
- 23. Assertion** : N_2^+ is more stable than N_2^-
Reason : N_2^+ has less electrons in antibonding orbitals.
 (1) A (2) B (3) C (4) D
- 24. Assertion** : SO_4^{2-} is square planar in shape
Reason : SO_4^{2-} has sp^3d hybridisation
 (1) A (2) B (3) C (4) D
- 25. Assertion** : NF_3 molecule is more polar than NH_3 molecule.
Reason : NF_3 is pyramidal while NH_3 is trigonal planar.
 (1) A (2) B (3) C (4) D
- 26. Assertion** : OF_4 does not exists.
Reason : Empty d-orbitals are absent in valence shell of oxygen.
 (1) A (2) B (3) C (4) D
- 27. Assertion** : BeCl_2 shows covalent character.
Reason : More polarization of Cl^- by Be^{+2} .
 (1) A (2) B (3) C (4) D
- 28. Assertion** : MgO and NaF are isomorphous
Reason : Crystal structure of MgO and NaF is identical.
 (1) A (2) B (3) C (4) D
- 29. Assertion** : NF_3 molecule is polar.
Reason : N-F bonds are polar.
 (1) A (2) B (3) C (4) D
- 30. Assertion** : NaCl is soluble in non polar solvents.
Reason : NaCl is a non-polar covalent compound.
 (1) A (2) B (3) C (4) D
- 31. Assertion** : Sigma bonds are stronger than π bonds.
Reason : Sigma bonds are covalent bonds.
 (1) A (2) B (3) C (4) D
- 32. Assertion** : CO_2 is non polar while SO_2 is polar molecule.
Reason : S-O bonds are polar while C-O non polar.
 (1) A (2) B (3) C (4) D
- 33. Assertion** : Both H_2O and SnCl_2 are bent molecules.
Reason : Both H_2O and SnCl_2 are sp^2 hybridised.
 (1) A (2) B (3) C (4) D
- 34. Assertion** : NO is paramagnetic in nature.
Reason : Bond order of NO is 2.5.
 (1) A (2) B (3) C (4) D
- 35. Assertion** : Nitrogen is unreactive at room temperature but becomes reactive at elevated temperature (on heating or in the presence of catalysts).
Reason : In nitrogen molecule, there is extensive delocalization of electrons.
 (1) A (2) B (3) C (4) D
- 36. Assertion** : NO_3^- is planar.
Reason : N in NO_3^- is sp^2 and no lone pair at central atom.
 (1) A (2) B (3) C (4) D
- 37. Assertion** : The electronic structure of O_3 is 
Reason :  Structure is not allowed because octet around 'O' can not be expanded.
 (1) A (2) B (3) C (4) D
- 38. Assertion** : LiCl is predominantly a covalent compound.
Reason : Electronegativity difference between 'Li' and 'Cl' is too small.
 (1) A (2) B (3) C (4) D

39. Assertion :- Ozone is a powerful oxidizing agent in comparison to O_2 .

Reason :- Ozone is diamagnetic but O_2 is paramagnetic.

(1) A (2) B (3) C (4) D

40. Assertion :- B_2 molecule is diamagnetic.

Reason :- The highest occupied molecular orbital is of σ type.

(1) A (2) B (3) C (4) D

41. Assertion :- BH_4^- is known while BH_6^{3-} is not.

Reason :- B has very small atomic size.

(1) A (2) B (3) C (4) D

42. Assertion :- Some molecules are polar.

Reason :- The centre of negative charge and positive charge do not coincide each other in some molecule.

(1) A (2) B (3) C (4) D

43. Assertion :- $R_3P=O$ exists but $R_3N=O$ does not exist

Reason :- P is more electronegative than N

(1) A (2) B (3) C (4) D

44. Assertion :- ClF_3 has T-shape structure.

Reason :- It has two lone pair arrange at 180° (Angle).

(1) A (2) B (3) C (4) D

45. Assertion :- Bond Dissociation energy is $F_2 > Cl_2$.

Reason :- Cl_2 have more electronic repulsion than F_2 .

(1) A (2) B (3) C (4) D

46. Assertion :- H_2O molecule can form four hydrogen bonds.

Reason :- Two lone pairs and two hydrogens are directly attached with oxygen atoms.

(1) A (2) B (3) C (4) D

47. Assertion :- Bond angle in  is more than expected.

Reason :- It is due to lp-lp repulsion.

(1) A (2) B (3) C (4) D

EXERCISE-IV (Assertion & Reason)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	4	2	2	3	2	3	3	3	3	3	3	3	1	2
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	2	4	3	1	3	3	4	1	4	4	1	1	1	2	4
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	2	3	3	2	3	1	1	3	2	4	2	1	3	3	4
Que.	46	47													
Ans.	1	3													

IMPORTANT NOTES

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